HP IO Accelerator User Guide

Abstract

This document describes software requirements for all relevant HP IO Accelerators using Linux, Windows, ESXi, and Solaris operating systems. This document is intended for system administrators who plan to install and use HP IO Accelerators. It is helpful to have previous experience with HP IO Accelerators. This user guide is intended for IO Accelerator VSL software release 4.1.2 or later.

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Contents

Introduction 7
- Overview 7

Software installation 8
- IO Accelerator VSL software installation for Linux 8
  - Installation overview for Linux 8
  - Installing RPM packages in Linux 9
  - Installing DEB packages in Linux 11
  - Loading the IO Accelerator VSL driver in Linux 13
- IO Accelerator VSL software installation for Windows 15
  - Installation overview for Windows 16
  - Installing software on a Windows system 16
  - Silent installation option in Windows 17
- IO Accelerator VSL software installation for ESXi 17
  - VMDirectPathIO 18
  - Command-line installation 18
  - Installation overview for ESXi 18
  - Downloading the ESXi software 19
  - Installing the IO Accelerator VSL on ESXi 5.x 20
- IO Accelerator VSL software installation for Solaris 21
  - Installation overview for Solaris 21
  - Installing the software in Solaris 22
  - Loading the IO Accelerator VSL driver 22
- Upgrading the firmware 23

Configuration 24
- Configuration overview 24
- Enabling additional PCIe power override 24
  - Automatic power draw 24
  - Enabling full slot power draw 24
  - Enabling the override parameter 25
- Discard (TRIM) support 27
  - Discard (TRIM) in Linux 27
  - Discard (TRIM) in Windows Server 2008 R2 and later 27
- Setting the IO Accelerator VSL options in Linux 28
  - One-time configuration in Linux 28
  - Persistent configuration in Linux 28
- Using the device as swap in Linux 28
- Using the Logical Volume Manager in Linux 30
  - Mounting filesystems using the init script in Linux 30
  - Configuring RAID using the Logical Volume Manager in Linux 30
- Configuring RAID using mdadm in Linux 31
  - Mounting arrays in Linux 31
  - RAID0 in Linux 32
  - RAID1 in Linux 32
  - RAID10 in Linux 33
- Setting the IO Accelerator VSL options in Windows 33

Device naming in Windows ... 33
Adding a filesystem in Windows ... 33
Creating a RAID configuration in Windows ... 34
Using the device as a page files store in Windows ... 34
Configuring the device to support VM disks in ESXi ... 35
Modifying a VMware resource pool to reserve memory in ESXi ... 36
Using the device as swap in Solaris ... 37
Configuring a ZFS pool in Solaris ... 37
Using a single device in Solaris ... 37
RAID0 in Solaris ... 37
RAID1 in Solaris ... 37
RAID10 in Solaris ... 38

Performance and tuning ... 39
Overview ... 39
Disabling CPU frequency scaling ... 39
Limiting ACPI C-states ... 39
Setting NUMA affinity in Linux and Windows ... 40
Setting the interrupt handler affinity in Linux and Windows ... 40
Linux IRQ balancing ... 40
Windows IRQ policy ... 41

Monitoring and managing devices ... 42
Overview ... 42
Management tools ... 42
Management tools for Linux, Windows, and Solaris ... 42
Management tools for ESXi ... 43
Example conditions to monitor ... 43
Device status ... 43
Temperature ... 44
Health reserves percentage ... 45
Write (health reserves) status ... 45
Device LED indicator ... 45

Maintenance ... 47
Overview ... 47
Uninstalling the IO Accelerator VSL software ... 47
Uninstalling the IO Accelerator VSL software in Linux ... 47
Uninstalling the IO Accelerator VSL software in Windows ... 48
Uninstalling the IO Accelerator VSL software in ESXi 5.x ... 49
Uninstalling the IO Accelerator VSL software in Solaris ... 49
Unloading or disabling the software driver ... 49
Unloading the software driver in Linux ... 49
Upgrading the kernel in Linux ... 49
Disabling the IO Accelerator VSL software in Linux ... 49
Unloading or disabling the software driver in ESXi ... 49
Disabling the IO Accelerator VSL software in Solaris ... 50
Loading or enabling the driver ... 50
Loading or enabling the IO Accelerator VSL driver in ESXi ... 50
Enabling the IO Accelerator VSL in Solaris ... 51
Disabling auto-attach ... 51
Disabling auto-attach in Linux ... 51
Disabling auto-attach in Windows ... 52
Disabling auto-attach in ESXi ... 52

Disabling auto-attach in Solaris 52
Unmanaged shutdown issues 53
Improving rescan times 53
Upgrading the software with a RAID configuration in Windows 56
Defragmentation in Windows 56
Detaching an IO Accelerator device in ESXi 56

Utilities 58
Utilities reference 58
fio-attach 59
fio-beacon 60
fio-bugreport 60
fio-bugreport - Linux 61
fio-bugreport - Windows 61
fio-bugreport - ESXi 62
fio-bugreport - Solaris 63
fio-config 64
fio-config options 64
fio-detach 66
fio-firmware 67
fio-format 68
fio-pci-check 69
fio-status 69
fio-sure-erase 72
fio-update-iodrive 74

Module parameters 76
Using module parameters 76

Monitoring IO Accelerator health 79
Health metrics 79
Health monitoring techniques 79
Software RAID and health monitoring 80

VMDirectPathIO 81
Working with IO Accelerator devices and VMDirectPathIO 81

NUMA configuration 82
Introduction to NUMA architecture 82
Using the numa_node_override parameter in Linux 82
Determining the device ID in Linux 82
numa_node_override parameter in Linux 83
Advanced configuration in Linux 83
Using the FIO_AFFINITY parameter in Windows 84
Determining the bus number in Windows 84
FIO_AFFINITY parameter in Windows 84
Advanced configuration in Windows 85
Checking the log for errors in Windows 85

Using Windows page files with the IO Accelerator 87
Introduction to Windows page files 87
Configuring IO Accelerator device paging support 87
RAM consumption 87
Non-paged memory pool 88
Enabling and disabling paging support 88

Windows page file management 89
Setting up page files 89
System drive paging file configuration 89
Guaranteeing minimum committable memory 90
Verifying page file operation 90
Virtual Memory performance 91

Troubleshooting event log messages 92
Troubleshooting event log messages in Windows 92
Verbose event log parameter 92
Viewing logs 92
Windows event log error messages 92
Windows event log informational messages 93

Manual installation 94
Overview 94
Manual installation in Windows Server 2008 R2 and Windows Server 2012 94
Installation wizard 95

Regulatory information 96
Safety and regulatory compliance 96
Turkey RoHS material content declaration 96
Ukraine RoHS material content declaration 96
Warranty information 96

Support and other resources 97
Before you contact HP 97
HP contact information 97
Customer Self Repair 97

Acronyms and abbreviations 105

Documentation feedback 107

Index 108

Introduction

Overview

The IO Accelerator platform combines IO Accelerator VSL software with IO Accelerator hardware to take enterprise applications and databases to the next level.

Performance

The IO Accelerator platform provides consistent microsecond latency access for mixed workloads, multiple gigabytes per second access and hundreds of thousands of IOPS from a single product. The sophisticated IO Accelerator architecture allows for nearly symmetrical read and write performance with best-in-class low queue depth performance, making the IO Accelerator platform ideal across a wide variety of real world, high-performance enterprise environments.

The IO Accelerator platform integrates with host system CPUs as flash memory to give multiple (and mostly idle) processor cores, direct and parallel access to the flash. The cut-through architecture of the platform gives systems more work per unit of processing, and continues to deliver performance increases as CPU power increases.

Endurance

The IO Accelerator platform offers best-in-class endurance in all capacities, which is crucial for caching and write-heavy databases and applications.

Reliability

The IO Accelerator platform eliminates concerns about reliability like NAND failures and excessive wear. The intelligent, self-healing feature called Adaptive Flashback provides complete, chip-level fault tolerance. Adaptive Flashback technology enables an IO Accelerator product to repair itself after a single chip or a multi-chip failure without interrupting business continuity.

Software installation

IO Accelerator VSL software installation for Linux

Before continuing with the installation of the software:

1. Confirm the operating system is one of the supported operating systems listed in the IO Accelerator release notes.
2. Before installing the IO Accelerator VSL software, make sure you have properly installed the IO Accelerator devices. For full details and hardware requirements, see the *IO Accelerator Hardware Installation Guide*.

All command require administrator privileges. Use `sudo` or log in as root to run the installation.

Installation overview for Linux

IMPORTANT: This version of the IO Accelerator VSL software only supports third generation devices, such as the SX300 and PX600 devices. It does not support devices that were compatible with IO Accelerator VSL software version 3.x.x or earlier.

IMPORTANT: If you plan to upgrade the kernel when the IO Accelerator VSL software is installed, you must follow the procedure in "Upgrading the kernel in Linux (on page 49)."

1. Download the latest version of the software on the HP website (http://www.hp.com/go/support).
2. If you have a previous version of the IO Accelerator VSL software installed, uninstall the IO Accelerator VSL software and the utilities. For more information, see "Uninstalling the IO Accelerator VSL software (on page 47)." After you uninstall the software, return to this procedure.
3. Install the IO Accelerator VSL software. Depending on the format your distribution uses for installation packages, follow the instructions in one of the following sections:
   - "Installing RPM packages in Linux (on page 9)"
   - "Installing DEB packages in Linux (on page 11)"

   The installation instructions help determine which of the package types is supported by your kernel:
   - A pre-compiled binary package
   - A source-to-build package
4. Install utilities and management software, included in the driver installation instructions.
5. Follow the instructions in "Loading the IO Accelerator VSL driver in Linux (on page 13)."
6. Consider the options in "Setting the IO Accelerator VSL options in Linux (on page 28)."
7. Determine if you need to upgrade the firmware to the latest version. For more information, see "Upgrading the firmware (on page 23)."

Installing RPM packages in Linux

To install the Linux IO Accelerator VSL software and utilities on SUSE, RHEL, and CentOS:

1. You need to install a version of the IO Accelerator VSL software that is built for your kernel. To determine what kernel version is running on your system, use the following command at a shell prompt:

   ```
   $ uname -r
   ```

2. Compare your kernel version with the binary versions of the software available on the HP website (http://www.hp.com/support).
   - If a binary version of the software corresponds to your kernel version, download that binary version. For example:

     ```
     iomemory-vsl4-<kernel-version>-<VSL-version>.x86_64.rpm
     ```

   - If no binary version of the software corresponds to your kernel, download the source package. For example:

     ```
     iomemory-vsl4-<VSL-version>.src.rpm
     ```

   Exact package names might vary, depending on the software and kernel version chosen.

   Use the source package that is made for your distribution. Source packages from other distributions are not guaranteed to work.

3. Download the support RPM packages you need from the HP website (http://www.hp.com/support). The packages provide utilities, firmware, and other files.

   The following table provides examples of RPM support packages.

Package	What is installed
fio-util-<VSL-version>.x86_64.rpm	IO Accelerator VSL utilities – Recommended
fio-firmware-fusion_<version>-<date>-1.0.noarch.rpm	Firmware archive package, installs the firmware archive file in a specific location (/usr/share/fio/firmware) – Optional
fio-firmware-fusion_<version>-<date>.fff	Firmware archive file (not an installer package) used to upgrade the firmware; make note of where you store this file – Recommended
libvsl-<version>.x86_64.rpm	SDK libraries needed for management tools – Recommended. For more information on available management tools, see "Monitoring and managing devices (on page 42)."
lib32vsl-<version>.i386.rpm	SDK libraries needed for working with 32-bit management applications – Optional and only available for certain distributions. You must have a full set of 32-bit system libraries installed (32-bit compatibility layer installed on your 64-bit system) before you can install this package.
fio-common-<VSL-version>.x86_64.rpm	Files required for the init script – Recommended
fio-sysvinit-<VSL-version>.x86_64.rpm	Init script – Recommended For more information, see "Loading the IO Accelerator VSL driver in Linux (on page 13)."

4. Change to the directory where you downloaded the installation packages.
5. If needed, build the IO Accelerator VSL software from source.
   - If you downloaded a binary version of the software: continue to the next step.

- If you downloaded the software source package: follow the instructions in "Building the software from source for RPM packages in Linux (on page 10)."

6. Enter the following command to install the custom-built software package. Use the package name that you just copied/downloaded into that directory.

```
$ rpm -Uvh iomemory-vsl4-<kernel-version>-<VSL-version>.x86_64.rpm
```

7. Enter the following commands to install the support files:

```
$ rpm -Uvh lib*.rpm
rpm -Uvh fio*.rpm
```

If the installation of the `fio-util-<VSL-version>.x86_64.rpm` package fails due to missing dependencies, you need to install the missing packages before you run the install command again. For example:

```
$ yum install lsof pciutils
```

The IO Accelerator VSL software and utilities are installed to the following locations.

Package type	Installation location
IO Accelerator VSL software	/lib/modules/<kernel-version>/extra/fio/iomemory-vsl4.ko
Utilities	/usr/bin

You may also install the IO Accelerator Management Tool software (optional GUI management software). IO Accelerator Management Tool software and documentation are available as a separate download in the IO Accelerator Management Tool Software download folder.

After the packages are installed, continue to "Loading the IO Accelerator VSL driver in Linux (on page 13)."

Building the software from source for RPM packages in Linux

You only need to perform the following additional instructions if you downloaded the source package.

To build the software for DEB packages from the source package:

1. Install the prerequisite files for your kernel version.

   Some of the prerequisite packages might already be in the default operating system installation. If the system is not configured to get packages over the network, then you might need to mount your install CD/DVD.

   - For RHEL/CentOS 5/6, you need `kernel-devel`, `rpm-build`, `GCC4`, `make`, and `rsync`.

     ```
     $ yum install rsync tar gcc make kernel-devel- 'uname -r' rpm-build
     ```

     The command forces `yum` to download the exact versions for the kernel. If the exact versions are no longer available in the repository, then you must manually search for them on the Internet.

     If you use the UEK kernel in Oracle Linux, replace `kernel -devel` with `kernel-uek-devel`.

   - For SLES 10/11/12, you need `kernel-syms`, `make`, `rpm-build`, `GCC4`, and `rsync`.

     ```
     $ zypper install kernel-syms make rpm gcc rsync
     ```

2. To build an RPM installation package for the current kernel, navigate to the directory with the downloaded source RPM file and run the following command:

   ```
   $ rpmbuild --rebuild iomemory-vsl4-<VSL-version>.src.rpm
   ```

   If you use the UEK kernel, you might need to use the `--nodeps` option.

   When using an RPM source package for a non-running kernel, run the following command:

```
$ rpmbuild --rebuild --define 'rpm_kernel_version <kernel-version>'
iomemory-vsl-<VSL-version>.src.rpm
```

3. The new RPM package is located in a directory that is indicated in the output from the `rpmbuild` command. To find it, look for the `Wrote` line. In the following example, the RPM packages are located in the `/usr/src/redhat/RPMS/x86_64/` directory.

```
...
Processing files: iomemory-vsl-source-<version>-1.0.x86_64.rpm
Requires(rpmlib): rpmlib(PayloadFilesHavePrefix) <=
4.0-1rpmlib(CompressedFileNames) <= 3.0.4-1
Obsoletes: iodrive-driver-source
Checking for unpackaged file(s):
/usr/lib/rpm/check-files/var/tmp/iomemory-vsl-<version>-root
Wrote:
/usr/src/redhat/RPMS/x86_64/iomemory-vsl-2.6.18-128.el5-<version>-1.0.x8
6_64.rpm
Wrote:
/usr/src/redhat/RPMS/x86_64/iomemory-vsl-source-<version>-1.0.x86_64.rpm
```

In this example, `iomemory-vsl4-2.6.18-128.el5-<version>-1.0.x86_64.rpm` is the package you use.

4. Copy the custom-built software installation RPM package into the directory where you downloaded the installation packages, and then navigate to that directory.
5. Continue to the next step in "Installing RPM packages in Linux (on page 9)."

Installing DEB packages in Linux

To install the Linux IO Accelerator VSL software and utilities on Ubuntu:

1. You need to install a version of the IO Accelerator VSL software that is built for your kernel. To determine what kernel version is running on your system, use the following command at a shell prompt:

```
$ uname -r
```

2. Compare your kernel version with the binary versions of the software available on the HP website (http://www.hp.com/support).
   - If a binary version of the software corresponds to your kernel version, download that binary version. For example:

     `iomemory-vsl4-<kernel-version>-<VSL-version>.amd65.deb`

   - If no binary version of the software corresponds to your kernel, download the source package. For example:

     `iomemory-vsl4-<VSL-version>.tar.gz`

   Exact package names might vary, depending on the software and kernel version chosen.

   Use the source package that is made for your distribution. Source packages from other distributions are not guaranteed to work.

3. Download the support packages you need from the HP website (http://www.hp.com/support). The packages provide utilities, firmware, and other files.

Package	What is installed
`fio-util-<VSL-version>.x86_64.deb`	IO Accelerator VSL utilities – Recommended

Package	What is installed
fio-firmware-fusion_<version>-<date>-<number>_all.deb	Firmware archive package, installs the firmware archive file in a specific location (/usr/share/fio/firmware) – Optional
fio-firmware-fusion_<version>-<date>.fff	Firmware archive file (not an installer package) used to upgrade the firmware; make note of where you store this file – Recommended
libvsl-<version>.x86_64.deb	SDK libraries needed for management tools – Recommended. For more information on available management tools, see "Monitoring and managing devices (on page 42)."
fio-common-<VSL-version>.x86_64.deb	Files required for the init script – Recommended
fio-sysvinit-<VSL-version>.x86_64.deb	Init script – Recommended For more information, see "Loading the IO Accelerator VSL driver in Linux (on page 13)."

4. Change to the directory where you downloaded the installation packages.
5. If needed, build the IO Accelerator VSL software from source.
   - If you downloaded a binary version of the software: continue to the next step.
   - If you downloaded the software source package: follow the instructions in "Building the software from source for DEB packages in Linux (on page 12)."
6. Use the following command to install each package you want to install, adding package names as necessary.

```
$ dpkg -i iomemory-vsl4-<kernel-version>_<VSL-version>_amd64.deb
fiofirmware_<firmware-version>_all.deb fio-util_<VSL-version>_amd64.deb
```

The IO Accelerator VSL software and utilities are installed to the following locations.

Package type	Installation location
IO Accelerator VSL software	/lib/modules/<kernel-version>/extra/fio/
Utilities	/usr/bin

You may also install the IO Accelerator Management Tool software (optional GUI management software). IO Accelerator Management Tool software and documentation are available as a separate download in the IO Accelerator Management Tool Software download folder.

After the packages are installed, continue to "Loading the IO Accelerator VSL driver in Linux (on page 13)."

Building the software from source for DEB packages in Linux

You only need to perform the following additional instructions if you downloaded the source package.

To build the software for DEB packages from the source package:

1. Install the prerequisite files for building the source IO Accelerator VSL package.

```
$ sudo apt-get install gcc fakeroot build-essential debhelper
linux-headers-$(uname --r) rsync
```

   You can also install packages separately by running the following command:

```
$ sudo apt-get install gcc rsync
```

2. Download the software source file for your target operating system from the HP website (http://www.hp.com/go/support).
3. Unpack the IO Accelerator VSL source file.

   ```
   $ tar zxvf iomemory-vsl4_<VSL-version>.tar.gz
   ```
4. Change the directory to the folder created in the previous step.

   ```
   $ cd iomemory-vsl4-source-<VSL-version>
   ```
5. Begin the software rebuild.

   ```
   $ sudo dpkg-buildpackage
   ```

   You can also build the IO Accelerator VSL as a non-root user.

   ```
   $ dpkg-buildpackage -rfakeroot -b
   ```

   The new DEB packages are located in the parent directory.

   A failure of any of these builds might be due to required packages missing from the system. The output from the failed command informs you which packages are missing. Try installing any missing packages, and then repeating the build process.

   You have build installation packages for your distribution and kernel.
6. Copy the custom-built software installation packages into the directory where you downloaded the installation packages.
7. Continue to the next step in "Installing DEB packages in Linux (on page 11)."

Loading the IO Accelerator VSL driver in Linux

To load the IO Accelerator VSL driver, run the following command:

```
$ modprobe iomemory-vsl4
```

The IO Accelerator VSL driver automatically loads at system boot. The IO Accelerator device is now available to the operating system as `/dev/fiox`, where `x` is a letter (for example, `a`, `b`, `c`, and so on).

To confirm the IO Accelerator device is attached, run the `fio-status` utility from the command line. The output lists each drive and its status (attached or not attached).

If the IO Accelerator device is not automatically attaching, check the `/etc/modprobe.d` files to ensure that the `auto_attach` option is set to `1`.

On SLES systems, you must also allow unsupported modules for this command to work. Modify the `/etc/modprobe.d/iomemory-vsl4.conf` file and uncomment the appropriate line:

```
# To allow the ioMemory VSL driver to load on SLES11, uncomment below
allow_unsupported_modules 1
```

Controlling driver loading in Linux

Control driver loading either through the init script or through udev.

In newer Linux distributions, users can rely on the udev device manager to automatically find and load drivers for their installed hardware at boot time, though udev can be disabled and the init script used in nearly all cases. HP recommends using the init script to load the IO Accelerator VSL driver if you are managing a RAID array using LVM, mdadm, or Veritas Storage Foundation.

For older Linux distributions without udev functionality, users must rely on a boot-time init script to load needed drivers.

Using the init script in Linux

On systems where udev loading of the driver doesn't work or is disabled, enable the init script to load the driver at boot. The init script might be enabled by default in some distributions.

The init script is part of the `fio-sysvinit` package, which must be installed before you can enable it.

You can disable the loading of the driver with the following command:

```
$ chkconfig --del iomemory-vsl4
```

To re-enable the driver loading in the init script, use the following command:

```
$ chkconfig --add iomemory-vsl4
```

The IO Accelerator VSL software installation process places an init script in `/etc/init.d/iomemory-vsl4`. In turn, the script uses the setting options found in the options file in `/etc/sysconfig/iomemory-vsl4`. The options file must have `ENABLED` set (non-zero) for the init script to be used:

```
ENABLED=1
```

The options file contains documentation for the various settings. For more information on two of the settings, `MOUNTS` and `KILL_PROCS_ON_UMOUNT`, see "Handling driver unloads in Linux (on page 15)."

Mounting filesystems when using the init script

Because the IO Accelerator VSL driver does not load by the standard means (in the `initrd`, or built into the kernel), using the standard method for mounting filesystems (`/etc/fstab`) for filesystems hosted on the IO Accelerator VSL software does not work.

To set up auto-mounting of a filesystem hosted on an IO Accelerator device:

1. Add the filesystem mounting command to `/etc/fstab` as normal.
2. Add the `noauto` option and the `0 0` flag to `/etc/fstab` as in the two following sample entries. Modify the samples to fit your use case, including your filesystem.

   ```
   /dev/fioa /mnt/fioa ext3 defaults,noauto 0 0
   /dev/fiob1 /mnt/ioDrive ext3 defaults,noauto 0 0
   ```

   where `a` in `fioa` can be `a`, `b`, `c`, and so on, depending on how many IO Accelerator devices are installed in the system.

> **CAUTION:** Failure to add `noauto 0 0` to `fstab` might cause a boot failure.

To have the init script mount these drives after the driver is loaded and unmounted and before the driver is unloaded, add a list of mount points to the options file using the procedure documented there.

For the filesystem mounts shown in the earlier example, the line in the options file would look like the following:

```
MOUNTS="/mnt/fioa /mnt/iodrive"
```

Using udev in Linux

On systems that rely on udev to load drivers, modify an IO Accelerator VSL software options to prevent udev from auto-loading the IO Accerlerator VSL at boot time. To do this, locate and edit the `/etc/modprobe.d/iomemory-vsl4.conf` file that already has the following line:

```
# blacklist iomemory-vsl4
```

To disable loading, remove the `#` from the line and save the file.

With the blacklist command in place, restart Linux. The IO Accelerator VSL driver is not loaded by udev.

To restore the udev-loading of the driver, replace the `#` to comment out the line.

Using either udev or init script systems in Linux

Disable loading the driver at boot time and prevent the auto-attach process for diagnostic or troubleshooting purposes on either udev or init script systems. To disable or re-enable the auto-attach functionality, see "Disabling auto-attach (on page 51)."

Alternatively, prevent the IO Accelerator VSL driver from loading by appending the following parameter at the kernel command line of the boot loader:

```
iomemory-vsl4.disable=1
```

However, this method is not preferred because it prevents the driver from functioning at all, thus limiting the amount of troubleshooting you can perform.

You can also disable auto-attach by appending the following parameter at the kernel command line of the boot loader:

```
iomemory-vsl4.auto_attach=0
```

IO Accelerator devices and multipath storage in Linux

If you use IO Accelerator devices along with multipath storage, blacklist the IO Accelerator devices to prevent `device-mapper` from trying to create a `dm-device` for each IO Accelerator device. This must be done prior to activating `dm-multipath` and/or loading the driver. If IO Accelerator devices are not blacklisted, they appear busy and you cannot attach, detach, or update the firmware on the devices.

To blacklist IO Accelerator devices, edit the `/etc/multipath.conf` file and include the following:

```
blacklist {
devnode          "^fio[a-z]"
}
```

Handling driver unloads in Linux

Special consideration must be taken during IO Accelerator VSL driver unload time. By default, the init script searches for any processes holding open a mounted filesystem and kills them, thus allowing the filesystem to be unmounted. This behavior is controlled by the option `KILL_PROCS_ON_UMOUNT` in the options file. If these processes are not killed, the filesystem cannot be unmounted. This may keep the IO Accelerator VSL software from unloading cleanly, causing a significant delay on the subsequent boot.

IO Accelerator VSL software installation for Windows

Before continuing with the installation of the software:

1. Confirm the operating system is one of the supported operating systems listed in the IO Accelerator release notes.
2. Before installing the IO Accelerator VSL software, make sure you have properly installed the IO Accelerator devices. For full details and hardware requirements, see the *IO Accelerator Hardware Installation Guide*.

If you have IO Accelerator devices installed and in a RAID configuration, before upgrading the software and the firmware, see "Upgrading the software with a RAID configuration in Windows (on page 56)."

Installation overview for Windows

IMPORTANT: This version of the IO Accelerator VSL software only supports third generation devices, such as the SX300 and PX600 devices. It does not support devices that were compatible with IO Accelerator VSL software version 3.x.x or earlier.

1. Download the latest version of the software on the HP website (http://www.hp.com/go/support).
2. If you have a previous version of the IO Accelerator VSL software installed, uninstall the IO Accelerator VSL software and the utilities. For more information, see "Uninstalling the IO Accelerator VSL software (on page 47)." After you uninstall the software, return to this procedure.
3. Install the IO Accelerator VSL software and command line utilities.
4. Determine if you need to upgrade the firmware to the latest version. For more information, see "Upgrading the firmware (on page 23)."
5. Configure the devices by following the configuration instructions. For more information, see "Configuration (on page 24)."

Installing software on a Windows system

1. Review the IO Accelerator release notes available for this version of the software for additional steps that might be needed to complete the installation.
2. For new device installations, make sure you have properly installed the devices before you install the IO Accelerator VSL software.
3. Log in as Administrator or have Administrator rights.
4. If needed, uninstall the existing IO Accelerator VSL software, utilities, and so on, using **Programs and Features** in the Control Panel.
5. Restart the system.

   The IO Accelerator VSL installation program attempts to remove previous versions of the software. However, if it fails and you remove a previous version after the latest version is installed, the IO Accelerator VSL software no longer loads after a restart. In this case, you need to:

   a. Run the Repair option in the installation program, from **Programs and Features** or **Add or Remove Programs** in the Control Panel.
   b. Restart the system.
6. Download the IO Accelerator VSL installation program for Windows to your desktop or a convenient directory from the HP website (http://www.hp.com/go/support).

   Also download the `fio-firmware-fusion_<version>-<date>.fff` firmware archive file for this release and save it in the same location.
7. Run the IO Accelerator VSL installation program. The installation program displays a window with an options button. The options button enables you to choose a custom installation directory. The default directory is `C:\Program Files\Fusion-io ioMemory VSL4`, and the utilities are installed by default in `C:\Program Files\Common Files\VSL Utils`.
8. Follow the onscreen prompts to complete the installation.
9. Click **Finish** on the finish screen of the installer.

If Windows does not recognize the IO Accelerator devices after rebooting, you might need to manually install the IO Accelerator VSL software for the devices. For information on manual installation, see "Manual installation (on page 94)."

If the IO Accelerator device is configured for page file support, you might need to reboot a second time before Windows creates a permanent page file. For more information, see "Using the device as a page files store in Windows (on page 34)."

You can also install the IO Accelerator Management Tool software, which is optional GUI management software. The IO Accelerator Management Tool software and documentation are available as a separate download.

IO Accelerator devices cannot be used as hibernation devices.

After the system restarts, continue to "Upgrading the firmware (on page 23)."

Silent installation option in Windows

Uninstalling the previous version

If you have a version of the HP IO Accelerator VSL software previously installed, you must uninstall it first. You must manually reboot the computer after installing the new version with the silent install option. This step must be performed prior to using any IO Accelerator VSL utilities or functionality.

Silently installing the IO Accelerator VSL software

If you are installing remotely or with scripts, you can use the silent install option (`/quiet`) when you run the installation program in the command line interface.

In the command line interface, navigate to the folder that contains the `.exe` installer file, and then run the following command:

```
<installname>.exe /quiet
```

Where `<installname>.exe` is the name of the installer file.

This option installs the IO Accelerator using the default settings and eliminates the need to click **Next** or select settings during installation.

Use the `/quiet` parameter. The command line quiet installation parameter changed and the installer no longer supports the abbreviated parameter (`/qn`). If you pass the `/qn` parameter to the installer, the installer ignores the parameter and the installer GUI launches.

Silent uninstallation in Windows

Silently uninstall the IO Accelerator VSL software with the following command:

```
<installname>.exe /uninstall /quiet
```

IO Accelerator VSL software installation for ESXi

Before continuing with the installation of the software:

1. Confirm the operating system is one of the supported operating systems listed in the IO Accelerator release notes.

2. Before installing the IO Accelerator VSL software, make sure you have properly installed the IO Accelerator devices. For full details and hardware requirements, see the *IO Accelerator Hardware Installation Guide*.

VMDirectPathIO

The ESXi IO Accelerator VSL software is only required if you plan to use the device with the host operating system (for example, as a VMFS Datastore or as a caching device). If you are passing the devices through using VMDirectPathIO (also known as PCI passthrough), you do not need to install the IO Accelerator VSL software on the ESXi system. Instead, install the IO Accelerator VSL on the guest operating system. For example, pass the device through to a Windows VM, and then install the Windows IO Accelerator VSL on that VM. For installation and user instructions, see "IO Accelerator VSL software installation for Windows (on page 15)."

When passing through an IO Accelerator device, you must be aware of certain constraints. For more information, see "Working with IO Accelerator devices and VMDirectPathIO (on page 81)."

Command-line installation

Unless you use a VUM, you must use a CLI to install the IO Accelerator VSL software. To install and manage the IO Accelerator devices and the IO Accelerator VSL software, you must use a CLI.

VMware provides the vCLI to run against your ESXi system. Install a vCLI package on a physical machine running Linux or Windows. For more information about the VMware vCLI, see the VMware website (http://www.vmware.com/support/developer/vcli/).

ESXi command line installation

HP does not recommend using the vCLI within a virtual machine that is hosted on an ESXi system. The IO Accelerator VSL software installation and configuration processes involve rebooting the host.

When installing the VSL, you can choose to use the TSM, also known as Shell or SSH (when used remotely), instead of the vCLI to install the IO Accelerator VSL software. The TSM/Shell might be required for managing or troubleshooting your device with the command-line utilities.

CAUTION: To avoid damage to the system, HP recommends using TSM only for the purposes of troubleshooting and remediation. HP recommends using the vSphere Client or any other VMware Administration Automation Product to perform routine ESXi host configuration tasks that do not involve troubleshooting. For more information on using TSM, see *Using Tech Support Mode in ESXi 4.1, ESXi 5.x, and ESXi 6.0* on the VMware website (http://kb.vmware.com/selfservice/microsites/search.do?language=en_US&cmd=displayKC&externalId=1017910).

Installation overview for ESXi

IMPORTANT: This version of the IO Accelerator VSL software only supports third generation devices, such as the SX300 and PX600 devices. It does not support devices that were compatible with IO Accelerator VSL software version 3.x.x or earlier.

1. Download the latest version of the software on the HP website (http://www.hp.com/go/support).

2. If you have a previous version of the IO Accelerator VSL software installed, uninstall the IO Accelerator VSL software and the utilities. For more information, see "Uninstalling the IO Accelerator VSL software (on page 47)." After you uninstall the software, return to this procedure.
3. Install the latest version of the IO Accelerator VSL and command line utilities.
4. Reboot the ESXi system. Rebooting loads the IO Accelerator VSL software driver and attaches the IO Accelerator devices.
5. Determine if you need to upgrade the firmware to the latest version. For more information, see "Upgrading the firmware (on page 23)."
6. Follow the instructions in "Configuring the device to support VM disks in ESXi (on page 35)."

CAUTION: HP does not support installing an ESXi operating system and booting from the IO Accelerator device. The IO Accelerator device is meant to be used as a data storage disk or caching device. An ESXi installation on an IO Accelerator device fails on reboot.

Downloading the ESXi software

Download the installation packages to a remote machine running the vCLI or the vSphere client.

The IO Accelerator VSL software is available as an offline bundle from the HP website (http://www.hp.com/support). Navigate to the appropriate folder for your operating system. An example bundle to download is `scsi-iomemory-vsl4_<version>.offline-bundle.zip`.

The offline bundle might be in a .zip file archive:

```
iomemory-vsl4-<version>.zip
```

IMPORTANT: The package you need to download depends on your version of ESXi. If you are using ESXi 5.5, download the packages with "`55`" in the file names. For all other versions, download files with "`5X`" in the file names.

Also download the firmware archive file. An example firmware archive file is `fio-firmware-fusion_<version>-<date>.fff`

HP recommends downloading and installing the VSL management library. This library is required if you decide to use a remote management tool based on the SMI-S interface. For more information about these tools, see "Monitoring and managing devices (on page 42)." An example file name is `libvsl-1.0.0-5X-offline-bundle.<version>.zip`.

Transferring the IO Accelerator VSL files to the ESXi server

Transfer the firmware file to the ESXi host. Depending on your ESXi version and your preferred installation method, you might need to transfer the two bundle installation files to the host as well. HP recommends transferring all of the files at this point, and then choosing the installation method later.

With any method that you select for transferring the files, HP recommends saving the files to a datastore on the host. The example paths to the bundles and firmware in this guide show them located in a `bundles` directory on a datastore:

```
/vmfs/volumes/<datastore>/bundles/
```

Where `<datastore>` is the name of the datastore where the files are saved.

Transfer methods

Transfer the files using one of several methods, including:

- vSphere Client (Datastore Browser, Upload Files)
- vCLI `vifs` command
- SCP (using SSH)

You can copy the files to the host from your remote system or from an NFS share.

vCLI example

To transfer files to the ESXi host using vCLI:

1. On your remote machine, be sure you have downloaded the appropriate files, and then record the file locations.
2. Choose an available datastore with at least 200 MB of available storage on the hypervisor that you will use to temporarily store the bundles.
3. Create a directory in the datastore named `bundles` using the `vifs` remote command:

   ```
   vifs --server <servername> --mkdir "[<datastore>]bundles"
   ```

   The brackets ([ ]) and quotes (" ") are required. Substitute your datastore name for the `<datastore>` variable.

   You are prompted to enter the username and password for the ESXi host. For convenience, you can add the following options to each command:

   ```
   --username <username> -- password <password>
   ```

> **IMPORTANT:** When using the vCLI on a Windows operating system, many of the commands are slightly different. Most of the commands end with `.pl`. Throughout this guide, when you run the vCLI on a Windows operating system, you must add `.pl` to the command. For a Windows operating systems, this command is:
> ```
> vifs.pl --server <servername> --mkdir "[<datastore>]bundles"
> ```

4. Use the following example command line to transfer each file individually to the `bundles` directory of the datastore:

   ```
   vifs --server <servername> --put "<path-on-local-machine>/<filename>"
   "[<datastore>]bundles/<filename>"
   ```

   Where `<filename>` is the full filename. For example:

   - `fio-firmware-fusion_<version>-<date>.fff`
   - `iomemory-vsl4_<version>.offline-bundle.zip`

Installing the IO Accelerator VSL on ESXi 5.x

These instructions describe how to install the IO Accelerator VSL on a single hypervisor. However, if you are familiar with the VUM plugin for the vCenter Server, you can use that method to install the IO Accelerator VSL on multiple hosts. For more details on VUM, see the vCenter Server documentation.

> **IMPORTANT:** Uninstall all previous versions of the IO Accelerator VSL software before installing the latest version.

You can choose to install the software using the vCLI or SSH. Whether you use the SSH or vCLI, you must first transfer the files to a datastore on the ESXi host.

The offline bundle might be within a .zip archive: `iomemory-vsl4-<version>.zip`. Unpack the offline bundle for installation.

Installing the IO Accelerator VSL on ESXi 5.x using vCLI

1. Install the bundle by running the following command against your ESXi 5.x system using the vCLI:

   `esxcli --server <servername> software vib install -d <offline-bundle>`

   Where `<offline-bundle>` is the absolute path to the offline bundle on the hypervisor host. For example, if the offline bundle is in the `bundles` directory of a datastore with the name of `datastore1`, the local path is:
   `/vmfs/volumes/datastore1/bundles/iomemory-vsl4_<version>.offline-bundle.zip`.

   This absolute path must begin with a forward slash (/) or ESXi returns an error message.
2. If you want to install the `libvsl` offline bundle, repeat the previous step using the path to the `libvsl` offline bundle.
3. Reboot your ESXi system.

The IO Accelerator VSL and command line utilities are installed on the host.

Installing the IO Accelerator VSL on ESXi 5.x using the command line

1. Install the bundle by running the following command against your ESXi 5.x system using TSM/SSH:

   `esxcli software vib install -d <offline-bundle>`

   Where `<offline-bundle>` is the absolute path to the offline bundle on the hypervisor host. For example, if the offline bundle is in the `bundles` directory of a datastore with the name `datastore1`, an example local path would be:
   `/vmfs/volumes/datastore1/bundles/iomemory-vsl4_<version>.offline-bundle.zip`
2. If you also want to install the `libvsl` offline bundle, repeat the previous step using the path to the `libvsl` offline bundle.
3. Reboot your ESXi system.

The IO Accelerator VSL and command-line utilities are installed on the host.

Continue to "Upgrading the firmware (on page 23)."

IO Accelerator VSL software installation for Solaris

Before continuing with the installation of the software:

1. Confirm the operating system is one of the supported operating systems listed in the IO Accelerator release notes.
2. Before installing the IO Accelerator VSL software, make sure you have properly installed the IO Accelerator devices. For full details and hardware requirements, see the *IO Accelerator Hardware Installation Guide*.

Installation overview for Solaris

Before continuing with the installation of this software, do the following:

1. Ensure that your operating system is included in the list of supported operating systems contained in the release notes for this release.

2. Before installing the IO Accelerator VSL, make sure you have properly installed the IO Accelerator devices. For more information, see the IO Accelerator hardware installation guide. Every IO Accelerator device in a system must be upgraded to the appropriate firmware requirements.

Installing the software in Solaris

To install the IO Accelerator VSL software and utilities:

1. Download the appropriate tarball and firmware file archive from the HP website (http://www.hp.com/go/support).

   Example:
   ```
   iomemory-vsl-<version>.pkg.tar.gz
   fusion_<version>-<date>.fff
   ```
2. Change to the directory containing the downloaded tarball.
3. Uncompress the `-vsl` package:
   ```
   $ gunzip -c iomemory-vsl*pkg.tar.gz|tar -xf -
   ```
4. Run the `pkgadd` command:
   ```
   $ pfexec pkgadd -d . iomemory-vsl
   ```

The software and utilities are installed to the following locations.

File type	Installation location
Driver	/usr/kernel/drv/amd64
Utilities	/opt/fusionio/bin
Driver config file	/usr/kernel/drv

Loading the IO Accelerator VSL driver

The driver is installed and automatically loaded during the package installation (`pkgadd`; see "Installing the software in Solaris (on page 22)").

To manually unload the driver, run:
```
$ pfexec rem_drv iomemory-vsl
```

To manually load the driver, run:
```
$ pfexec add_drv -i "pci1aed,3001 pciex1aed,3001 pcie1aed,3002
pciex1aed,3002 pcie1aed,3003 pciex1aed,3003" -m "* 0666 root sys"
iomemory-vsl4
```

The load command might be too long for the shell. You might need to add a \ (backslash) character at the end of the first line to signal that the line is continued on the next line.

The driver automatically loads at system boot. The IO Accelerator device is now available to the OS as the following:
```
/dev/rdsk/c*d0p0
```

Attached IO Accelerator devices show up as block devices in the file system in three places:
```
/dev/fio*
/dev/rdsk/c*d0*
```

Where `c*d0p0` is the master partition covering the entire raw block device
```
/dev/dsk/c*d0*
```

Where `c*d0s2` is the master slice covering the entire block device

The driver creates `/dev/fiox` block device nodes, where `x` is a letter (a, b, c, and so forth).

CAUTION: Only use the traditional disk interfaces (`/dev/rdsk/*` and `/dev/dsk/*`). Do not use the `/dev/fiox` block device nodes.

To confirm the IO Accelerator device is attached, use the following command:

```
$ pfexec /opt/fusionio/bin/fio-status
```

The output indicates the attach status of the device: attached or not attached.

On SLES systems, enable unsupported modules for the command to work. Modify the `/etc/modprobe.d/iomemory-vsl4.conf` file and uncomment the appropriate line:

```
# To allow the ioMemory VSL driver to load on SLES11, uncomment below
allow_unsupported modules 1
```

Upgrading the firmware

After the IO Accelerator driver is loaded, be sure that the IO Accelerator device firmware is up to date by running the `fio-status` (on page 69) command line utility.

All command-line utilities require root permission for Linux, Administrator rights for Windows, or TSM enabled for ESXi. If the output shows that the device is running in minimal mode, download the latest firmware from the HP website (http://www.hp.com/support). After you download the latest firmware, upgrade the firmware using `fio-update-iodrive` (on page 74).

CAUTION: Do not attempt to downgrade the firmware on any IO Accelerator device. Doing so might void your warranty.

CAUTION: When installing a new IO Accelerator device along with existing devices, upgrade all of the currently installed devices to the latest available versions of the firmware and IO Accelerator VSL software before installing the new devices. For more information, see the release notes for this version of the IO Accelerator VSL for any upgrade considerations.

IMPORTANT: Download the firmware archive file for this version of the IO Accelerator VSL software.

For more information on the command-line utilities, see "Utilities (on page 58)."

To upgrade the firmware:

1. Run the `fio-status` utility and examine the output. For usage information, see "`fio-status` (on page 69)."
    - If a device is in minimal mode and the reason is outdated firmware, update the firmware.
    - If a device is not in minimal mode but the firmware listed for the device is an earlier firmware version than the latest firmware version available with this version of the IO Accelerator VSL software, then the firmware is old but not outdated.
2. If the firmware is old or outdated, update the firmware using the `fio-update-iodrive` utility. For complete information and warnings, see "`fio-update-iodrive`."

When using an IO Accelerator device within a guest operating system (for example, using VMDirectPathIO), you must power cycle the host server after upgrading the device. Restarting the virtual machine does not apply the firmware update.

Configuration

Configuration overview

After installing the IO Accelerator device, loading the IO Accelerator VSL software, and ensuring the firmware on the device is current, you might need to configure the device and the software. This section outlines some of the common configurations to consider.

Enabling additional PCIe power override

Newer IO Accelerator devices, such as SX300 and PX600 devices, benefit from more power than the minimum 25 W provided by PCIe Gen2 x8 slots. The devices still function properly with the minimum 25 W of power, but device performance is limited by the amount of available power.

For improved performance, use the IO Accelerator device in a PCIe slot that provides additional power, and then enable full power draw on the device. For more information about additional power requirements, see the product specification sheet for each device.

Automatic power draw

Some PCIe slots provide additional power, often up to 75 W. If the slot is rated to provide more power than the standard 25 W provided by PCIe Gen2 x8 slots and the system BIOS correctly reports the power configuration of the slot, then the IO Accelerator device automatically draws additional power up to the power needed for peak performance or to the power limit of the slot, whichever is lower.

The IO Accelerator VSL software reports the additional power draw in the system logs. The following example shows the software reporting the power limit on a 75 W slot.

```
fioinf <device> 0000:11:00.0: PCIe Slot reported power limit: 75000mWatts
fioinf <device> 0000:11:00.0: PCIe Adapter power limit: 75000mWatts
fioinf <device> 0000:11:00.0: PCIe Adapter power Throttle point: 74750mWatts
```

Enabling full slot power draw

If the slot is rated to provide additional power (for maximum power required for your device, see the product specification sheet), but the BIOS does not correctly report the power configuration, then enable the device to draw additional power from the PCIe slot by setting a VSL module parameter.

The parameter overrides the internal device setting that prevents devices from drawing more than 25 W from the PCIe slot. The parameter is enabled per device using device serial numbers. When the setting is overridden, each device is able to draw up to the power amount specified. For specific peak power draw numbers, see the product specification sheet.

CAUTION: If the PCIe slot is not capable of providing the needed amount of power, then enabling full power draw from the PCIe slot might result in malfunction or even damage to server hardware. The user is responsible for any damage to equipment due to improper use of the override parameter. HP expressly disclaims any liability for damage arising from improper use of the override parameter. To confirm the power limits and capabilities of each slot, as well as the entire system, contact the server manufacturer. For information about using the override parameter, contact HP Customer Support.

Before enabling the override parameter, be sure that each PCIe slot is rated to provide enough power for all slots, devices, and server accessories. To determine the slot power limits, see the server documentation, BIOS interface, setup utility, or (if available) use "`fio-pci-check` (on page 69)."

Contact the server manufacturer to confirm the power limits and capabilities of each slot, as well as the entire system.

Important considerations

- When installing more than one IO Accelerator device and enabling the override parameter for each device, be sure the motherboard is rated to provide adequate power to each slot used.

  IMPORTANT: Consult with the manufacturer to determine the total PCIe slot power available. Some motherboards safely provide up to 75 W to any one slot, but have power constraints when multiple slots are used to provide that much power. Installing multiple devices in this situation might also result in server hardware damage or device malfunction.

- When enabled correctly, the override parameter persists in the system and enables additional power draw on an enabled device, even if you remove the device and place it in a different slot within the same system. If you place the device in a slot that is not rated to provide adequate additional power, you might cause the device to malfunction or damage the server hardware.
- The override parameter is a setting for the IO Accelerator VSL software per server, and it is not stored in the device. When moved to a new server, the device defaults to the 25 W power limit or the maximum power rating of the new slot until the override parameter is enabled for that device in the new server. Consult with the manufacturer to determine the total PCIe slot power available for the new server.

Enabling the override parameter

Determining the serial number

Before enabling the parameter, determine the adapter serial number for the device you put in a compatible slot. To determine the adapter serial number, use the `fio-status` command line utility.

You can also inspect the adapter serial number label on the device to determine the serial number. However, as a best practice, confirm each serial number is an adapter serial number by using `fio-status`.

Run the `fio-status` command line utility.

```
fio-status
...
Adapter: ioMono
     ioMemory PX600-5200, Product Number:F14-005-5200-CS-0001, SN:1331G0009
, FIO SN:1331G0009
     External Power: NOT connected
```

```
PCIe Power limit threshold: 24.75W
Connected ioMemory modules:
  fct0: Product Number:F14-005-5200-CS-001, SN:1331G0009
```

In this example, `1331G0009` is the adapter serial number.

If you have multiple devices installed, you can use `fio-beacon` to verify where each device is physically located. For more information, see "`fio-beacon` (on page 60)."

Setting the parameter in Linux

In Linux, set the module parameter by editing the `/etc/modprobe.d/iomemory-vsl4.conf` file and changing the value of the `external_power_override` parameter.

```
options iomemory-vsl4 external_power_override=<SN-value>:<mW-value>
```

The `<SN-value>:<mW-value>` for the parameter is a comma-separated list of value pairs with adapter serial numbers and the maximum amount of power each device should pull in milliwatts.

For example, `1149D0969:40000,1159E0972:40000,1331G0009:40000` enables the three devices with serial numbers `1149D0969`, `1159E0972`, and `1331G0009` to each draw a maximum of approximately 40 W.

Reboot or reload the driver to enforce any parameter changes.

Setting the parameter in Windows

In Windows, set the module parameter by using the `fio-config` utility and specifying a new value for the `external_power_override` parameter.

```
fio-config -p FIO_EXTERNAL_POWER_OVERRIDE <SN-value>:<mW-value>
```

The `<SN-value>:<mW-value>` for the parameter is a comma-separated list of value pairs with adapter serial numbers and the maximum amount of power each device should pull in milliwatts.

For example, `1149D0969:40000,1159E0972:40000,1331G0009:40000` enables the three devices with serial numbers `1149D0969`, `1159E0972`, and `1331G0009` to each draw a maximum of approximately 40 W.

Reboot or reload the driver to enforce any parameter changes.

Setting the parameter in ESXi

In ESXi, set the module parameter by using the `esxcfg-module` command and changing the value of the `external_power_override` parameter.

```
esxcfg-module --server <server-name> iomemory-vsl4 -s
'external_power_override=<SN-value>:<mW-value>'
```

The remote option (`--server`) is only required for the vCLI.

The `<SN-value>:<mW-value>` for the parameter is a comma-separated list of value pairs with adapter serial numbers and the maximum amount of power each device should pull in milliwatts.

For example, `1149D0969:40000,1159E0972:40000,1331G0009:40000` enables the three devices with serial numbers `1149D0969`, `1159E0972`, and `1331G0009` to each draw a maximum of approximately 40 W.

Reboot the ESXi system to enforce any parameter changes.

Setting the parameter in Solaris

In Solaris, set the module parameter by editing the `/usr/kernel/drv/iomemory-vsl4.conf` file and changing the value of the `external_power_override` parameter.

```
external_power_override="<SN-value>:<mW-value>";
```

The `<SN-value>:<mW-value>` for the parameter is a comma-separated list of value pairs with adapter serial numbers and the maximum amount of power each device should pull in milliwatts.

For example, `1149D0969:40000,1159E0972:40000,1331G0009:40000` enables the three devices with serial numbers `1149D0969`, `1159E0972`, and `1331G0009` to each draw a maximum of approximately 40 W.

Reboot or reload the driver to enforce any parameter changes.

Discard (TRIM) support

In this version of the IO Accelerator VSL software, Discard, also known as TRIM, is enabled by default.

Discard addresses an issue unique to solid-state storage. When a user deletes a file, the device is not aware that it can reclaim the space. The device must assume the data is valid.

Discard is a feature on most modern filesystem releases. It informs the device of logical sectors that no longer contain valid user data. This enables the wear-leveling software to reclaim that space as reserve to handle future write operations.

Discard (TRIM) in Linux

Discard is enabled by default in the IO Accelerator VSL software release. However, for discard to be implemented, the Linux distribution must support this feature, and discard must be enabled for the file system mount point. TRIM commands are sent to the IO Accelerator device if the discard mount option is added to the filesystem, or by periodically running `fstrim` against the mount point.

In other words, if the Linux distribution supports discard, and if discard is enabled on the system, then discard is implemented on the IO Accelerator device.

In Linux, discards are not limited to being created by the filesystem. Userspace applications using the discard ioctl of the kernel can also directly generate discard requests.

CAUTION: A known issue is that ext4 in Kernel.org 2.6.33 or earlier might silently corrupt data when Discard is enabled.
The issue has been fixed in many kernels provided by distribution vendors. Check with your kernel provider to ensure that your kernel properly supports Discard. For more information, see the release notes for this version of the driver.

In some Linux distributions, MD and LVM do not pass discards to underlying devices, and any IO Accelerator devices that are part of an MD or LVM array do not receive discards sent by the filesystem. For details, see the documents of your distribution.

Starting in RHEL 6.5, MD supports discard for RAID0, RAID1, and RAID10.

Discard (TRIM) in Windows Server 2008 R2 and later

IMPORTANT: Windows does not support TRIM with a RAID5 configuration.

Windows Server 2008 R2 and later have built-in TRIM support. With these operating systems, IO Accelerator devices work with Windows TRIM commands by default.

Setting the IO Accelerator VSL options in Linux

Set IO Accelerator VSL software options in Linux for a one-time configuration or for a persistent configuration. For more information about setting specific options, see "Using module parameters (on page 76)."

One-time configuration in Linux

In Linux, IO Accelerator VSL software options can be set at the time of installation in the command line of either `insmod` or `modprobe`. For example, to set the `auto_attach` option to `0`, run the following command:

```
$ modprobe iomemory-vsl4 auto_attach=0
```

This option takes effect only for this load of the IO Accelerator VSL software; subsequent calls to `modprobe` or `insmod` do not have this option set.

Persistent configuration in Linux

In Linux, to maintain a persistent setting for an option, add the option to `/etc/modprobe.d/iomemory-vsl4.conf` or a similar file. To prevent IO Accelerator devices from auto-attaching, add the following line to the `iomemory-vsl4.conf` file.

```
options iomemory-vsl4 auto_attach=0
```

This option takes effect for every subsequent IO Accelerator VSL software load, as well as on autoload of the IO Accelerator VSL software during boot time.

Using the device as swap in Linux

In Linux, to safely use the IO Accelerator device as swap space requires passing the `preallocate_memory` kernel module parameter. To set this parameter, use the optional IO Accelerator Management software, or add the following line to the `/etc/modprobe.d/iomemory-vsl4.conf` file (for more information on using parameters, see "Using module parameters (on page 76)"):

```
options iomemory-vsl4
preallocate_memory=1149D2717-1121,1149D2717-1111,1331G0009
```

`1149D2717-1121`,`1149D2717-1111`, and `1331G0009` are serial numbers obtained from `fio-status` (on page 69).

A 4 KB sector size format is required for swap. This reduces the IO Accelerator VSL software memory footprint. Use `fio-format` to format the IO Accelerator device with 4 KB sector sizes.

Be sure to provide the serial numbers for the IO Accelerator device, not an adapter, when applicable.

IMPORTANT: Be sure to have enough RAM available to enable the IO Accelerator device with pre-allocation enabled for use as swap. Attaching an IO Accelerator device with preallocation enabled without sufficient RAM might result in system instability and the loss of user processes. For RAM requirements with this version of the IO Accelerator VSL software, see the release notes.

The `preallocate_memory` parameter is recognized by the IO Accelerator VSL software at load time, but the requested memory is not allocated until the specified device is attached.

Setting the amount of preallocated memory

In Linux and Windows, if you enable devices for preallocation, the IO Acclerator VSL software automatically preallocates the amount of memory based on your formatted sector size. If you have a sector size that is less than 4 KiB (for example, 512 B sectors), then the IO Accelerator VSL software preallocates a very large amount of memory (calculated for worst-case scenarios based on 512 B sectors).

For worst-case RAM requirements based on formatted sizes, see the *HP IO Accelerator Driver and Management Software Release Notes*. The IO Accelerator VSL software preallocates enough memory for worst-case scenarios in order to avoid an out-of-memory situation where the IO Accelerator device no longer functions due to insufficient memory (resulting in a system crash).

Operating systems normally use at least 4 KiB blocks of data for virtual memory, so it would be safe in most cases for the IO Accelerator VSL software to preallocate enough memory for a worst-case scenario based on 4KiB sectors. There are two ways to force the IO Accelerator VSL software to preallocate based on 4 KiB block sizes:

- **Format the device to 4 KiB sectors.** With preallocation enabled, the IO Accelerator VSL software automatically preallocates memory based on 4 KiB blocks of data. Use `fio-format` to format the IO Accelerator device with 4 KiB sector sizes.
- **Use the `PREALLOCATE_MB` parameter.** The parameter sets the amount of memory that the IO Accelerator VSL software preallocates for every IO Accelerator device in the system that is enabled for preallocation (as described in the previous section).

IMPORTANT: Be sure to have enough RAM available to enable the IO Accelerator device with pre-allocation enabled for use as swap. Attaching an IO Accelerator device with preallocation enabled without sufficient RAM might result in system instability and the loss of user processes. For RAM requirements with this version of the IO Accelerator VSL software, see the release notes.

1. Determine the amount of system memory needed for every device in a worst-case scenario.
   a. For worst-case RAM requirements based on sector sizes, see the *HP IO Accelerator Driver and Management Software Release Notes.*
   b. Base the calculation on the sector size that aligns with the data block size your operating system uses for virtual memory.
2. Set the parameter.

   Linux

   In Linux, set the parameter by adding the following line to the `/etc/modprobe.d/iomemory-vsl.conf` file. For more information on using parameters, see "Using module parameters."

   ```
   options iomemory-vsl preallocate_mb=<value>
   ```

   Windows

   In Windows, set the parameter by using the `fio-config` command line utility. For full utility instructions, see "`fio-config` (on page 64)."

   ```
   fio-config -p PREALLOCATE_MB=<value>
   ```

   In Linux and Windows, `<value>` is the amount of system memory in MB that the IO Accelerator VSL software should preallocate for every IO Accelerator device enabled for preallocation. For example, if you enter a value of `3500`, then the IO Accelerator VSL software preallocates about 3.5 GB of RAM for every IO Accelerator device enabled for preallocation in the system.

In order for the preallocation of memory to be effective, the value should be larger than the default memory usage of the device, as reported by running `fio-status -a`.

Using the Logical Volume Manager in Linux

In Linux, the LVM volume group management application handles mass storage devices like IO Accelerator devices if you add the IO Accelerator device as a supported type.

1. Edit the `/etc/lvm/lvm.conf` configuration file.
2. Add an entry to the file similar to the following example:

   ```
   types = [ "fio", 16 ]
   ```

   The parameter `16` represents the maximum number of partitions supported by the device.

If using LVM or MD, do not use udev to load the IO Accelerator VSL driver. The init script ensures that the LVM volumes and MD devices are detached before attempting to detach the IO Accelerator device.

Mounting filesystems using the init script in Linux

If using the init script to load the IO Accelerator VSL software, after you configure the device you need to follow the instructions in "Mounting filesystems using the init script in Linux (on page 30)."

Configuring RAID using the Logical Volume Manager in Linux

The simplest way to use the IO Accelerator devices with LVM is to use the entire block device with it: for example, `/dev/fioa` rather than `/dev/fioa1`, `/dev/fioa2`, and so on. This way the block device does not need to be partitioned ahead of time, though partitioning is also supported. The following examples assume the entire block device is used.

Whether you plan to stripe (RAID0) or mirror (RAID1) the devices, the first two steps are the same.

1. Create physical volumes.

   ```
   $ pvcreate /dev/fioa /dev/fiob
   ```

2. Add the physical volumes to a volume group.

   ```
   $ vgcreate iomemory_vg /dev/fioa /dev/fiob
   ```

Creating a striped volume (RAID0) using LVM

With the volume group created, create logical volumes within the volume group. In the following example, only one is created and named `iomemory_lv`.

1. Create the striped volume using the `-i2` options for two stripes.

   ```
   $ lvcreate -l 100%VG -n iomemory_lv -i2 iomemory_vg
   ```

2. Create a file system on the newly created volume.

   ```
   $ mkfs.ext4 /dev/iomemory_vg/iomemory_lv
   ```

3. In order to make this persistently mount at boot time, edit the `/etc/sysconfig/iomemory-vsl` file by adding the volume group path under the example line.

   ```
   Example: LVM_VGS="/dev/vg0 /dev/vgl"
   LVM_VGS="/dev/iomemory_vg"
   ```

Be sure to only add the volume group path and not the logical volume as well. For more information on using the init script and the `/etc/sysconfig/iomemory-vsl4` file, see "Loading the IO Accelerator VSL driver in Linux (on page 13)."

Creating a mirrored volume (RAID1) using LVM

With the volume group created, create logical volumes within the volume group. In the following example, only one is created and named `iomemory_lv`.

1. Create the mirrored volume using the `-m1` option for an original linear volume plus one copy.

   ```
   $ lvcreate -l 50%VG -n iomemory_lv -m1 --corelog iomemory_vg
   ```

   Monitor the progress of the initial mirror synchronization using the `lvs` command.

   ```
   lvs -a -o +devices
   ```

   With the \ option, LVM uses an in-memory region or log to track the state of the mirror legs. Because it is in memory and lost at reboot, it must be regenerated at boot by scanning the mirror legs. For alternative configurations, see the LVM documentation.

2. Create a file system on the newly created volume.

   ```
   $ mkfs.ext4 /dev/iomemory_vg/iomemory_lv
   ```

3. To make this persistently mount at boot time, edit the `/etc/sysconfig/iomemory-vsl4` file by adding the volume group path under the example line.

   ```
   # Example: LVM_VGS="/dev/vg0 /dev/vg1"
   LVM_VGS="/dev/iomemory_vg"
   ```

   Be sure to only add the volume group path and not the logical volume as well. For more information on using the init script and the `/etc/sysconfig/iomemory-vsl4` file, see "Loading the IO Acccelerator VSL driver in Linux ("Loading the IO Accelerator VSL driver in Linux" on page 13)."

Configuring RAID using mdadm in Linux

In Linux, configure two or more IO Accelerator devices into a RAID array using software-based RAID solutions.

If using RAID1/mirroring and one device fails, be sure to run `fio-format` on the replacement device (not the remaining good device) after rebuilding the RAID.

This section provides some examples of common RAID configurations using the `mdadm` utility.

The Linux kernel RAID5 implementation performs poorly at high data rates. This is an issue in the Linux kernel. Alternatives include using RAID10 or a third-party RAID stack.

Mounting arrays in Linux

After creating the array using a configuration example, edit the `/etc/sysconfig/iomemory-vsl4` file and add the array path under the example line. This makes the array mount at boot time.

In the provided examples, the created arrays are named `md0`. The examples show how to edit the `/etc/sysconfig/iomemory-vsl4` file to include the array using `md0` as an example. Add the array immediately after the example line in the file.

```
# Example: MD_ARRAYS="/dev/md0 /dev/md1"
MD_ARRAYS="/dev/md0"
```

If you are using the init script to load the IO Accelerator VSL software, after you configure the devices you need to follow the instructions in "Mounting filesystems using the init script in Linux (on page 30)."

RAID0 in Linux

To create a striped set where `fioa` and `fiob` are the two IO Accelerator devices to stripe, run the following command:

```
$ mdadm --create /dev/md0 --chunk=256 --level=0 --raid-devices=2 /dev/fioa
/dev/fiob
```

Making the array persistent (existing after restart)

In some versions of Linux, the configuration file is in `/etc/mdadm/mdadm.conf`, not in `/etc/mdadm.conf`.

Inspect `/etc/mdadm.conf`. If there are one or more lines declaring the devices to inspect, make sure one of the lines specifies `partitions` as an option. If not, add a new `DEVICE` line to the file specifying partitions:

```
DEVICE partitions
```

Also add a device specifier for the fio IO Accelerator devices:

```
DEVICE /dev/fio*
```

To see if any updates are needed to `/etc/mdadm.conf`, run the following command:

```
$ mdadm --examine --scan
```

Compare the output of this command to what currently exists in `mdadm.conf` and add any needed sections to `/etc/mdadm.conf`.

For example, if the array consists of two devices, there are three lines in the output of the command that are not present in the `mdadm.conf` file: one line for the array, and two device lines (one line for each device). Be sure to add those lines to the `mdadm.conf` file so it matches the output of the command.

For more information, see the `mdadm` and `mdadm.conf` man pages for your distribution.

With these changes, on most systems the RAID0 array is created automatically upon restart. However, if you have problems accessing `/dev/md0` after restart, run the following command:

```
$ mdadm --assemble --scan
```

Also, you might want to disable udev loading of the IO Accelerator VSL driver if needed, and use the init script provided for driver loading. For more information on using the init script, see "Using the init script in Linux (on page 14)."

In SLES 11, you might need to run the following commands to make sure the services run on boot:

```
chkconfig boot.md on chkconfig mdadmd on
```

RAID1 in Linux

To create a mirrored set using the two IO Accelerator devices `fioa` and `fiob`, run the following command:

```
$ mdadm --create /dev/md0 --level=1 --raid-devices=2 /dev/fioa /dev/fiob
```

When creating an MD RAID other than RAID0 on devices immediately following a device format, add `--assume-clean` to the command to reduce the RAID creation time.

RAID10 in Linux

To create a striped, mirrored array using four IO Accelerator devices (`fioa`, `fiob`, `fioc`, and `fiod`), run the following command:

```
$ mdadm --create /dev/md0 -v --chunk=256 --level=raid10 --raid-devices=4
/dev/fioa /dev/fiob /dev/fioc /dev/fiod
```

Setting the IO Accelerator VSL options in Windows

In Windows, configure the IO Accelerator VSL software using various module parameters. Individual module parameters are described in this guide. For a complete list of all parameters and how to implement them, see "`fio-config` (on page 64)."

Device naming in Windows

In Windows, the IO Accelerator device receives a name and number as part of the installation process for identification. The syntax is `fctx`, where `x` indicates the device number: 0, 1, 2, and so on. For example, `/dev/fct0` indicates the first IO Accelerator device installed on the system.

To view the bus number, use the IO Accelerator Management Tool software, the `fio-status` utility, or perform the following steps:

1. Click **Start** > **Control Panel** > **System** > **Hardware** > **Device Manager**.
2. Select **Disk Drives**.
3. Click the HP PCIe Workload Accelerator device in the list. The **Properties** dialog box appears.

   The Location field shows the PCIe bus number for the device.

Adding a filesystem in Windows

In Windows, with IO Accelerator devices and IO Accelerator VSL software installed, use the Windows Disk Management utility to make the device available to applications. Typically, Windows detects the new device, initalizes, it, and displays it in Disk Management. Add partitions, format a volume, or create a RAID configuration on the IO Accelerator device using the standard Windows procedures. For more information, see the Windows Disk Management Utility documentation.

Devices with capacities greater than 2 TB require the following partition types:

- Single device: GPT (GUID partition table)
- Multiple devices (for a RAID configuration): dynamic disk

These devices also require sector sizes greater than 512 B. HP recommends 4 KB sectors. When formatting these devices using `fio-format`, the default sector size is 4 KB.

If Windows does not initialize the device, do so manually.

To manually initialize an IO Accelerator device:

1. Click **Start** > **Control Panel**.
2. Click **Administrative Tools**.
3. Click **Computer Management**.

4. Click **Disk Management** in the Storage section of the console tree.
5. Right-click the IO Accelerator device in the list of storage devices on the right. If the IO Accelerator device does not appear in the list, select **Rescan Disks** from the Action menu. You might need to restart your computer to display the IO Accelerator device in the list.
6. Click **Initialize Disk**.

You can now use the Disk Management Utility to add a file system to the IO Accelerator device.

Creating a RAID configuration in Windows

In Windows, use the IO Accelerator device as part of a RAID configuration with one or more additional IO Accelerator devices. To do so, format the IO Accelerator devices as dynamic volumes. In turn, use the dynamic volumes to create multi-disk RAID configurations (spanned, striped, or mirrored).

For specific steps to perform a RAID configuration, see the Windows Disk Management Utility documentation.

If using RAID1/mirroring and one device fails, be sure to run `fio-format` on the replacement device (not the remaining good device) after rebuilding the RAID.

Using the device as a page files store in Windows

In Windows, to safely use the IO Accelerator device with page files (also known as Virtual Memory) requires passing the `preallocate_memory` kernel module parameter. To set this parameter, use the command line utility `fio-config` (on page 64).

```
fio-config -p FIO_PREALLOCATE_MEMORY 1149D2717-1121,1149D2717-1111,10345
```

`1149D2717-1111`,`1149D2717-1111`, and `10345` are serial numbers obtained from `fio-status` (on page 69).

A 4 KB sector size format is required for swap. This reduces the IO Accelerator VSL software memory footprint. Use `fio-format` to format the IO Accelerator device with 4 KB sector sizes.

After setting the parameter, go into the system settings and use the IO Accelerator device to store the paging files. For more information, see "Using Windows page files with the IO Accelerator (on page 87)."

Be sure to provide the serial numbers for the IO Accelerator device, not an adapter, when applicable.

IMPORTANT: Be sure to have enough RAM available to enable the IO Accelerator device with pre-allocation enabled for use as swap. Attaching an IO Accelerator device with preallocation enabled without sufficient RAM might result in system instability and the loss of user processes. For RAM requirements with this version of the IO Accelerator VSL software, see the release notes.

The `preallocate_memory` parameter is recognized by the IO Accelerator VSL software at load time, but the requested memory is not allocated until the specified device is attached.

Setting the amount of preallocated memory

In Linux and Windows, if you enable devices for preallocation, the IO Acclerator VSL software automatically preallocates the amount of memory based on your formatted sector size. If you have a sector size that is less than 4 KiB (for example, 512 B sectors), then the IO Accelerator VSL software preallocates a very large amount of memory (calculated for worst-case scenarios based on 512 B sectors).

For worst-case RAM requirements based on formatted sizes, see the *HP IO Accelerator Driver and Management Software Release Notes*. The IO Accelerator VSL software preallocates enough memory for worst-case scenarios in order to avoid an out-of-memory situation where the IO Accelerator device no longer functions due to insufficient memory (resulting in a system crash).

Operating systems normally use at least 4 KiB blocks of data for virtual memory, so it would be safe in most cases for the IO Accelerator VSL software to preallocate enough memory for a worst-case scenario based on 4KiB sectors. There are two ways to force the IO Accelerator VSL software to preallocate based on 4 KiB block sizes:

- **Format the device to 4 KiB sectors.** With preallocation enabled, the IO Accelerator VSL software automatically preallocates memory based on 4 KiB blocks of data. Use `fio-format` to format the IO Accelerator device with 4 KiB sector sizes.
- **Use the** `PREALLOCATE_MB` **parameter.** The parameter sets the amount of memory that the IO Accelerator VSL software preallocates for every IO Accelerator device in the system that is enabled for preallocation (as described in the previous section).

IMPORTANT: Be sure to have enough RAM available to enable the IO Accelerator device with pre-allocation enabled for use as swap. Attaching an IO Accelerator device with preallocation enabled without sufficient RAM might result in system instability and the loss of user processes. For RAM requirements with this version of the IO Accelerator VSL software, see the release notes.

1. Determine the amount of system memory needed for every device in a worst-case scenario.
   a. For worst-case RAM requirements based on sector sizes, see the *HP IO Accelerator Driver and Management Software Release Notes*.
   b. Base the calculation on the sector size that aligns with the data block size your operating system uses for virtual memory.
2. Set the parameter.

   Linux

   In Linux, set the parameter by adding the following line to the `/etc/modprobe.d/iomemory-vsl.conf` file. For more information on using parameters, see "Using module parameters."

   ```
   options iomemory-vsl preallocate_mb=<value>
   ```

   Windows

   In Windows, set the parameter by using the `fio-config` command line utility. For full utility instructions, see "`fio-config` (on page 64)."

   ```
   fio-config -p PREALLOCATE_MB=<value>
   ```

   In Linux and Windows, `<value>` is the amount of system memory in MB that the IO Accelerator VSL software should preallocate for every IO Accelerator device enabled for preallocation. For example, if you enter a value of `3500`, then the IO Accelerator VSL software preallocates about 3.5 GB of RAM for every IO Accelerator device enabled for preallocation in the system.

In order for the preallocation of memory to be effective, the value should be larger than the default memory usage of the device, as reported by running `fio-status -a`.

Configuring the device to support VM disks in ESXi

IMPORTANT: ESXi VMFS requires 512 B physical sector sizes. IO Accelerator PX600 devices and IO Accelerator SX300 devices come formatted with 4 KiB sectors. You must format these devices to 512 B sectors. You might want to run the `fio-format` utility against all of the IO Accelerator devices in the system all at once.

1. If needed, format the device to 512 B physical sector sizes.

   CAUTION: Formatting the device erases all user data. For more information, see "`fio-format` (on page 68)."

2. Run the following command in the SSH/TSM:

   ```
   fio-format /dev/fct*
   ```

3. Within the vSphere client, click the **Configuration** tab.
4. Under **Hardware**, click **Storage**.
5. Click **Add Storage** in the top right corner. The Add Storage wizard appears.
6. Use the Add Storage wizard to configure the device.

For more information and an explanation of options (including setting the VM File System Block Size), see the vSphere documentation.

You can also create a VMFS datastore using `fdisk` and `vmkfstools` in the TSM directly on the ESXi host. This method is not supported by VMware.

HP recommends a thick virtual disk. A thin provisioning conserves space, but might significantly degrade performance.

After adding and configuring the storage, you can store virtual machines on the IO Accelerator device.

Modifying a VMware resource pool to reserve memory in ESXi

Under certain circumstances, the ESXi operating system might temporarily require most or all of the RAM available on the system, leaving no memory for the IO Accelerator VSL software.

For example, a host running VMware View might need to rapidly provision multiple VDI images. This might happen so quickly that the host memory is temporarily exhausted.

If the VMs starve the IO Accelerator VSL software of RAM, the IO Accelerator device might go offline or stop processing requests. To address this issue, limit the memory consumed by the VMs.

As a starting point, HP recommends limiting RAM available to the VMs equal to the Total Host RAM - RAM equivalent to 0.5% of the total IO Accelerator device capacity. Set the limit by modifying the user pool. For more information on the calculation, see the following example scenario.

The exact amount to limit depends on workload and requires tuning for specific use cases. If 0.5% is not enough, consult the RAM requirements in the IO Accelerator VSL release notes for an idea of how much RAM the VSL might use in worst-case scenarios.

To modify the user pool using the vSphere client:

1. Click the **Summary** tab in the vSphere client to view the current memory usage and capacity.

   The total IO Accelerator device datastore capacity is also visible. Make a note of the capacity.

2. Navigate to the **user Resource Allocation** window.

   a. Select the host > **Configuration** tab > **Software** pane > **System Resource Allocation** link > **Advanced** link. The System Resource Pools appear.
   b. Select the user node under the host tree. The details for the user appear below.
   c. Click the **Edit settings** link. The **user Resource Allocation** window appears.
3. Limit the memory allocated to the VMs.
   a. Under **Memory Resources**, clear the **Unlimited** checkbox.
   b. Set the limit on VM memory consumption.

Example scenario

An ESXi host has:

- Memory capacity of 36852 MB
- Total IO Accelerator device datastore capacity of 1000 GB (or approximately 1000000 MB)

1000000MB device capacity x 0.5% of device capacity ~ 5000 MB of RAM equivalent.

36852 MB total memory capacity - 5000 MB free = 31852 MB of memory limited to the host. The new value under **Limit in Memory Resources** is 31852 MB.

Using the device as swap in Solaris

In Solaris, HP currently does not recommend using the IO Accelerator device as a swap space because deadlocked memory issues can occur.

Configuring a ZFS pool in Solaris

In Solaris, configure two or more IO Accelerator devices into a RAID array using a single ZFS pool.

If using RAID1/mirroring and one device fails, be sure to run `fio-format` on the replacement device (not the remaining good device) after rebuilding the RAID.

In the RAID configurations in the following examples, `c12d0p0`, `c13d0p0`, `c14d0p0`, and `c15d0p0` are examples. The examples do not necessarily match your RAID installation.

Using a single device in Solaris

```
$ pfexec zpool create mypool c12d0p0
```

RAID0 in Solaris

Create a striped set where `c12d0p0` and `c13d0p0` are the two IO Accelerator devices to stripe.

```
$ pfexec zpool create mypool c12d0p0 c13d0p0
```

RAID1 in Solaris

Create a mirrored set using the two IO Accelerator devices `c12d0p0` and `c13d0p0`.

```
$ pfexec zpool create mypool mirror c12d0p0 c13d0p0
```

RAID10 in Solaris

Create a striped, mirrored array using four IO Accelerator devices.

```
$ pfexec zpool create mypool mirror c12d0p0 c13d0p0 mirror c14d0p0 c15d0p0
```

Performance and tuning

Overview

IO Accelerator devices provide high bandwidth, high IOPS, and are specifically designed to achieve low latency.

As IO Accelerator devices improve IOPS and low latency, the device performance might be limited by operating system settings and BIOS configuration. The settings might need to be tuned to take advantage of the performance of IO Accelerator devices.

While IO Accelerator devices generally perform well out of the box, there are some common areas where tuning might help achieve optimal performance.

Disabling CPU frequency scaling

DVFS is a power management technique that adjusts the CPU voltage and frequency to reduce power consumption by the CPU. This technique helps conserve power and reduce the heat generated by the CPU, but it adversely affects performance while the CPU transitions between low-power and high-performance states.

This power-savings technique is known to have a negative impact on I/O latency and IOPS. When tuning for performance, you might benefit from reducing or disabling DVFS completely, even though this might increase power consumption.

If available, DVFS is often configurable as part of the operating system power management features, as well as the system's BIOS interface. Within the operating system and BIOS, DVFS features are often found under the ACPI sections. For more information, see your computer documentation.

Limiting ACPI C-states

Newer processors have the ability to go into lower power modes when they are not fully utilized. These idle states are known as ACPI C-states. The C0 state is the normal, full power, operating state. Higher C-states (C1, C2, C3, and so on) are lower power states.

While ACPI C-states save on power, they can have a negative impact on I/O latency and maximum IOPS. With each higher C-state, typically more processor functions are limited to save power, and it takes time to restore the processor to the C0 state.

When tuning for maximum performance, you might benefit from limiting the C-states or disabling them completely, even though this might increase power consumption.

Setting ACPI C-state options

If the processor has ACPI C-states available, limit or disable them in the BIOS interface. ACPI C-states might be part of the ACPI menu. For more information, see the computer documentation.

C-states in Linux

In Linux, newer kernels have drivers that might attempt to enable ACPI C-states even if they are disabled in the BIOS. Limit the C-state in Linux, with or without the BIOS setting, by adding the following line to the kernel boot options:

```
intel_idle.max_cstate=0 processor.max_cstate=0
```

In this example, the maximum C-state allowed is C0 (disabled).

Setting NUMA affinity in Linux and Windows

In Linux and Windows, severs with a NUMA architecture might require special installation instructions in order to maximize IO Accelerator device performance. This includes most multi-socket servers.

During system boot on some servers with NUMA architecture, the BIOS does not associate PCIe slots with the correct NUMA mode. Incorrect mappings result in inefficient I/O handling that can significantly degrade performance.

For more information about parameters for setting this affinity, see "NUMA configuration (on page 82)."

Setting NUMA affinity in Linux

The IO Accelerator VSL software automatically reassigns devices with the appropriate NUMA node. However, you might still use deprecated NUMA node parameters to manually assign devices to the available NUMA nodes.

Setting NUMA affinity in Windows

To prevent incorrect mappings, manually assign IO Accelerator devices optimally among the available NUMA nodes.

Setting the interrupt handler affinity in Linux and Windows

Device latency can be affected by placement of interrupts on NUMA systems. HP recommends placing interrupts for a given device on the same NUMA node from which the application issues I/O. If the CPUs on the node are overwhelmed with user application tasks, in some cases it might benefit performance to move the interrupt to a remote node to help load-balance the system.

Many operating systems attempt to dynamically place interrupts across the modes, and generally make good decisions.

Linux IRQ balancing

In Linux, the dynamic placement of interrupts is called IRQ balancing. Check if the IRQ balancer is effective by checking `/proc/interrupts`. If the interrupts are unbalanced (too many interrupts on one node) or on an overwhelmed node, you might need to stop the IRQ balancer and manually distribute the interrupts in order to blance the load and improve performance.

Restarting the IRQ balancer after loading the IO Accelerator VSL software and attaching the IO Accelerator devices might resolve interrupt affinity issues. Run one of the following commands, depending on your distribution:

```
/etc/init.d/irqbalance restart
```

```
/etc/init.d/irq_balancer restart
```

If restarting the IRQ balancer does not resolve the affinity issues, HP recommends manually pinning the device interrupts to specific nodes.

Hand-tuning interrupt placement in Linux is an advanced option that requires application performance profiling on any given hardware. For more information on pinning specific device interrupts to specific nodes, see your operating system documentation.

Windows IRQ policy

By default, Windows uses a policy of IrqPolicyAllCloseProcessors and a priority of IrqPriorityNormal, which works best for most applications.

If manual tuning is needed, Microsoft provides the Interrupt Affinity Policy Tool. For information about this tool, see the Microsoft MSDN website (http://msdn.microsoft.com/en-us/windows/hardware/gg463378). The settings the tool changes are also listed on the Microsoft MSDN website (http://msdn.microsoft.com/en-us/library/ff547969(v=vs.85).aspx).

On a machine with more than 64 processors running Windows Server 2008 or later, an additional GroupPolicy parameter can be set through the registry to set the affinity to a different processor group. For more information, see the Microsoft MSDN website (http://msdn.microsoft.com/en-us/windows/hardware/gg463349).

Monitoring and managing devices

Overview

HP provides tools for managing IO Accelerator devices. The tools enable monitoring the devices for errors, warnings, and potential issues. They also enable managing the devices, including performing the following functions:

- Firmware upgrades
- Low-level formats
- Attach and detach actions
- Device status and performance information
- Swap and paging configuration
- Bug report generation

Management tools

HP provides several tools for monitoring and managing IO Accelerator devices.

The IO Accelerator VSL software prints some error messages to the system logs. While the messages are useful for troubleshooting purposes, the IO Accelerator VSL software log messages are not designed for continual monitoring purposes (as each is based on a variety of factors that could produce different log messages depending on environment and use case). For best results, use the tools described in this section to regularly monitor your devices.

Management tools for Linux, Windows, and Solaris

Tools for monitoring and managing IO Accelerator devices include stand-alone tools that require no additional software and data-source tools that can be integrated with other applications.

Stand-alone tools

Stand-alone tools do not require additional software.

Command line utilities

These utilities are installed with the IO Accelerator VSL software and are run manually in a terminal. The `fio-status` utility provides status for all devices within a host. The other utilities allow you to perform other management functions. For more information, see "Utilities (on page 58)."

IO Accelerator Manager Management Tool software (Linux)

In Linux, he GUI browser-based IO Accelerator Manager Management Tool software enables you to monitor and manage every IO Accelerator device installed in multiple hosts across the network. It collects all of the alerts for all IO Accelerator devices, and then displays them in the Alert tab. You can set up the IO Accelerator Manager Management Tool software to send email or SMS messages for specific types of alerts or all alerts. The IO Accelerator Manager Management Tool software packages and documentation are available as separate downloads.

Data-source tools

The data-source tools provide comprehensive data, just like the stand-alone tools, but they do require integration with additional software. At a minimum, some tools can interface with a browser. However, the benefit of the tools is that they can be integrated into existing management software that is customized for your organization.

The tool packages and documentation are available as separate downloads from the IO Accelerator VSL software packages.

SNMP subagent

The SNMP AgentX subagent enables you to monitor and manage your IO Accelerator devices using SNMP. Use a normal SNMP browser, or customize an existing application to interface with the subagent.

SMI-S CIM provider

The CIM provider enables you to monitor and manage the devices using the Common Information Model. Use a normal CIM browser, or customize an existing application to interface with the CIM provider.

IO Accelerator VSL Management SDK (Linux)

In Linux, the C programing API enables you to write customize applications for monitoring and managing IO Accelerator devices.

Management tools for ESXi

TSM command line utilities are installed with the IO Accelerator VSL software and are run manually in a terminal. In order to use these utilities on ESXi, the Shell/TSM must be enabled. The `fio-status` utility provides status for all devices within a host. The other utilities enable you to perform other management functions. For more information, see "Utilities (on page 58)."

Example conditions to monitor

This section provides examples of conditions to monitor. It is intended as an introduction and not as a comprehensive reference. The conditions have slightly different names, states, and values, depending on the tool you choose. For example, an SNMP MIB might have a different name than an SMI-S object or an API function.

In order to properly monitor these conditions, become familiar with the tool you implement and read the documentation for that tool. You might discover additional conditions you want to frequently monitor.

The possible states/values of the conditions are described as Normal (green), Caution/Alert (yellow), or Error/Warning (red). Implement your own ranges of acceptable states/values, especially if you use a data-source tool.

Device status

All monitoring tools return information on the status of the IO Accelerator devices.

Status	Description
Normal (green)	Attached
Caution/Alert (yellow)	Detached, Busy (including Detaching, Attaching, Scanning, Formatting, and Updating)
Error/Warning (red)	Minimal mode, Powerloss Protect disabled

If the device is in minimal mode, the monitoring tool displays the reason for the minimal mode status.

Required actions

If the device is in minimal mode, the action depends on the reason. For example, if the reason is outdated firmware, then update the firmware.

Temperature

IO Accelerator devices require adequate cooling. In order to prevent thermal damage, the IO Accelerator VSL software starts throttling write performance when the on-board controller reaches a specified temperature. If the controller temperature continues to rise, the software shuts down the device when the controller temperature reaches the maximum operating temperature.

The temperatures depend on the device. Newer IO Accelerator devices have higher thermal tolerances. To determine the thermal tolerances of all devices you monitor, see the *IO Accelerator Hardware Installation Guide*. The following table uses the controller thermal tolerances for newer devices (78°C throttling, 85°C shutdown).

Status	Temperature
Normal (green)	< 78° C
Caution/Alert (yellow)	78-84° C
Error/Warning (red)	85° C

You might want to shift the conditions by a few degrees so the Caution/Alert (yellow) condition exists before throttling occurs.

Status	Temperature
Normal (green)	< 75° C
Caution/Alert (yellow)	75-83° C
Error/Warning (red)	83° C

IO Accelerator devices report the temperature of the NAND, which is also an important temperature to monitor. To determine the NAND temperature thresholds and if the device reports the NAND temperature, see the *IO Accelerator Hardware Installation Guide*.

Required actions

If the temperature is at or approaching the Caution/Alert (yellow) condition, increase the cooling for the system. This might include increasing the fan speed, bringing down the ambient temperature, reducing write load, or moving the device to a different slot.

Health reserves percentage

IO Accelerator devices are highly fault-tolerant storage subsystems with many levels of protection against component failure and the loss nature of solid-state storage. As in all storage subsystems, component failures might occur.

By proactively monitoring device age and health, ensure reliable performance over the intended product life. The following table describes the Health Reserve conditions.

Status	Description
Normal (green)	> 10%
Caution/Alert (yellow)	0-10%
Error/Warning (red)	0%

At the 10% healthy threshold, a one-time warning is issued. At 0%, the device is considered unusable. At 3% it enters write-reduced mode. After the 1% threshold, the device enters read-only mode.

For complete information on Health Reserve conditions and their impact on performance, see "Monitoring IO Accelerator health (on page 79)."

Required actions

The device needs close monitoring as it approaches 0% reserves and goes into write-reduced mode, which results in reduced write performance. Prepare to replace the device soon.

Write (health reserves) status

In correlation with the health reserves percentage, the management tools return write states similar to the following states.

Status	Description
Normal (green)	Device is healthy.
Caution/Alert (yellow)	Device is getting close to entering write-reduced mode.
Error/Warning (red)	Device has entered write-reduced mode or read-only mode to preserve the flash from further wearout.

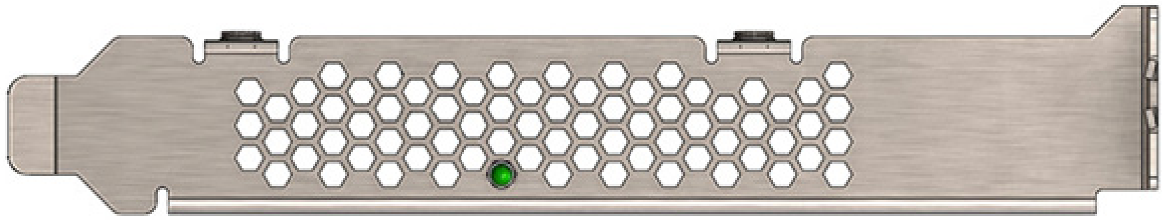
Required actions

The device needs close monitoring as it approaches 0% reserves and goes into write-reduced mode, which results in reduced write performance. Prepare to replace the device soon.

Device LED indicator

If you have physical access to the device and depending on your device configuration, use the LED indicator on the bracket to monitor the device status.

If the device has one LED, it might be similar to the following configuration.



LED	Indication	Notes
Illuminated	Power is on and driver is working.	-
Flashing (fast)	Read and/or write activity.	The faster flashing indicated activity; it does not reflect the amount of data that is read or written. the flashing might not indicate reads from empty sectors (all zeros).
Flashing (slow)	Location beacon.	Use the `fio-beacon` utility to initiate this behavior.
Off	Indicates one of the following: Power is off driver is not loaded, or driver is not working.	Check `fio-status` to see if the device is attached and there are no errors.

Maintenance

Overview

This section explains additional software maintenance functions not covered in the sections "Configuration (on page 24)" and "Monitoring and managing devices (on page 42)."

In Linux, all commands require administrator privileges. Log in as root or use sudo to run the commands.

In ESXi, some of the maintenance tasks are accessible only through the VMware tech support mode (Shell/SSH).

VMware suggests using TSM only "for the purposes of troubleshooting and remediation." VMware recommends using the vSphere Client or any other VMware Administration Automation Product to perform routine ESXi host configuration tasks that do not involve a troubleshooting scenario. For more information, see the article about using TSM in the VMware knowledge base (http://kb.vmware.com/selfservice/microsites/search.do?language=en_US&cmd=displayKC&externalId=1017910).

Uninstalling the IO Accelerator VSL software

IMPORTANT: This version of the IO Accelerator VSL software only supports third generation devices, such as the SX300 and PX600 devices. It does not support devices that were compatible with IO Accelerator VSL software version 3.x.x or earlier.

IMPORTANT: Uninstall all previous versions of the IO Accelerator VSL software before installing the latest version.

Uninstalling the IO Accelerator VSL software in Linux

Uninstalling the IO Accelerator VSL utilities and other support packages in Linux

Uninstalling 2.x support packages

To uninstall the support RPM packages, run the following command (adding or removing package names as necessary):

```
$ rpm -e fio-util fio-snmp-agentx fio-common fio-firmware iomanager-gui
iomanager-jre libfio libfio-doc libfusionjni fio-sysvinit fio-smis fio-snmpmib
libfio-dev
```

To uninstall the support DEB packages, run the following command (adding or removing package names as necessary):

```
$ dpkg -r fio-util fio-snmp-agentx fio-common fio-firmware iomanager-gui
iomanager-jre libfio libfio-doc libfusionjni fio-sysvinit fio-smis fio-snmpmib
libfio-dev
```

Uninstalling 3.x support packages

To uninstall the support RPM packages, run the following command (adding or removing package names as necessary):

```
$ rpm -e fio-util fio-snmp-agentx fio-common fio-firmware libvsl libvsl-doc
fio-sysvinit fio-smis fio-snmp-mib libvsl-dev
```

To uninstall the support DEB packages, run the following command (adding or removing package names as necessary):

```
$ dpkg -r fio-util fio-snmp-agentx fio-common fio-firmware libvsl libvsl-doc
fio-sysvinit fio-smis fio-snmp-mib libvsl-dev
```

Uninstalling the IO Accelerator VSL software RPM package in Linux

With versions 2.x and later (including 3.x releases) of the IO Accelerator VSL software, you must specify the kernel version of the package you are uninstalling. Run the following command to find the installed driver packages:

```
$ rpm -qa | grep -i iomemory
```

Sample output:

```
iomemory-vsl-2.6.18-194.el5-2.2.2.82-1.0
```

Uninstall the IO Accelerator VSL software by running a command similar to the following example (specify the kernel version of the driver to uninstall):

```
$ rpm -e iomemory-vsl-2.6.18-194.el5-2.2.0.82-1.0
```

Uninstalling the IO Accelerator VSL software DEB package in Linux

With versions 2.x and later (including 3.x releases) of the IO Accelerator VSL software, specify the kernel version of the package to uninstall. Run the following command to find the installed packages:

```
$ dpkg -l | grep -i iomemory
```

Sample output:

```
iomemory-vsl-2.6.32-24-server
```

Uninstall the IO Accelerator VSL software by running a command similar to the following example (specify the kernel version of the software to uninstall):

```
$ dpkg -r iomemory-vsl-2.6.32-24-server
```

Uninstalling the IO Accelerator VSL software in Windows

To uninstall the IO Accelerator VSL software:

1. Click **Start** > **Control Panel**.
2. Click **Programs & Features**.
3. Select the entry for the HP PCIe Workload Accelerator.
4. Click **Uninstall**.

Windows uninstalls the IO Accelerator VSL software folder along with all files and folders.

Uninstalling the IO Accelerator VSL software in ESXi 5.x

To uninstall the IO Accelerator VSL package, remove the VIB containing the driver and utilities by running the following command using the vCLI from a remote machine (remove `--server <servername>` if you are on the host CLI):

```
esxcli --server <servername> software vib remove -n block-iomemory-vsl
```

Uninstalling the IO Accelerator VSL software in Solaris

To uninstall the IO Accelerator VSL software and utilities, run the following command:

```
$ pfexec pkgrm iomemory-vsl4
```

Unloading or disabling the software driver

Unloading the software driver in Linux

In Linux, to unload the driver, run the following command:

```
$ modprobe -r iomemory-vsl4
```

Upgrading the kernel in Linux

If you plan to upgrade the kernel when the IO Accelerator VSL software is installed, you must:

1. Unload the IO Accelerator VSL driver.
2. Uninstall the IO Accelerator VSL software.
3. Upgrade the kernel.
4. Install an IO Accelerator VSL software package that is compiled for the new kernel.

Failure to follow this procedure might result in driver load issues.

Disabling the IO Accelerator VSL software in Linux

The IO Accelerator VSL software automatically loads by default when the operating system starts. Disable the IO Accelerator VSL software for diagnostic or troubleshooting purposes.

To disable auto-load, do one of the following:

- Uninstall the IO Accelerator VSL software to keep it from loading.
- Move the IO Accelerator VSL software out of the `/lib/modules/<kernel_version>` directory.

Unloading or disabling the software driver in ESXi

If you need to diagnose or troubleshoot a problem, you might need to unload or disable the IO Accelerator VSL software. Both methods take the IO Accelerator VSL software offline; however, HP recommends disabling the IO Accelerator VSL software autoload and rebooting rather than unloading the IO Accelerator VSL driver.

You must properly unmount and detach all IO Accelerator devices before unloading the IO Accelerator VSL driver. However, if you disable autoload and then reboot, the devices safely unmount and detach on shutdown and then do not auto-attach or mount on boot.

If you must unload the driver and detach an IO Acclerator device, before running the detach utility, carefully read all of the warnings in the optional remote management documentaiton or in "`fio-detach` (on page 66)." Failure to follow the instructions may cause errors, data loss, or corruption.

HP recommends disabling autoload rather than directly unloading the driver.

Disabling the IO Accelerator VSL software autoload

The IO Accelerator VSL driver automatically loads by default when the operating system starts. By disabling the autoload and rebooting, the IO Accelerator VSL software is offline.

In ESXi, to disable driver auto-load, run the following command in TSM, and then reboot the system:

```
$ esxcfg-module --disable iomemory-vsl4
```

The command prevents the IO Accelerator VSL driver from loading on boot, so the device is not available to users. However, all other services and applications are available.

If you disable the driver autoload in ESXi 5.1 or later, you also disable the ability to directly load the driver. You need to re-enable the driver to load the driver.

Unloading the IO Accelerator VSL driver

If you need to unload the driver for diagnostic or troubleshooting purposes, you must properly unmount and detach all IO Accelerator devices.

In ESXi, to unload the IO Accelerator VSL driver, run the following command in TSM:

```
vmkload_mod –u iomemory-vsl4
```

Disabling the IO Accelerator VSL software in Solaris

In Solaris, the IO Accelerator VSL software automatically loads by default when the operating system starts. Disable the IO Accelerator VSL software for diagnostic or troubleshooting purposes.

To disable IO Accelerator VSL software auto-load:

1. Remove the IO Accelerator VSL.

   ```
   $ pfexec rem_drv iomemory-vsl4
   ```

   The IO Accelerator VSL software does not load, so the device is not available to users.
2. Troubleshoot to correct the issue.

Loading or enabling the driver

Loading or enabling the IO Accelerator VSL driver in ESXi

In ESXi, if you previously disabled driver autoload, you might want to re-enable it and then reboot. Otherwise, you might bring the driver online by loading directly. If you load the driver, you need to re-attach the IO Accelerator devices in your system. For more information, see "`fio-detach` (on page 66)."

If you disable the driver autoload in ESXi 5.1 or later, you also disable the ability to directly load the driver. You need to re-enable the driver to load the driver.

If you enable driver autoload and then reboot, the IO Accelerator devices attach automatically unless you have disabled auto attach.

Depending on the situation, you might want to both re-enable driver autoload (to ensure that the IO Accelerator VSL driver loads on the next boot) and directly load the driver to bring it online immediately.

Enabling IO Accelerator VSL driver autoload

To enable the IO Accelerator VSL driver on boot after maintenance, run the following command in TSM and reboot the system.

```
$ esxcfg-module --enable iomemory-vsl4
```

After a reboot, if the driver is enabled, then it appears in the modules listed when you run the following command:

```
$ esxcfg-module --query
```

Loading the IO Accelerator VSL driver

To immediately load the IO Accelerator VSL software, run the following command:

```
vmkload_mod iomemory-vsl4
```

The `vmkload_mod` command loads the driver with default parameters, even if you modified the parameters. You can force the driver to load with modified parameters.

```
vmkload_mod iomemory-vsl4 auto_attach=0
```

For a list of parameters, see "Using module parameters (on page 76)."

Enabling the IO Accelerator VSL in Solaris

To enable the the IO Accelerator VSL software after maintenance, run the following command:

```
$ pfexec /usr/sbin/add_drv -i "pci1aed,1005" "pciex1aed,1005" "pci1aed,1003"
"pciex1aed,1003" "pci1aed,1010" "pciex1aed,1010" "pci1014,3c3" "pci103c,324d"
"pci103c,324e" "pci103c,178b" "pci103c,178c" "pci103c,178d" "pci103c,178e"
"pci103c,176f" "pci103c,1770" -m "*0666 root sys" iomemory-vsl4
```

Disabling auto-attach

When the IO Accelerator VSL software is installed, it is configured to automatically attach any devices when the IO Accelerator VSL software is loaded. Sometimes you might want to disable the auto-attach feature to assist in troubleshooting or diagnostics.

Disabling auto-attach in Linux

To disable the auto-attach feature in Linux:

1. Edit the following file:
   `/etc/modprobe.d/iomemory-vsl4.conf`
2. Uncomment the following line in the file:
   `options iomemory-vsl4 auto_attach=0`
3. Save the file.

To re-enable auto-attach, edit the file and remove the line, comment it out, or change it to the following:

```
options iomemory-vsl4 auto_attach=1
```

Disabling auto-attach in Windows

To disable the auto-attach feature in Windows:

1. Open the command line interface with Administrator permissions.
2. Run the following command:

   ```
   fio-config -p AUTO_ATTACH 0
   ```

   For more information on setting parameters, see "fio-config (on page 64)."

After you restart the system, the IO Accelerator device no longer attaches automatically until you re-enable auto-attach.

When you finish troubleshooting the IO Accelerator VSL software issue, use the `fio-attach` utility to attach the IO Accelerator devices and make them available to Windows.

To re-enable the auto-attach feature in Windows:

1. Open the command line interface with Administrator permissions.
2. Run the following command:

   ```
   fio-config -p AUTO_ATTACH 0
   ```

   For more information on setting parameters, see "fio-config (on page 64)."

The next time you restart the Windows system, the IO Accelerator device automatically attaches.

Disabling auto-attach in ESXi

In ESXi, to load the IO Accelerator VSL software on boot with auto-attach disabled, set the `auto_attach` parameter to `0` using the vCLI.

```
esxcfg-module -s 'auto_attach=0' iomemory-vsl4
```

The change is not enforced until you reboot the system.

To re-enable auto-attach, set the parameter to `1`.

Disabling auto-attach in Solaris

To disable the auto-attach feature in Solaris:

1. Edit the following file:

   ```
   /usr/kernel/drv/iomemory-vsl4.conf
   ```

2. Add the following line to the file:

   ```
   auto_attach=0;
   ```

3. Save the file.

To re-enable auto-attach in Solaris, remove the `auto_attach=0;` line from the `/usr/kernel/drv/iomemory-vsl4.conf` file or change it to the following:

```
auto_attach=1;
```

Unmanaged shutdown issues

Unmanaged shutdowns due to power loss or other circumstances force the IO Accelerator to perform a consistency check during the reboot. This reboot might take several minutes or more to complete.

Although data written to the IO Accelerator is not lost due to unmanaged shutdowns, important data structures might not have been properly committed to the drive. This consistency check repairs these data structures.

In ESXi, check `fio-status` after a crash to determine if the devices are in an "Attaching" state.

In Windows, the reboot is indicated by a progress percentage during a startup. You can cancel this consistency check by pressing the **Esc** key during the first 15 seconds after the **Consistency Check** message appears at the prompt. However, if you choose to cancel the check, the IO Accelerators remain unavailable to users until the check is completed. You can perform this check later using the IO Accelerator Management Tool **Attach** function.

Improving rescan times

The rescan of the device (also called a consistency check) the IO Accelerator VSL software performs after an unmanaged shutdown may take an extended period of time depending on the total capacity of the devices that the IO Accelerator VSL software needs to scan.

Default fast rescan

By default, all IO Accelerator devices formatted with the `fio-format` utility are formatted to have improved rescan times. Disable the default fast rescan by reformatting the device and using the `-R` option. Disabling the default fast rescan feature reclaims some reserve capacity that is normally set aside to help improve rescan times.

If you leave the default fast rescan feature in place, you can take further steps to improve rescan times by implementing one of the following module parameters.

Faster rescans using module parameters

The RMAP and RSORT parameters require the default fast rescan formatting structure, and they also use RAM to help improve rescan times. The extra memory enables the rescan process to complete faster, which reduces downtime after a hard shutdown. The memory allocation is only temporary and is freed up after the rescan process completes.

If you use one of the parameters, you need to set the upper limit of RAM used by that parameter. To do so, determine how much RAM to allow each parameter use in your scenario, how much system RAM is available, and therefore which parameter is more suited for your use case.

For more information on setting module parameters in Linux, ESXi, and Solaris, see "Using module parameters (on page 76)."

For more information on setting module parameters in Windows, see "`fio-config` (on page 64)."

The following is a brief comparison of the two parameters.

RMAP parameter

- Fastest: The parameter improvement results in the fastest rescan times.

- Less scalable: All or nothing. The parameter requires enough RAM to function. If the RAM limit is set too low, then the IO Accelerator VSL software does not use RMAP at all, and it reverts back to the default fast rescan process.
- Target scenario: The parameter improves any use case if enough RAM is available for the parameter. It is more suited for smaller capacity IO Accelerator devices or systems with fewer IO Accelerator devices installed. HP recommends it for devices that have been used for many small, random writes.

RSORT parameter

- Faster: The parameter improves rescan times over the default fast rescan process.
- Scalable: With the RSORT parameter, the IO Accelerator VSL software works with the system RAM to improve rescan times until it reaches the RAM limit set in the parameter. At that point, the software reverts back to the default fast rescan process.
- Target scenario: The parameter improves rescan times in any use case. It is especially useful in systems with multiple IO Accelerator devices or larger-capacity IO Accelerator devices. HP recommends it when IO Accelerator devices are used to store databases.

RMAP parameter

The `RMAP_MEMORY_LIMIT_MiB` parameter sets the upper RAM limit (in mebibytes) used by the IO Accelerator VSL software to perform the RMAP rescan process. Use this option only if you have enough memory for all of your IO Acclerator devices in the system. If you do not have enough memory to use this option, use the RSORT parameter instead.

Because the RMAP parameter requires a set amount of memory, it often works best with fewer IO Accelerator devices or smaller-capacity IO Acclerator devices in a system, but the determining factor is how much memory is in the system and whether there is enough to set the appropriate memory limit.

The RMAP parameter requires 4.008 B of RAM per block of IO Accelerator device capacity.

1. Determine the number of blocks formatted for each device.
   - This information is visible when formatting the device using the `fio-format` utility.
   - Estimate the number of blocks using the device capacity and the formatted sector size.

     The following is a quick estimation of the number of blocks on a 1000 GB device with 512 B size sectors (2 blocks per KB).

     ```
     1000GB * 1000MB/GB * 1000KB/MB * 1000B/KB * 1 block/512B = 1,953,125,000
     blocks
     ```
2. To determine the memory limit required for the RMAP parameter to function, multiply the number of blocks by 4.008 bytes of RAM per block, and then convert the result into MiB.

   The example had about 1.95 billion blocks.

   ```
   1,953,125,000 blocks * 4.008 B/block * 1KiB/1024 B * 1 MiB/1024 KiB = ~74656
   MiB of RAM
   ```

   In this example, you need about 7465 MiB of RAM available in your system for a 1000 GB IO Accelerator device formatted for 512 B sectors, and you need to set the RMAP parameter to 7500.

   By default, the RMAP parameter is set to 3100. The low default value is to prevent the rescan process from using all of the RAM in systems with less available memory.

   - If the RMAP value is too low for the number of IO Accelerator device blocks in the system, then the IO Accelerator VSL software does not use the RMAP process to improve rescan times. It uses the default fast rescan process. RMAP is an all-or-nothing setting.

- If the system does not have enough system memory to use the RMAP parameter, consider using the RSORT parameter. The RSORT parameter uses its RAM limit to improve the rescan process, and then the IO Accelerator VSL software reverts to the default fast rescan process to finish the consistency check.

3. Set the module parameter to the value you determined.

   For more information on setting module parameters in Linux, ESXi, and Solaris, see "Using module parameters (on page 76)."

   For more information about setting parameters in Windows, see "`fio-config` (on page 64)."

RSORT parameter

The `RSORT_MEMORY_LIMIT_MiB` parameter sets the RAM limit used by the IO Accelerator VSL software to perform the RSORT rescan process. The RSORT rescan process is faster than the default rescan process. HP recommends using it to rescan devices that are used as datastores for databases.

If the RSORT parameter is given any memory limit, the IO Accelerator VSL software uses the RSORT process until either the rescan is done or it consumes the memory limit. If the process runs out of memory, it reverts to the default fast rescan process. However, in order to optimize the use of this process, calculate the target RAM usage and set the limit based on that target. There is no penalty for setting a high limit; the RSORT process only uses the RAM it needs (up to the set limit).

This target is based on 32 bytes per write extent. For example, if your database writes 16 KB at a time, there is one write extent per 16KB of IO Accelerator device capacity.

One measure of the the benefits of the RSORT process is to see how many blocks are written per write extent. The RSORT process improves rescan times over the default fast rescan process when a device has eight or more blocks written per extent. For example, if your IO Accelerator device is formatted to 512 B sector sizes (2 blocks per KB) and your database writes in 8 KB chunks, then your database writes 16 blocks per write extent and RSORT would improve the rescan times.

1. Determine the number of blocks formatted for each device.
   - This information is visible when formatting the device using the `fio-format` utility.
   - Estimate the number of blocks using the total device capacities and the formatted sector sizes.

     The following example shows a brief estimation of the number of blocks on 1000 GB of IO Accelerator device capacity with 512 B size sectors (2 sectors per KB):

     ```
     1000 GB * 1000 MB/GB * 1000 KB/MB * 1000 B/KB * 1 block/512 B = 1,953,125,000
     blocks
     ```

2. Divide the number of blocks by the write extents per block to determine the total possible number of write extents on the devices.

   The example had 1.95 billion blocks. Assume 16 KB write extents (32 blocks per write on 512B sectors).

   ```
   1,953,125,000 blocks * 1 write extent/32 blocks = 61,035,156 writes
   ```

3. To determine the memory target for the parameter, multiply the number of writes by 32 bytes of RAM per write, and then convert the result into MiB.

   The example had 61 million write extents.

   ```
   61,035,156 writes * 32 B/write * 1 KiB/1024 B * 1 MiB/1024 KiB = ~1863 MiB
   of RAM
   ```

   In this example, set the RSORT limit to approximately 2300 MiB of RAM available in your system for 1000 GB of IO Accelerator device capacity formatted for 512 B sectors.

   By default, the RMAP parameter is set to 0m and has a maximum of 100000 (100 GiB).

4. Set the module parameter to the value you determined.

   For more information on setting module parameters in Linux, ESXi, and Solaris, see "Using module parameters (on page 76)."

   For more information about setting parameters in Windows, see "`fio-config` (on page 64)."

Upgrading the software with a RAID configuration in Windows

> **IMPORTANT:** Read the *HP IO Accelerator Driver and Management Software Release Notes* that come with each new release as well as the procedure instructions to prevent data loss during upgrades.

To upgrade the IO Accelerator VSL software with a RAID configuration in place in Windows:

1. Close any applications accessing the IO Accelerator devices.
2. Open the IO Accelerator VSL utilities folder. The default location for this release is `C:\Program Files\Common Files\VSL Utils`.
3. Use the `fio-config` utility to disable auto attach.

   `fio-config -p AUTO_ATTACH 0`

   The IO Accelerator device no longer automatically attaches the next time you restart the system.
4. Uninstall the IO Accelerator VSL software in **Windows Add/Remove Programs**.
5. Restart the system.
6. Download the latest IO Accelerator VSL software package.
7. Unzip and install the IO Accelerator VSL software. While finishing installation, click **No** to select a manual restart.
8. Open the IO Accelerator VSL utilities folder. The default location for this release is `C:\Program Files\Common Files\VSL Utils`.
9. Use the `fio-config` utility to re-enable auto attach.

   `fio-config -p AUTO_ATTACH 1`

   The IO Accelerator device automatically attaches the next time you restart the system.
10. Update the device firmware. Follow the steps in "Upgrading the firmware (on page 23)."
11. Restart the computer after the firmware upgrade is complete. The IO Accelerator VSL Check Utility runs at the next boot.

Windows now detects the devices in the RAID configuration with the upgraded software.

Defragmentation in Windows

The IO Accelerator device does not need to be defragmented. However, some Windows versions automatically run defragmentation as a scheduled task. Turn off automatic defragmentation.

Detaching an IO Accelerator device in ESXi

In ESXi, HP does not recommend detaching IO Accelerator devices used as datastores.

The best practice is:

1. Disable auto-attach. For more information, see "Disabling auto-attach (on page 51)."
2. Reboot.
3. Perform the necessary maintenance operations.
4. Re-enable auto-attach.
5. Reboot.

If you must detach an IO Accelerator device, before running the detach utility, carefully read all of the warnings in the optional remote management documentation or "`fio-detach` (on page 66)." Failure to follow the instructions might cause errors, data loss, or corruption.

Utilities

Utilities reference

The IO Accelerator installation packages include various command line utilities. These utilities provide a number of useful manners to access, test, and manipulate your device.

IMPORTANT: Additional utilities are installed in the default utility installation folder. The additional utilities are dependencies used by the main IO Accelerator utilities. Do not use them unless HP Support advises you to do so.

Utility	Purpose
`fio-attach`	Makes an IO Accelerator available to the OS.
`fio-beacon`	Lights the IO Accelerator external LEDs.
`fio-bugreport`	Prepares a detailed report for use in troubleshooting issues.
`fio-config` (Windows only)	In Windows, enables configuration parameters for device operation.
`fio-detach`	Temporarily removes an IO Accelerator from OS access.
`fio-firmware` (Linux only)	In Linux, reports the firmware version and revision without loading the driver.
`fio-format`	Performs a low-level format of an IO Accelerator.
`fio-pci-check`	In Linux, Windows, and ESXi, checks for errors on the PCI bus tree, specifically for IO Accelerator devices.
`fio-status`	Displays information about the device.
`fio-sure-erase`	In Linux and Windows, clears or purges data from the device.
`fio-update-iodrive`	Updates the IO Accelerator firmware.

All utilities have `-h` (Help) and `-v` (Version) options. Also, `-h` and `-v` cause the utility to exit after displaying the information.

Linux

The default utility installation folder in Linux is `/usr/bin`.

Windows

The default utility installation folder in Windows is `C:\Program Files\Common Files\VSL Utils`.

The command line utilities require administrator rights to run in Windows. Right-click the **Command Prompt** menu item, and then select **Run as administrator**.

To run these utilities from a command line, change to the directory that contains them (by default, `C:\Program Files\Common Files\VSL Utils`) or add that directory to your system path. As a convenience, if you used the Windows installer, the utilities directory was added to the system path. See the documentation for your Windows version for information about adding a directory to the system path.

ESXi

The default utility installation folder in ESXi is `/usr/bin`.

The command line utilities are only accessible through VMware Tech Support Mode (Shell/TSM) in ESXi and the Console Operating System (COS) in ESX.

Solaris

The default utility installation folder in Solaris is `/opt/fusion-io/bin`.

IMPORTANT: VMware suggests that TSM only be used for the purposes of troubleshooting and remediation. VMware recommends using the vSphere Client or any other VMware Administration Automation Product to perform routine ESXi host configuration tasks that do not involve a troubleshooting scenario. For more information on using TSM, see the VMware Knowledge Base article (http://kb.vmware.com/selfservice/microsites/search.do?language=en_US&cmd=displayKC&externalId=1003677).

You can use the SMI-S remote management tools instead of TSM command line utilities. The SMI-S remote management tools provide a management experience similar to the command line utilities. For more information, see "Management tools (on page 42)."

fio-attach

Description

Attaches the IO Accelerator device and makes it available to the operating system. The command displays a progress bar and percentage as it operates.In Linux and Windows, the `fio-attach` utility creates a block device in `/dev` named `fiox` (where `x` is `a`, `b`, `c`, and so on). Then, you might partition or format the IO Accelerator device, or set it up as part of a RAID array.

NOTE: In most cases, the IO Accelerator automatically attaches the device on load and does a scan. You only have to run `fio-attach` if you ran `fio-detach` or if you set the IO Accelerator `auto_attach` parameter to `0`.

If the IO Accelerator device is in minimal mode, then `auto-attach` is disabled until the cause of the device being in minimal mode is fixed.

Syntax

```
fio-attach <device> [options]
```

where `<device>` is the name of the device node (`/dev/fctx`), where `x` indicates the device number: `0, 1, 2`, and so on. For example, `/dev/fct0` indicates the first IO Accelerator device detected on the system.

For Linux and Windows, you can specify multiple IO Accelerator devices. For example, `/dev/fct1 /dev/fct2` indicates the second and third IO Accelerator devices installed on the system.

Option	Description
`-r`	Force a metadata rescan. This might take an extended period of time, and is not normally required.*
`-c`	Attach only if clean.

Option	Description
`-q`	Quiet: disables the display of the progress bar and percentage.
`-Q`	Quiet: disables the display of the progress bar only.

*Only use this option when directed by HP Support (http://www.hp.com/support).

In ESXi, if a device attaches, but the claiming process hangs, then one or more of the devices might not have been properly unclaimed when they were previously detached. The improperly claimed devices are preventing other devices from being claimed.

To correct this issue, attempt to attach each of the other devices individually. This claims the devices that were improperly unclaimed and allows the hung device to proceed with attaching. If you want, run `fio-detach` on any devices to detach them again.

fio-beacon

Description

Lights the LED of specified IO Accelerator to locate the device. Detach the IO Accelerator, and then run the `fio-beacon`.

Syntax

```
fio-beacon <device> [options]
```

Where `<device>` is the name of the device node (`/dev/fctx`), where *x* indicates the card number: 0, 1, 2, and so on. For example, `/dev/fct0` indicates the first IO Accelerator installed on the system. The device numbers are visible using `fio-status` (on page 69).

Options

Option	Description
`-0`	Off: (Zero) Turns off the beacon.
`-1`	On: Lights the LED beacon.
`-p`	Prints the PCI bus ID of the device at `<device>` to standard output. Usage and error information might be written to standard output rather than to standard error.

fio-bugreport

Description

Prepares a detailed report of the device for use in troubleshooting problems. The results are saved in a file that indicates the date and time the utility was run.

Syntax

```
fio-bugreport
```

NOTE: If the utility recommends that you contact Fusion-io support, disregard that message and contact HP support (http://www.hp.com/support) instead.

fio-bugreport - Linux

Linux

The results of the `fio-bugreport` utility are saved in the `/tmp` directory.

Example

```
/tmp/fio-bugreport-20100121.173256-sdv9ko.tar.bz2
```

Sample output

```
-bash-3.2# fio-bugreport
BUGREPORT_VERSION 2.0
Running fio-read-lebmap /dev/fct0
Running fio-read-lebmap -m /dev/fct0
Running fio-get-erase-count /dev/fct0
Running fio-get-erase-count -b /dev/fct0
Running fio-kinfo -i /dev/fct0
Running fio-kinfo -i --driver-data /dev/fct0
Running dc-status -v
Running dc-status -b -fj -U
Collecting /proc/buddyinfo
Collecting /proc/cgroups
Collecting /proc/cmdline
...
Collecting /etc/init.d/iomemory-vsl4
Collecting /etc/modprobe.d/iomemory-vsl4.conf
Collecting /etc/sysconfig/iomemory-vsl4
Building tar file...
Please attach the bugreport tar file
/tmp/fio-bugreport-20090921.173256-sdv9ko.tar.bz2
to your support case, including steps to reproduce the problem.
If you do not have an open support case for this issue, please open a support
case with a problem description and then attach this file to your new case.
```

For example, the filename for a bug report file named

`/tmp/fiobugreport-20090921.173256-sdv9k0.tar.bz2` indicates the following:

- Date (20090921)
- Time (173256, or 17:32:56)
- Misc. information (sdv9ko.tar.bz2)

fio-bugreport - Windows

Windows

This utility captures the current state of the device. When a performance or stability issue occurs with the device, run the `fio-bugreport` utility. For assistance with troubleshooting, contact HP Support (http://www.hp.com/support).

The `fio-bugreport` utility runs several information-gathering utilities and combines the resulting data into a text file. The results are saved in the current working directory in a `.zip` file that indicates the date and time the utility ran.

Sample output

```
C:\Users\username>"\Program Files\Fusion-io\Utils\fio-bugreport.exe"

Generating bug report. Please wait, this may take a while...

----------------------------------------------

Gathering all Windows Event Logs...DONE

Gathering Fusion-io Windows Event Logs...DONE

Gathering System Information...DONE

Running fio utilities...DONE

Compressing to CAB file...DONE

Bug report has successfully been created:

fio-bugreport-20090921-173256.zip.

Please attach this file to your support case.

If you do not have an open support case for this issue, please open a support
case with a problem description and then attach this file to your new case.
```

For example, the filename for a bug report file named `fiobugreport-20090921.173256.zip` indicates the following:

- Date: 20090921 (YYYY:MM:DD)
- Time: 173256, or 17:32:56

fio-bugreport - ESXi

ESXi

This utility captures the current state of the device. When a performance or stability issue occurs with the device, run the `fio-bugreport` utility. For assistance with troubleshooting, contact HP Support (http://www.hp.com/support).

The output indicates where the bugreport is saved.

IMPORTANT: The `fio-bugreport` utility uses `vm-support` to gather information including system logs. If `vm-support` fails, the utility does not gather the system logs.

Sample output

```
~ # fio-bugreport

VMkernel-5.0.0

Report output: /var/tmp/fio-bugreport-20090921.173256-sdv9ko.tar.gz

OS: VMware-ESXi-5.0.0

BUGREPORT_VERSION 2.0
```

```
Running esxcli software vib list
Running fio-read-lebmap /dev/fct0
Running fio-read-lebmap -m /dev/fct0
Running fio-get-erase-count /dev/fct0
Running fio-get-erase-count -b /dev/fct0
Running fio-kinfo -i /dev/fct0
Running fio-kinfo -i --driver-data /dev/fct0
Running uname -a
Running vib-env
Running lspci
Running lspci -p
Running lspci -vd
Running vmchkdev -L
Running df -k
Running vdf
Running cim-diagnostic.sh
...
Building tar file...
Please attach the bugreport tar file
/var/tmp/fio-bugreport-20090921.173256-sdv9ko.tar.bz2
to your support case, including steps to reproduce the problem.
If you do not have an open support case for this issue, please open a support
case with a problem description and then attach this file to your new case.
```

For example, the filename for a bug report file named

`/tmp/fiobugreport-20090921.173256-sdv9k0.tar.bz2` indicates the following:

- Date (20090921)
- Time (173256, or 17:32:56)
- Misc. information (sdv9ko.tar.bz2)

fio-bugreport - Solaris

Solaris

This utility captures the current state of the device. When a performance or stability issue occurs with the device, run the `fio-bugreport` utility. For assistance with troubleshooting, contact HP Support (http://www.hp.com/support).

Sample output

For example, the filename for a bug report file named

`/tmp/fiobugreport-20090921.173256-sdv9k0.tar.bz2` indicates the following:

- Date: 20090921 (YYYY:MM:DD)

- Time (173256, or 17:32:56)

fio-config

Description

In Windows, sets and gets configuration parameters for device operation. For a list of parameters, see "fio-config options (on page 64)."

In order for the parameter values to be enforced, either reboot the system or first disable and then re-enable all IO Accelerator devices in the Device Manager. This reloads the IO Accelerator VSL software with the values enabled. Use the `-p` option if you plan to reboot.

Syntax

```
fio-config [options] [<parameter>] [<value>]
```

where <parameter> is the IO Accelerator VSL software parameter to set, and <value> is the value to set for the parameter. name given to the IO Accelerator during the installation process.

Options

Option	Description
`-e`	Enumerate configuration parameter names and values.
`-g <name>`	Get the configuration parameter.
`-p <name>`	Set and persist the configuration parameter.
`-s <name>`	Set the configuration parameter in memory only.
`-V`	Print verbose information.
`-v`	Print version information.

fio-config options

The following table describes the driver parameters you can set with the `fio-config` utility.

IMPORTANT: `fio-config` options must be entered in uppercase to function properly.

IMPORTANT: MSI (Message Signaled Interrupts) is enabled by default and it cannot be disabled using fio-config.

Other than `FIO_PREALLOCATE_MEMORY` and `FIO_EXTERNAL_POWER_OVERRIDE`, all `fio-config` options are global—they apply to all IO Accelerator devices in the computer.

IMPORTANT: Setting the `FIO_PREALLOCATE_MEMORY` and `FIO_EXTERNAL_POWER_OVERRIDE` parameters overwrites previous values. To add additional serial numbers to the list, list the new serial numbers as well as the previously entered numbers. To clear the list, set the parameter without any values.

Option	Default (min/max)	Description
`AUTO_ATTACH`	1 (0,1)	Always attach the device on driver load (1).

IODRIVE_TINTR_HW_WAIT	0 (0, 255)	Interval (microseconds) to wait between hardware interrupts.
FIO_EXTERNAL_POWER_OVERRIDE	No devices selected	Enables selected devices to draw full power from the PCIe slot.*
FORCE_MINIMAL_MODE	0 (0,1)	Forces minimal mode on the device (1). This parameter is set to false (0) by default.
PARALLEL_ATTACH	0 (0,1)	Enables parallel attach of multiple devices (1). This parameter is set to false (0) by default.
FIO_PREALLOCATE_MEMORY	0	For the selected device, preallocates all memory necessary to have the drive usable as swap space. For example: `fio-config /dev/fct0 -p FIO_PREALLOCATE_MEMORY "1234,54321"` where `1234` and `54321` are serial numbers obtained from `fio-status`.
PREALLOCATE_MB	No value selected	The `<value>` is the amount of system memory in MB that the IO Accelerator VSL software should preallocate for every IO Accelerator device enabled for preallocation. For example, if you enter a value of `3500`, the IO Accelerator VSL software preallocates about 3.5 GB of RAM for every IO Accelerator device enabled for preallocation in that system. For more information, see "Using the device as a page files store in Windows (on page 34)."
rmap_memory_limit_MiB	3100 (0, 100000)	Amount of system memory in MiB allocated for improving rescan times after an unclean shutdown. For more information, see "Improving rescan times (on page 53)."
rsort_memory_limit_MiB	0 (0, 100000)	Amount of system memory in MiB allocated for improving rescan times after an unclean shutdown. For more information, see "Improving rescan times (on page 53)."
WIN_LOG_VERBOSE	0 (0, 1)	If enabled (1), the IO Accelerator VSL software writes additional messages to the event log. This assists HP Support in troubleshooting any device issues or failures.
WIN_DISABLE_ALL_AFFINITY	0 (0, 1)	When `WIN_DISABLE_ALL_AFFINITY` is set to 0, the driver enables interrupt and worker thread affinity in the driver. When `WIN_DISABLE_ALL_AFFINITY` is set to 1, the driver disables all affinity settings. This is an override of any other affinity settings. The driver must be reloaded for this parameter to take effect.
WIN_DISABLE_DEFAULT_NUMA_AFFINITY	0 (0, 1)	When `WIN_DISABLE_DEFAULT_NUMA_AFFINITY` is set to 0, during initialization, the driver queries Windows for the affinity settings assigned to the adapter by the operating system. This setting is known as the "default NUMA affinity." After the affinity is queried correctly, the driver sets the affinity of the adapter interrupt and associated worker threads to the default operating system

		setting. This action has the effect of setting the affinity of the interrupt and worker threads to all processors on a single NUMA node in the system. When `WIN_DISABLE_DEFAULT_NUMA_AFFINITY` is set to 1, the driver ignores the affinity settings assigned to the adapter by the operating system. The driver must be reloaded for this parameter to take effect.
`FIO_AFFINITY`	N/A	`FIO_AFFINITY` is a list of `<affinity specification>` triplets to specify the affinity settings of all adapters in the system. Each item in the triplet is separated by a comma, and each triplet set is separated by a semicolon. For syntax information and examples showing the use of this parameter, see "NUMA configuration (on page 82)."
`WIN_SCSI_BUS_ID`	0 (0, 254)	The `WIN_SCSI_BUS_ID` parameter sets the Windows SCSI ID number for all IO Accelerator devices in the system to avoid conflicts with other SCSI device IDs. The default value, 0, is off and no IDs are set. Any value between 1-254 sets the SCSI IDs for all IO Accelerator devices in the system to that number. IO Accelerator devices do not directly use SCSI IDs, so any non-conflicting number is acceptable.

*Use carefully. For more information, see "Enabling additional PCIe power override (on page 24)."

fio-detach

Description

Detaches the IO Accelerator device and removes the corresponding `fctx` IO Accelerator block device from the operating system. By default, the command displays a progress bar and percentage as it completes the detach.

In Linux and Windows, the `fio-detach` command waits until the device completes all read/write activity before executing the detach process.

CAUTION: Before using this utility, be sure that the IO Accelerator to be detached is not currently mounted or otherwise in use (for example, as a datastore). The driver cannot determine whether the drive is in use at the time of detaching. Detaching a drive while it is in use might cause errors, data loss, and corruption.

Windows

The Windows operating system refuses the request to detach the drive associated with the IO Accelerator device because it is part of a RAID volume and might cause the volume to fail. This does not occur with simple volumes (such as a single IO Accelerator device). To detach a simple volume, take the volume offline using the Disk Management MMC plug-in application.

Unmounting the device in ESXi

Detaching a device while mounted or in use can cause errors, data loss, and/or corruption.

In most cases, HP does not recommend using the `fio-detach` utility to ensure a device is detached. Instead, as a best practice, as a safe detach workaround follow the instructions in "Disabling auto-attach (on page 51)."

Before you use this utility, make sure each device is unmounted. HP recommends also putting the host system in maintenance mode.

Syntax

```
fio-detach <device> [options]
```

where `<device>` is the name of the device node, where `x` indicates the card number: 0, 1, 2, and so on. For example, `/dev/fct0` indicates the first IO Accelerator device detected on the system.

You can specify multiple IO Accelerator devices. For example, `/dev/fct1 /dev/fct2` indicates the second and third devices installed on the system. You can also use a wildcard to indicate all IO Accelerator devices on the system. For example, `/dev/fct*`.

Options

Option	Description
`-f` (Windows only)	Force: Causes an immediate detach (does not save metadata).*
`-q`	Quiet: Disables the display of the progress bar and percentage.
`-Q`	Quiet: Disables the display of the progress bar only.

*Although the `-f` (force) option causes the IO Accelerator device to detach, even in a RAID setup, HP strongly recommends to take the drives/volume offline using the Windows Disk Management plug-in and then perform the detach. Forcing the detach might result in loss of data.

Notes

Attempting to detach an IO Accelerator device might fail with an error indicating that the device is busy. This might occur if the device is part of a software RAID (0,1,5) volume, is part of a mounted RAID, is mounted, is in use by VM, or some process has the device open.

In Linux, the tools `fuser`, `mount`, and `lsof` can help determine what is holding open the device.

fio-firmware

Description

In Linux, reports the firmware version and revision on all IO Accelerator devices without requiring the loading of the IO Accelerator VSL software driver. The utility assists diagnosing system issues by indicating the firmware versions on installed devices, helping determine which version of the IO Accelerator VSL software to load.

Output is in CSV format, with each IO Accelerator device on a separate line.

```
<PCIe address>, <FW version>, <FW revision>
```

Syntax

```
fio-firmware [options]
```

Option	Description
`-f`	Prints only x.y.z firmware version.
`-n`	Disables printing header information on columns.

Option	Description
-p	Prints pending version of firmware to be activated on next reboot.
-r	Prints only firmware revision.

fio-format

Description

Performs a low-level format of the IO Accelerator device. By default, the `fio-format` utility displays a progress bar and percentage as it completes the format.

For Linux and Windows, IO Accelerator devices ship pre-formatted, so `fio-format` is generally not required except to change the logical size or block size of a device, or to erase user data on a device. To ensure the user data is truly erased, use `fio-sure-erase` (on page 72).

> **CAUTION:** Use this utility with care since it deletes all user information on the IO Accelerator.

If the `-s` or `-o` options are not included, the device size defaults to the advertised capacity. If used, the `-s` and `-o` options must include the size or percentage indicators.

HP recommends adding power backup to the system to prevent power failures during formatting.

In Linux or Windows, using a larger block (sector) size, such as 4,096 bytes, can significantly reduce worst-case IO Accelerator host memory consumption. However, some applications are not compatible with non-512-byte sector sizes.

VMFS, the filesystem used by ESXi, requires 512 B sector size. If your device ships with 4 KiB sector sizes, you must format the device to 512 byte sectors.

Syntax

```
fio-format [options] <device>
```

where `<device>` is the name of the device node (`/dev/fctx`), where `x` indicates the device number: 0, 1, 2, and so on. For example, `/dev/fct0` refers to the first IO Accelerator device detected on the system. To view this number, use `fio-status` (on page 69).

Options

Option	Description
-b <size B|K>	Sets the block (sector) size, in bytes or kibibytes (base 2). Only 512 B or 4 KiB sector sizes are supported. For example: `-b 512B` or `-b 4K` (`B` in `512B` is optional). SX300 devices and PX600 devices ship with 4 KiB sector sizes. If you do not specify a sector size, the utility formats the sectors to the default size of 4 KiB sector sizes.[1, 2]
-f	Forces the format size, bypassing normal checks and warnings. This option might be needed when `fio-format` does not proceed properly. The "Are you sure?" prompt still appears unless you use the `-y` option.
-q	Quiet: Disables the display of the progress and percentage indicators.
-Q	Quiet: Disables the display of the progress bar only.

Option	Description
-s <size M|G|T|%>	Sets the device capacity as a specific size (in TB, GB, or MB) or as a percentage of the advertised capacity: • T Number of terabytes (TB) to format • G Number of gigabytes (GB) to format • M Number of megabytes (MB) to format • % Percentage, such as 70% (the percent sign must be included)
-R	Disables fast rescan on unclean shutdown to reclaim some reserve capacity.
-y	Auto-answers "yes" to all queries from the application (bypass prompts).

[1]ESXi only supports 512 B sector sizes for use in VMFS datastores. Do not format the IO Accelerator device with any other sector size if you plan to use VMFS. If you pass the device through to a VM using VMDirectPathIO, the guest VM can use any sector size appropriate for the guest operating system. In this case, formatting is done in the guest.
[2]Solaris only supports 512 B sector sizes. Do not format the IO Accelerator device with any other sector size.

Re-attach the device in order to use the IO Accelerator device. For details, see fio-attach (on page 59).

fio-pci-check

Description

Checks for errors on the PCI bus tree, specifically for IO Accelerator devices. This utility displays the current status of each IO Accelerator device. It also prints the standard PCI Express error information and resets the state.

It is normal to see a few correctable errors when fio-pci-check is initially run. Subsequent runs reveal only one or two errors during several hours of operation.

Syntax

```
fio-pci-check [options]
```

Options	Description
-d <value>	1 = Disables the link; 0 = enables the link (not recommended).
-e	Enables PCIe error reporting.
-f	Scans every device in the system.
-n	Does not perform any writes to config space. Prevents errors from being cleared.
-o	Optimizes the IO Accelerator device PCIe link settings by increasing the maximum read request size if it is too low.
-r	Forces the link to retrain.
-v	Verbose: Prints extra data about the hardware.

fio-status

Description

Provides detailed information on installed devices. This utility operates on either fctx or fiox devices. The utility runs from root level and having the IO Accelerator VSL driver loaded. If no IO Accelerator driver is loaded, a smaller set of status information is returned.

Enabling `fio-status` provides alerts for certain error modes, such as a minimal mode, read-only mode, and write-reduced mode, describing what is causing the condition.

Syntax

```
fio-status [<device>] [<options>]
```

where `<device>` is the name of the device node (`/dev/fctx`), where `x` indicates the card number: `0`, `1`, `2`, and so on. For example, `/dev/fct0` indicates the first IO Accelerator device installed on the system.

If `<device>` is not specified, `fio-status` displays information for all cards in the system. If the IO Accelerator VSL driver is not loaded, this parameter is ignored.

Options	Description
`-a`	Report all available information for each device.
`-e`	Show all errors and warnings for each device. This option is for diagnosing issues, and it hides other information such as format sizes.
`-c`	Count: reports only the number of IO Accelerator devices installed.
`-d`	Show basic information set plus the total amount of data read and written (lifetime data volumes). This option is not necessary when the \\`-a` option is used.
`-fj`	Format JSON: creates the output in JSON format.
`-fx`	Format XML: creates the output in XML format.
`-u`	Show unavailable fields. Only valid with `-fj` or `-fx`.
`-U`	Show unavailable fields and details why. Only valid with `-fj` or `-fx`. Some `fio-status` fields are unavailable depending on the operating system or device. For example, some legacy fields are unavailable on newer IO Accelerator devices.
`-F<field>`	Print the value for a single field (for field names, see the next option). Requires that a device be specified. Multiple `-F` options may be specified.
`-l`	List the fields that can be individually accessed with `-F`.
`-L`	List all available IO Accelerator devices on the system.

Basic information: If no options are used, `fio-status` reports the following basic information:

- Number and type of devices installed in the system
- IO Accelerator software version

Adapter information:

- Adapter type
- Product number
- Product UUID
- PCIe power limit threshold (if available)
- Connected IO Accelerator devices

Block device information:

- Attach status
- Product name
- Product number
- Serial number

- PCIe address and slot
- Firmware version
- Size of the device, out of total capacity
- Internal temperature (average and maximum, since IO Accelerator VSL software load) in degrees Centigrade
- Health status: healthy, nearing wearout, write-reduced, or read-only
- Reserve capacity (percentage)
- Warning capacity threshold (percentage)

Data volume information: If the `-d` option is used, the following data volume information is reported in addition to the basic information:

- Physical bytes written
- Physical bytes read

All information: If the `-a` option is used, all information is printed, which includes the following information in addition to basic and data volume information.

Adapter information:

- Manufacturer number
- Part number
- Date of manufacture
- Power loss protection status
- PCIe bus voltage (average, minimum, maximum)
- PCIe bus current (average, maximum)
- PCIe bus power (average, maximum)
- PCIe power limit threshold (watts)
- PCIe slot available power (watts)
- PCIe negotiated link information (lanes and throughput)
- Product UUID

Block device information:

- Manufacturer's code
- Manufacturing date
- Vendor and sub-vendor information
- Format status and sector information (if device is attached)
- Controller ID and Low-level format GUID
- PCIe slot available power
- PCIe negotiated link information
- Card temperature, in degrees Centigrade
- Internal voltage (average, maximum)

- Auxiliary voltage (average, maximum)
- Percentage of good blocks, data, and metadata
- Lifetime data volume statistics
- RAM usage

Error mode information: If the IO Accelerator VSL software is in minimal mode, read-only mode, or write-reduced mode when `fio-status` is run, the following differences occur in the output:

- Attach status is "Status unknown: Driver is in MINIMAL MODE:"
- The reason for the minimal mode state appears (for example, "Firmware is out of date. Update firmware.")
- "Geometry and capacity information not available." appears.
- No media health appears.
- No media health information appears.

fio-sure-erase

CAUTION: Before using this utility, back up any important data.

IMPORTANT: As a best practice, do not use this utility if you have any IO Accelerator devices installed in the system that you do not want to clear or purge. First remove any devices that you do not want to accidentally erase. After the data is removed with this utility it is not recoverable.

IMPORTANT: If the device is in Read-only mode, perform a format using `fio-format` before running `fio-sure-erase`. If the device is in Minimal mode, then `fio-sure-erase` cannot erase the device. Updating the firmware may take the device out of Minimal Mode. If the device remains in Minimal mode, contact HP Customer Support for further assistance.

IMPORTANT: After using `fio-sure-erase`, format the device using `fio-format` before using the device again.

To run the `fio-sure-erase` utility, the block device must be detached. For more information, see "`fio-detach` (on page 66)."

Description

In Linux and Windows, the `fio-sure-erase` is a command-line utility that securely removes data from IO Accelerator devices. It complies with the Clear and Purge level of destruction from the following standards:

- DOD 5220.22-M–Comply with instructions for Flash EPROM
- NIST SP800-88–Comply with instructions for Flash EPROM

For information regarding certifications, see the Fusion-io website (http://www.fusionio.com/overviews/fusionsureerase/).

Registry requirement for Windows

To create a registry key in Windows to configure the driver for ECC-bypass mode:

1. Locate the following key:

```
HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\fiodrive\Parameters
```

2. Create a DWORD key underneath it named `BypassECC` and set the value to `1`.
3. Restart the computer before running the utility.

Syntax

```
fio-sure-erase [options] <device>
```

where `<device>` is the name of the device node (`/dev/fctx`), where `x` indicates the card number: 0, 1, 2, and so on. For example, `/dev/fct0` indicates the first IO Accelerator device detected on the system. To view this device node, use `fio-status` (on page 69).

Options

If `fio-sure-erase` is run with no options, a Clear operation is performed.

Option	Description
`-p`	Purge instead of Clear: Performs a write followed by an erase. Purging the device might take several hours to complete, depending on the size of the device.
`-y`	No confirmation: Does not require a yes/no response to execute the utility.
`-t`	Does not preserve current format parameters, including device and sector size (reset to default).
`-q`	Quiet: Does not display the status bar.

Clear support

A Clear is the default state of running `fio-sure-erase` (with no options), and is the act of performing a full low-level erase (every cell pushed to 1) of the entire NAND media, including retired erase blocks.

Metadata that is required for operation is not destroyed (media event log, erase counts, physical bytes read/written, performance and thermal history), but any user-specific metadata is destroyed.

A Clear operation performs the following:

1. Creates a unity map of every addressable block (this allows `fio-sure-erase` to address every block, including previously unmapped bad blocks).
2. Performs an erase cycle for each block (every cell is pushed to 1).
3. Restores the bad block map.
4. Formats the device to make the device usable again. The utility erases all of the headers during the clear.

Purge support

A Purge operation is implemented by using the `-p` option with `fio-sure-erase`. Purge is the act of first overwriting the entire NAND media (including retired erase blocks) with a single character (every cell written to logical 0), and then performing a full chip erase (every cell pushed to 1) across all media (including retired erase blocks).

Metadata that is required for operation is not destroyed (media event log, erase counts, physical bytes read/written, performance and thermal history), but any user-specific metadata is destroyed.

A Purge operation performs the following:

1. Creates a unity map of every addressable block (this allows `fio-sure-erase` to address every block, including previously unmapped bad blocks).
2. Performs a write cycle for each block (every cell written to 0).
3. Performs an erase cycle for each block (every cell pushed to 1).

4. Restores the bad block map.
5. Formats the drive to make the drive usable again. The utility erases all of the headers during the clear.

fio-update-iodrive

> **CAUTION:** HP strongly recommends that data is backed up on any IO Accelerator device before performing a firmware upgrade.

Description

Updates the IO Accelerator device firmware. This utility scans the PCIe bus for all IO Accelerator devices and updates them.

A progress bar and percentage are shown for each device as the update completes.

> **CAUTION:**
>
> - Do not turn off the power during a firmware upgrade, as this might cause device failure. Consider adding a UPS to the system prior to performing a firmware upgrade if one is not already in place.
> - It is critical to load the IO Accelerator VSL driver after each firmware upgrade step when performing multiple firmware upgrades in sequence (for example: 1.2.7 to 2.1.0 to 2.3.1). Otherwise the on-drive format is not changed and there is data loss. For more information on staged upgrades, see the *HP IO Accelerator Driver and Management Software Release Notes*.
> - Do not use this utility to downgrade the IO Accelerator device to an earlier version of the firmware. Do so might result in data loss and void the warranty. If you have issues with your upgrade, contact HP Support (http://www.hp.com/support).
> - The default utility action (without using the `-d` or `-s` option) is to upgrade all IO Accelerator devices with the firmware in the `fio-firmware-fusion_<version>-<date>.fff` firmware archive file. Confirm that all devices need the firmware upgrade prior to running the update. Use the `-p` (Pretend) option to view the possible results of the update.
> - You must follow a specific upgrade path when upgrading IO Accelerator devices. Before upgrading any IO Accelerator devices, see the *HP IO Accelerator Driver and Management Software Release Notes* for this IO Accelerator VSL software release.

> **IMPORTANT:** If you receive an error message when updating the firmware that instructs you to update the midprom information, contact HP Customer Support (http://www.hp.com/support).

To update one or more specific devices:

If the IO Accelerator is loaded, use the `-d` option with the device number.

In ESXi, when using VMDirectPathIO, when you upgrade the firmware on an IO Accelerator device, cycle the server power to apply the change. Restarting the virtual machine does not apply the change.

Online firmware updates

The utility upgrades compatible IO Accelerator devices while attached. The utility checks to make sure the devices support live firmware upgrades while attached.

The firmware update does not take effect until you restart the system. If you have updated the firmware but not restarted, the fio-status utility reports the following:

```
...
Firmware vX.X.X, rev 115781 Public
    Unactivated Firmware vX.X.Y, rev XXXXXX -- Reboot required to
activate  <--------
```

If a device is not compatible, the fio-update-iodrive utility returns an error that the device is not supported. In this case, first detach the device, and then run the utility again. In either case, you must restart the system before the updated firmware takes effect.

Syntax

```
fio-update-iodrive [options] <firmware-path>
```

where <firmware-path> is the full path to the firmware archive file fio-firmware-fusion_<version>-<date>.fff provided by HP.

In Linux and Windows, if you downloaded the .fff firmware archive file, then the firmware is most likely with the other downloaded packages.

In Linux, if you installed the firmware from the firmware package, the default path is /usr/share/fio/firmware/. This parameter is required.

In ESXi, the firmware archive path depends on where it is located on the ESXi host. For example, you can transfer the archive to a folder on a datastore, and then use the path to the file in that folder.

Options	Description
-d	Updates the specified devices (by fctx, where x is the number of the device shown in fio-status). If this option is not specified, all devices are updated.[1]
-f	Forces upgrade. Use only when directed by HP Support (http://www.hp.com/support). In Linux, if the IO Accelerator VSL driver is not loaded, this option also requires the -s option.[2]
-l	Lists the firmware available in the archive.
-p	Pretend: Shows what updates would be done. However, the actual firmware is not modified.
-c	Clears locks placed on a device.
-q	Quiet: Disables the display of the progress bar or percentage.
-Q	Quiet: Disables the display of the progress bar only.
-y	Confirms all warning messages.
-s	In Linux, updates the devices in the specified slots using * as a wildcard for devices. The slots are identified in the following PCIe format (as shown in lspci): [[[[<domain>]:]<bus>]:][<slot>][.[<func>]]

[1]Use the -d option carefully, as updating the incorrect IO Accelerator device could damage your device.
[2]Use the -f option carefully, as it could damage your card.

If you arrived at this section from the firmware upgrade section, return to "Upgrading the firmware (on page 23)."

Module parameters

Using module parameters

The following table describes the module parameters you can set in Linux, ESXi, and Solaris.

Linux

In Linux, edit the module parameter values in the `/etc/modprobe.d/iomemory-vsl4.conf` file.

Each module parameter in the configuration file must be preceded by `options iomemory-vsl4`. The `/etc/modprobe.d/iomemory-vsl4.conf` file has some example parameters that are commented out. Use the examples as templates or uncomment them to use them.

In order for the changes to take effect, complete the changes before the IO Accelerator VSL software loads.

ESXi

In ESXi, edit the module parameter values using the `esxcfg-module` command.

The remote option (`--server`) is only required for the vCLI.

The following is an ESXi sample command:

```
esxcfg-module --server <server-name> iomemory-vsl4 -s '<parameter>=<value>'
```

To query the current module parameters, use the following command:

```
esxcfg-module --server <server-name> iomemory-vsl -g
```

You must reboot the ESXi system to enforce any parameter changes.

Solaris

In Solaris, edit the module parameter values in the `/usr/kernel/drv/iomemory-vsl4.conf` file.

In order for the changes to take effect, complete the changes before the IO Accelerator VSL software loads.

Module parameter	Default (min/max)	Description
`auto_attach`	1 (0, 1)	1 = Always attach the device on driver load. 0 = Do not attach the device on driver load.
`background_attach` (ESXi only)	1 (0, 1)	1 = The host scans and attaches IO Accelerator devices in the background while the system continues to boot. 0 = Forces the host server to wait until all IO Accelerator devices are ready and attached before continuing to fully boot and become operational. The default setting saves time on startup. However, in some instances the IO Accelerator devices might not be ready when the host system scans for datastores. If this occurs, set the parameter to `0` and reboot.

Module parameter	**Default (min/max)**	**Description**
`disable_msi`	0 (0, 1)	1 = Disables message signal interrupts as well as MSI-X. 0 = Enables message signal interrupts as well as MSI-X.
`disable_msix`	0 (0, 1)	1 = Disables only MSI-X interrupts. 0 = Enables MSI-X interrupts.
`external_power_override*`	No devices selected	Allows selected devices to draw full power from the PCIe slot.
`fio_dev_wait_timeout_secs` (Linux only)	30	Number of seconds to wait for `/dev/fio*` files to appear during driver load. For systems not using `udev`, set the parameter to 0 to disable the timeout and avoid an unnecessary pause during driver load.
`force_minimal_mode`	0 (0, 1)	1 = Force minimal mode on the device. 0 = Do not force minimal mode on the device.
`numa_node_override` (Linux only)	Nothing selected	A list of `<affinity specification>` couplets that specify the affinity settings of all devices in the system. each item in the couplet is separated by a colon, and each couplet set is separated by a comma. Each `<affinity specification>` couplet has the following syntax: `<device-id>=<node-number>` For more information, see "NUMA configuration (on page 82)."
`preallocate_memory` (Linux only)	No devices selected	For the selected devices, preallocate all memory necessary to have the drive usable as swap space. The `<value>` for the parameter is a comma-separated list of device serial numbers.
`preallocate_mb` (Linux only)	No value selected	Where `<value>` is the amount of system memory in MB that the IO Accelerator VSL software should preallocate for every IO Accelerator device enabled for preallocation. For example, if you entered a value of `3500`, then the IO Accelerator VSL software preallocates about 3.5GB of RAM for every IO Accelerator device enabled for preallocation in that system. For more information, see "Using the device as swap in Linux (on page 28)."
`rmap_memory_limit_MiB`	3100 (0, 100000)	Amount of RAM in MiB allocated for improving rescan times after an unclean shutdown. For more information, see "Improving rescan times (on page 53)."
`rsort_memory_limit_MiB`	0 (0, 100000)	Amount of RAM in MiB allocated for improving rescan times after an unclean shutdown. For more information, see "Improving rescan times (on page 53)."

Module parameter	Default (min/max)	Description
use_workqueue (Linux only)	0 (0 or 3)	3 = Use stardard operating system I/O elevators. 0 = Bypass.

*Use this parameter carefully. For more information, see "Enabling PCIe power override."

Except for the `preallocate_memory` parameter (Linux) and the `external_power_override` parameter, module parameters apply to all IO Accelerator devices in the system.

Monitoring IO Accelerator health

Health metrics

The IO Accelerator VSL software manages block retirement using pre-determined retirement thresholds. The HP IO Accelerator Management Tool and the `fio-status` utilities show a health indicator that starts at 100 and counts down to 0. As certain thresholds are crossed, various actions are taken.

At the 10% healthy threshold, a one-time warning is issued. For more information, see "Health monitoring techniques."

At 0%, the device is considered unhealthy. At 3% it enters write-reduced mode, which somewhat prolongs its lifespan so data can be safely migrated off. In this state the IO Accelerator device behaves normally, except for the reduced write performance.

After the 1% threshold, the device will soon enter read-only mode, and any attempt to write to the IO Accelerator device causes an error. Some filesystems might require special mount options to mount a read-only block device in addition to specifying that the mount must be read-only.

For example, under Linux, `ext3` requires that `-o ro, noload` is used. The `noload` option tells the filesystem to not try and replay the journal.

Consider the read-only mode as a final opportunity to migrate data off the device, as device failure is more likely with continued use.

The IO Accelerator device might enter failure mode. In this case, the device is offline and inaccessible. This can be caused by an internal catastrophic failure, improper firmware upgrade procedures, or device wearout.

IMPORTANT:

- For service or warranty-related questions, contact the company you purchased the device from.
- For products with multiple IO Accelerator devices, these modes are maintained independently for each device.

Health monitoring techniques

fio-status -a

Output from the `fio-status` utility (using the `-a` option) shows the health percentage and device state. These items are referenced as `Reserve space status` in the following sample output.

```
Found 1 ioMemory device in this system

Driver version: 4.x.x build xxxx

...

Reserve space status: Healthy; Reserves: 100.00%, warn at 10.00%

Lifetime data volumes:
```

```
Physical bytes written: 6,423,563,326,064
Physical bytes read   : 5,509,006,756,312
```

The following Health Status messages are produced by the `fio-status` utility:

- Healthy
- Read-only
- Reduced-write
- Unknown

HP IO Accelerator Management Tool: Find information about device health within the IO Accelerator Management Tool. For more information, see the IO Accelerator Manager Management Tool software documentation.

Software RAID and health monitoring

Software RAID stacks are typically designed to detect and mitigate the failure modes of traditional storage media. The IO Accelerator attempts to fail as gracefully as possible, and its new failure mechanisms are compatible with existing software RAID stacks. A drive in write-reduced mode participating in a write-heavy workload is evicted from a RAID group for failure to receive data at a sufficient rate. A drive in read-only mode is evicted when write I/Os are returned from the device as failed. Catastrophic failures are detected and handled just as though they were on traditional storage devices.

VMDirectPathIO

Working with IO Accelerator devices and VMDirectPathIO

In ESXi, each IO Accelerator device can either be used directly by the host operating system as a VMFS datastore or as a caching device, or it can be passed through directly to a VM. In VMware documentation, passing through to a VM is referred to as VMDirectPathIO.

If you are passing the devices through, you do not have to install the VSL on the ESXi system. Instead, install the software on the guest system. Install the driver on the host only if you plan to create a VMFS on the devices.

When using VMDirectPathIO, if you upgrade the firmware on an IO Accelerator device, you must cycle the power on the ESXi host server to have the change take place. Restarting the VM does not apply the change.

NUMA configuration

Introduction to NUMA architecture

IMPORTANT: The NUMA node parameters are deprecated. The NUMA node parameters enable you to manually set the NUMA node affinity, but they are not needed in many are cases. The IO Accelerator VSL software now automatically creates interrupt affinity between IO Accelerator devices and NUMA nodes. However, you might still use the following information and parameters to manually configure NUMA node affinity.

Servers with NUMA architecture might require special installation instructions in order to maximize IO Accelerator device performance. Most multi-socket servers have NUMA architecture.

On some servers with NUMA architecture, during system boot the BIOS does not associate PCIe slots with the correct NUMA node. Incorrect mappings result in inefficient I/O handling that can significantly degrade performance.

In Linux, use the `numa_node_override` parameter to map devices with specific NUMA nodes.

In Windows, use the `fio_affinity` parameter to map devices with specific NUMA nodes.

Using the numa_node_override parameter in Linux

Use the numa_node_override parameter to map devices with specific NUMA nodes.

The following example shows the final implementation of custom affinity settings. The implementation requires an analysis of the specific system; the analysis includes the system architecture, type and number of IO Accelerator devices installed, and the particular PCIe slots used. Your particular circumstances require a custom analysis of your setup. The analysis requires understanding of the NUMA architecture of your system compared to your particular installation.

Your actual settings might be different than the following example, depending on your server configuration. In order to create the correct settings for your specific system, use `fio-status` to list all of the devices and determine the `<device-id>`. To determine the `<device-id>`, see "Determining the device ID in Linux (on page 82)." Use the following example of setting the `numa_node_override` parameter as a template and modify it for your particular system.

Determining the device ID in Linux

Present each `<device-id>` in the following format:

```
<domain>:<bus>:<device>.<function>
```

Typically the domain is `0000`; consult the server documentation to determine the device ID. The remainder of the device ID string is visible in the `fio-status` output.

Example

```
# fio-status
```

```
Found 2 ioMemory devices in this system
...
    PCI:04:00.0
...
    PCI:0F:00.0
```

In the example, the device IDs are 0000:04:00.0 and 0000:15:00.0 on a system that had a domain of 0000.

IMPORTANT: The PCI device ID, including the bus number, might change if you change any of the PCI devices in the system. For example, the PCI device ID might change if you add a network card or another IO Accelerator device. If the device ID changes, you must update the configuration.

numa_node_override parameter in Linux

Configuring the IO Accelerator devices for servers with NUMA architecture requires using the numa_node_override parameter by modifying the iomemory-vsl4.conf file.

The numa_node_override parameter is a list of <affinity specification> couplets that specify the affinity settings of all devices in the system. Each item in the couplet is separated by an equal sign (=), and each couplet set is separated by a comma.

Syntax

```
numa_node_override=<affinity specification>[,<affinity specification>...]
```

Each <affinity specification> has the following syntax:

```
<device-id>=<node-number>
```

Example

```
numa_node_override=0000:04:00:0=1,0000:1d:00.0=0,0000:05:00.0=2,0000:1e:00.0
=3
```

The example creates the following:

<device-id>	**Node/group**	**Processor affinity**
0000:04:00.0	node 1	all processors in node 1
0000:1d:00.0	node 0	all processors in node 0
0000:05:00.0	node 2	all processors in node 2
0000:1e:00.0	node 3	all processors in node 3

Advanced configuration in Linux

If the server has multiple NUMA nodes and multiple IO Accelerator devices installed, make sure that the IO Accelerator devices are spread out among the various nodes.

While it might be optimal to pair devices to nodes that are electronically closer to the PCIe slot of each device, which requires an advanced understanding of the NUMA architechure of the server and an analysis of the device installation, simply spreading out all of the node affinities of the devices among the available nodes typically results in improved performance.

In the example, the device IDs are 0000:04:00.0 and 0000:15:00.0 on a system with a domain of 0000.

IMPORTANT: The PCI device ID, including the bus number, might change if you change any of the PCI devices in the system. For example, the PCI device ID might change if you add a network card or another IO Accelerator device. If the device ID changes, you must update the configuration.

Using the FIO_AFFINITY parameter in Windows

Use the FIO_AFFINITY parameter to map devices with specific NUMA nodes.

The following example shows the final implementation of custom affinity settings. The implementation requires an analysis of the specific system; the analysis includes the system architecture, type and number of IO Accelerator devices installed, and the particular PCIe slots used. Your particular circumstances require a custom analysis of your setup. The analysis requires understanding of the NUMA architecture of your system compared to your particular installation.

Your actual settings might be different than the following example, depending on your server configuration. In order to create the correct settings for your specific system, use fio-status to list all of the devices and determine the <device-id>. To determine the <device-id>, see "Determining the bus number in Windows (on page 84)." Use the following example of setting the FIO_AFFINITY parameter as a template and modify it for your particular system.

Determining the bus number in Windows

In order to create the correct settings for your specific system, use fio-status to list all bus numbers for the devices.

```
fio-status

Found 2 ioMemory devices in this system

...

    PCI:04:00.0

...

    PCI:0F:00.0
```

IMPORTANT: In Windows, the bus number appears in hex, but you must enter the number in decimal. For example, 04 is 4 and 0F is 15.

IMPORTANT: The PCI device ID, including the bus number, might change if you change any of the PCI devices in the system. For example, the PCI device ID might change if you add a network card or another IO Accelerator device. If the device ID changes, you must update the configuration.

FIO_AFFINITY parameter in Windows

Configuring the IO Accelerator devices for servers with NUMA architecture requires the use of the FIO_AFFINITY parameter with the fio-config utility.

The `FIO_AFFINITY` parameter is a list of `<affinity specification>` triplets that specify the affinity settings of all devices in the system. Each item in the triplet is separated by a comma, and each triplet set is separated by a semicolon.

Syntax

```
fio-config -p FIO_AFFINITY <affinity specification>[;<affinity
specification>...]
```

Each `<affinity specification>` has the following syntax:

```
<[domain number:]bus number>,[g|n]<group or node number>[,<hex mask>]
```

If domain number is not specified, it is set to `0` (most common).

If there is no `g` or `n` character before the group/node number, then the number is assumed to be a group number.

The hex mask is optional. If it is not present, the mask is assumed to be `0xffffffffffffffff`. Also, the `0x` prefix is optional.

If the hex mask is a node mask, then the mask is relative to the node, not the group to which the node belongs.

Example

```
fio-config -p FIO_AFFINITY 1:7,n0,0xf;20,n1;80,7;2:132,4,0xff0
```

The example creates the following:

PCI address (domain:bus)	Node/group	Processor affinity
1:7	node 0	processors 0 to 3 in the node (mask 0xf)
0:20	node 1	all processors in the node (no hex mask)
0:80	group 7	all processors in the group (no hex mask)
2:132	group 4	processors 4 to 11 in the group (mask 0xff0)

Advanced configuration in Windows

If the server has multiple NUMA nodes and multiple IO Accelerator devices installed, make sure that the IO Accelerator devices are spread out among the various nodes.

While it might be optimal to pair devices to nodes that are electronically closer to the PCIe slot of each device, which requires an advanced understanding of the NUMA architechure of the server and an analysis of the device installation, simply spreading out all of the node affinities of the devices among the available nodes typically results in improved performance.

Checking the log for errors in Windows

If you enter a configuration that is not valid, then the settings are disabled and an error is available in the system logs.

Example

```
fio-config -p FIO_AFFINITY 5,g0,0xf;6,0xf
```

In the example, the affinity for device `fct6` is set incorrectly, because there is no group/node number before the mask. The following errors appear in the system log:

```
2011-09-09T12:22:15.176086800Z - ERROR - FusionEventDriver - FIO_AFFINITY:
Invalid group or node number
```

```
2011-09-09T12:22:15.176086800Z - ERROR - FusionEventDriver - Invalid
FIO_AFFINITY parameter syntax at character 13: "5,g0,0xf;6,0x". Manual affinity
settings are disabled!
```

When using the `FIO_AFFINITY` parameter with couplets, note the following information.

```
# fio-status

Found 2 ioMemory devices in this system

...

    PCI:04:00.0

...

    PCI:0F:00.0
```

IMPORTANT: In Windows, the bus number appears in hex, but you must enter the number in decimal. For example, `04` is `4` and `0F` is `15`.

In the example, the device IDs are `0000:04:00.0` and `0000:15:00.0` on a system with a domain of `0000`.

IMPORTANT: The PCI device ID, including the bus number, might change if you change any of the PCI devices in the system. For example, the PCI device ID might change if you add a network card or another IO Accelerator device. If the device ID changes, you must update the configuration.

Using Windows page files with the IO Accelerator

Introduction to Windows page files

This section describes how to effectively use paging (swap) files on IO Accelerator devices with Windows® operating systems.

Using a page file with a traditional disk drive places practical limits on the usable size of the page file and virtual memory due to the poor performance of disk drives in relation to RAM. Placing the operating system paging file on one or more IO Accelerators enables much larger page files and usable virtual memory. This is due to the much faster response times and bandwidth on IO Accelerators versus hard disks. IO Accelerator software prior to version 2.0.2 does not support paging files.

Configuring IO Accelerator device paging support

The IO Accelerator VSL software can be configured to support paging files on one or more IO Accelerator devices. Configuring paging support requires that each IO Accelerator device used with a paging file preallocates the worst-case amount of memory it might need in any possible I/O scenario. Memory is pre-allocated on a per-device instance.

Because of the extra host RAM memory use, paging must be enabled only on IO Accelerator devices that actually hold a paging file. You can place a single paging file on more than one IO Accelerator device. In this case, Windows stripes paging I/O across all available paging files, possibly providing additional performance to the Virtual Memory subsystem.

RAM consumption

The amount of RAM preallocated per IO Accelerator depends on the total size of the device and the sector (block) size selected when formatting the drive (with `fio-format`).

For the RAM usage per GB of IO Accelerator device, see the release notes for this version of the software.

Host RAM Consumption in Mebibytes (MiB); 1024 x 1024 = 1 MiB

Using a larger sector size significantly reduces the amount of host memory consumption needed for paging support. HP recommends using a 4K sector size be used because:

- That is generally the natural size of a host memory page.
- It minimizes overall host memory consumption.

On Windows operating systems, NTFS generally uses a cluster size of 4 K, so formatting to 512 is not useful except for applications that are compatible only with 512 B sector sizes (such as Windows XP and Windows Server 2003).

The indicated amount is needed per IO Accelerator that supports paging. You must carefully plan which IO Accelerators are used to hold a paging file.

Non-paged memory pool

Pre-allocated memory for the IO Accelerator comes from the Windows kernel non-paged memory pool. This pool dynamically grows as system components consume additional kernel memory. The maximum size of this pool is restricted to 75% of RAM up to a maximum of 128 GB.

The amount of in-use non-paged pool memory should be noted when planning page file usage. This is because the IO Accelerator preallocates RAM and that reduces the available physical non-paged memory. The driver fails to load if the total preallocated memory plus the in-use non-paged memory exceeds the maximum non-paged memory pool.

To determine the total non-paged memory pool use for two IO Accelerators, use the following example:

- One IO Accelerator requires 850 MB or RAM, and the other requires 1700 MB of RAM

  For RAM requirements, see the release notes for this version of the IO Accelerator VSL software.
- Both are formatted with a 4 K sector size
- Both will support paging files

The current allocated non-paged pool is obtained from Task Manager and, in this example, has a value of 576 MiB. (Values shown in Task Manager are in MiB (1024x1024 = 1 MiB)). The total RAM on the system is 8000 MB and the operating is Windows Server 2008 R2.

First, convert the 576 MiB into MB: 576 MiB * (1 MB/1.048576 MiB) = ~549 MB

To calculate the total available non-paged pool, use the following formula:

```
(8000 MB x 0.75) - 549 - 850 - 1700
```

This still leaves 2901 MB available for the non-paged pool.

Enabling and disabling paging support

Memory preallocation occurs during IO Accelerator driver initialization. To enable paging support, you must enable the `FIO_PREALLOCATE_MEMORY` configuration item. This can be done using the `fio-config` command-line utility. This parameter is assigned a string with a list of decimal serial numbers of the IO Accelerators that will support a paging file. The driver performs memory preallocation for those instances.

The following example illustrates using the `fio-config` utility, which enables paging and preallocation on two IO Accelerators with serial numbers `1149D2717` and `1331G0009`. Serial number information can be obtained using the `fio-status` utility.

```
fio-config -p FIO_PREALLOCATE_MEMORY "1149D2717,1331G0009"
```

To disable paging support on all drives, use a value of `0` for `FIO_PREALLOCATE_MEMORY`:

```
fio-config -p FIO_PREALLOCATE_MEMORY "0"
```

To query the current value, run this command:

```
fio-config -g FIO_PREALLOCATE_MEMORY
```

An alternate method to manage (enable or disable) paging support is to use the IO Accelerator Management Tool.

NOTE: You must reload the IO Accelerator driver for the new preallocation setting to take effect. Typically this can be done by restarting the machine or using **disable/enable** within **Device Manager** for each IO Accelerator instance.
Using the **Windows System Properties** to change paging file configuration requires a system

restart before the properties are applied. Therefore, you can change both FIO_PREALLOCATE_MEMORY and the system page file configuration and then apply both with a single restart.

Windows page file management

By default, the IO Accelerator driver disables support for page files. The previous section described how to enable support for page files on one or more IO Accelerators. The following section describes how to work with the built-in Windows® control panels to configure and set up paging files on IO Accelerators.

Setting up page files

To set up page files in Windows® operating systems:

1. In the **Control Panel,** double-click **System.**
2. On the **Task** pane, click **Advanced system settings.**
3. On the **Advanced** tab, click **Settings.** The **Performance Options** dialog box opens.
4. On the **Advanced** tab, click **Change.** The **Virtual Memory** dialog box opens.
5. Using this dialog box, configure a page file for each available drive in the system. Selecting the **Automatically manage paging file size for all drives** checkbox causes Windows® operating systems to create a single page file on the system drive, which is the drive the operating system starts from. This checkbox should be cleared when using an IO Accelerator with a paging file.

Windows® operating systems support up to 16 distinct paging files. To enable a page file on an IO Accelerator:

1. Choose the IO Accelerator from the device list.
2. Select the **Custom size** button.
3. Provide values in the **Initial size** and **Maximum size** fields.
4. Click **Set** to save the setting. Do not omit this step, or your changes will be lost.
5. Click **OK.**
6. When prompted to restart, click **Yes.** This is necessary for the new page file settings to take effect.

To remove a paging file on the drive, follow the steps above but select **No paging file.** For performance reasons, typically you will remove all paging files on any system hard disk.

NOTE: The Virtual Memory dialog box enables pages files to be configured on available IO Accelerators, even if the IO Accelerator has not been configured to support a page file. Even though the dialog box enables the page file, following the required restart, no page file is created on the device. Follow the directions in "Enabling and disabling paging support (on page 88)" to properly enable page file support on one or more IO Accelerators.

System drive paging file configuration

By default, Windows® operating systems create and manage a page file on the system boot drive (typically a hard disk), which is where the operating system is installed. Maintaining a regular page file on the system hard disk is typically not optimal, because the hard disk I/O performance is many orders of magnitude slower than an IO Accelerator. To resolve this issue, you can eliminate or minimize the size of the system boot drive page file, as explained later. Enabling page files on IO Accelerators but not the system drive improves VM subsystem performance since the VM manager stripes I/O across all available page files. Additionally, the IO Accelerators act as a large memory store, which can greatly improve memory usage for large applications.

The Windows® kernel uses the system disk page file to store crash dumps. Crash dumps might be small (mini-dumps) or large (full-kernel memory dumps). Typically, running without dump file support or with a small dump file is adequate. There are several possible system drive page file configurations:

- Eliminate all page files on any hard disks, including the system boot drive. Although this maximizes paging I/O on IO Accelerators, no post-mortem crash dump file will be available if a system crash occurs. However, you might be able to re-enable a page file on the system drive and then reproduce the crash scenario.
- Create a minimal-size page file on the system boot drive. The recommended minimum size is 16MB, although the operating system might warn that a minimum 400MB page file is needed.
- Create a page file large enough for a full-kernel memory dump. This typically requires a page file at least the size of installed RAM, with some recommending the size equal to RAM x 1.5.

To view or change the crash dump configuration:

1. Go to the **System Properties** dialog box.
2. Click the **Advanced** tab.
3. In the **Startup and Recovery** section, click **Settings.** The **Startup and Recovery** dialog box opens.
4. In the **System Failure** section, you can change settings to handle the system log, restart, and debugging information.

Guaranteeing minimum committable memory

If you enable **System managed size** or set a **Custom size** in the **Virtual Memory** dialog box, do so with care. If the initial size is less than the desired amount of committable virtual memory, this size might cause an application to have memory allocation failures if the amount of committed memory exceeds the currently allocated page file size or the initial size value. When committed memory exceeds the current page file size, a request to allocate additional memory will fail. The Windows Virtual Memory manager will slowly increase the size of the paging file up to the available size of its drive or to the **Maximum size** custom setting, whichever is smaller.

If you want to use a large amount of committed virtual memory (more than 1.5 times the amount of RAM) and avoid application memory allocation errors, the initial and maximum committed memory must be explicitly set for the expected application committed memory usage. These values should generally be the same.

The following articles explain in detail how to size the page file appropriately.

- Main Article Link: Pushing the Limits of Windows (http://blogs.technet.com/b/markrussinovich/archive/2008/07/21/3092070.aspx)
- Specific section that documents virtual memory: Pushing the Limits of Windows: Virtual Memory (http://blogs.technet.com/b/markrussinovich/archive/2008/11/17/3155406.aspx)

Verifying page file operation

To verify that a page file is actively placed on an IO Accelerator, you can browse for hidden files at the drive root. For example, run the following command at a prompt:

```
dir c: /ah
```

In the output listing there should be a file called `pagefile.sys`. If no page file is present, recheck the page file configuration in the **Virtual Memory** dialog box to verify that page file support has been enabled on the queried IO Accelerator.

Virtual Memory performance

Using the IO Accelerator as the paging store can improve overall Virtual Memory system performance. Actual benefits will vary widely with the application virtual memory usage and with hardware platform and performance.

Troubleshooting event log messages

Troubleshooting event log messages in Windows

The Windows System Event Log displays informational, warning, and error messages concerning the IO Accelerator device.

Each IO Accelerator device is numbered from 0 upwards. Use the `fio-status` utility to view the number for your device.

While the Windows System Event Log messages are useful for troubleshooting, the IO Accelerator VSL log messages are not designed for continual monitoring purposes, as each message is based on a variety of factors that produce different log messages depending on environment and use case. For best results to regularly monitor the devices, use the tools described in "Monitoring and managing devices (on page 42)."

Verbose event log parameter

The `WIN_LOG_VERBOSE` IO Accelerator VSL parameter (enabled by default) expands the extent of the IO Accelerator VSL error log messages in the event log and provides additional information for troubleshooting issues. You can disable this parameter.

For more information on enabling parameters, see "`fio-config` (on page 64)."

Viewing logs

To open the Windows Event Viewer:

1. Click **Start.**
2. Right-click **Computer,** and select **Manage** from the list.
3. Expand **System Tools.**
4. Expand **Event Viewer.**
5. Expand **Windows Logs.**
6. Select **System.**

Windows event log error messages

The following table lists common Event Log error messages and suggested solutions.

Message	Suggested solution
`Error: ioDrive(x) firmware is too old. The firmware must be updated.`	To update the firmware, see "Upgrading the firmware (on page 23)."
`Error: ioDrive initialization`	1 Reinstall the Windows IO Accelerator VSL. 2 Remove and reseat the IO Accelerator device.

Message	Suggested solution
failed with error code 0xerrorcode	3 Remove the IO Accelerator device and insert it in a different slot. For the 0xerrorcode values, see the following table.
Error: ioDrive was not attached. Use the fio-attach utility to rebuild the drive.	This error might appear after an unmanaged shutdown. You can use either the fio-attach command line utility or the IO Accelerator Management Tool software to re-attach the IO Accelerator device. The attach process might take up to ten minutes as the utility performs a consistency check on the devices.
Warning: ioDrive was not loaded because auto-attach is disabled.	The IO Accelerator device must attach to the Windows operating system to be available to users and applications. The attach normally occurs at boot time. As part of the attach process, the IO Accelerator VSL software verifies the presence of an AutoAttach parameter in the Windows registry. If you create the registry parameter to disable auto-attach, the attach operation does not complete. To attach an unattached IO Accelerator device: 1 Run the IO Accelerator Management Tool software. 2 Select the unattached IO Accelerator device from the Device Tree. 3 Click **Attach**. 4 Confirm the Attach operation. Your IO Accelerator device now attaches to the Windows operating system. To re-enable auto attach at boot time, see "Enabling auto attach."

0xerrorcode is one of the following.

0x **error code**	**Description**
0xFFFFFC00	Uncorrectable ECC error
0xFFFFFBFF	Uncorrectable ECC error
0xFFFFFBFE	Invalid media format
0xFFFFFBFD	Unknown error
Other code	Standard Windows errno definition (https://msdn.microsoft.com/en-us/library/t3ayayh1(v=vs.110).aspx)

The error code is converted to a negative number, and then reported in hexadecimal format. For example, errno=1 is converted to -1, and then is represented as 0xFFFFFFFF; and errno=1024 is converted to -1024, and then is represented as 0xFFFFFC00.

Windows event log informational messages

The following example is a common event long informational message.

Message	Additional information
Affinity not set for ioMemory VSL device fct119 because either WIN_DISABLE_ALL_AFFINITY is set to true or "SetWorkerAffinity119" does not exist in the registry and WIN_DISABLE_DEFAULT_NUMA_AFFINITY is set to true.	When WIN_DISABLE_ALL_AFFINITY is set to 0, the driver enables the interrupt and worker thread affinity in the driver. When WIN_DISABLE_ALL_AFFINITY is set to 1, the driver disables all affinity settings. This is an override of any other affinity settings. For more information about affinity settings, see "fio-config (on page 64)."

Manual installation

Overview

The Windows Setup program installs IO Accelerator VSL software on the Windows operating system. However, in some instances you might need to manually install the software for a particular IO Accelerator device, including:

- IO Accelerator devices do not appear in `fio-status` after a software installation (including an upgrade).
- Installing new IO Accelerator devices on a system with previously installed IO Accelerator devices and IO Accelerator VSL software.

Manual installation of the IO Accelerator VSL software ensures the software is installed for a particular device. If needed, repeat the manual installation for each device.

Manual installation in Windows Server 2008 R2 and Windows Server 2012

Before manually installing the IO Accelerator VSL software, download and run the IO Accelerator VSL Windows Setup program. The program installs the IO Accelerator VSL software on the system, enabling the installation of the IO Accelerator VSL software for each IO Accelerator device.

The Windows Driver Wizard might automatically detect the new IO Accelerator device and start to locate its IO Accelerator VSL software after you restart the system. If this happens, skip to the Installation Wizard procedure.

1. Launch the Device Manager.
   - In Windows Server 2008, click **Start** > **Control Panel** > **Device Manager**.
   - In Windows Server 2012, from the **Server Manage**r select **Tools** (in the upper right) > **Computer Management** > **Device Manager**.
2. Select **Storage Controllers**.
3. Click the IO Accelerator device in the list. The Properties dialog box appears.

   The device might be titled `Mass Storage Controller`.
   - If the Device Status has `This device is working properly`, then the IO Accelerator VSL software is installed.
   - If the device is not working correctly, manually install the software for the device. Continue with the manual installation.
4. Close the Properties dialog box.
5. Right-click the device and select **Update Driver**.
6. Follow the instructions in "Installation wizard (on page 95)."

Installation wizard

1. Click **Browse** next to the path field to locate the software driver.
2. Select the folder with the IO Accelerator VSL software. The default location is `C:\Program Files\Fusion-io\ioMemory VSL4\<VSL-Version>\Driver`.
3. Click **OK**.
4. Click **Next**. Windows locates the correct software and installs the device software.
5. When the driver installation completes, restart the system.
6. Proceed to "Upgrading the firmware (on page 23)."

Regulatory information

Safety and regulatory compliance

For safety, environmental, and regulatory information, see *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at the HP website (http://www.hp.com/support/Safety-Compliance-EnterpriseProducts).

Turkey RoHS material content declaration

Türkiye Cumhuriyeti: EEE Yönetmeliğine Uygundur

Ukraine RoHS material content declaration

Обладнання відповідає вимогам Технічного регламенту щодо обмеження використання деяких небезпечних речовин в електричному та електронному обладнанні, затвердженого постановою Кабінету Міністрів України від 3 грудня 2008 № 1057

Warranty information

HP ProLiant and X86 Servers and Options (http://www.hp.com/support/ProLiantServers-Warranties)

HP Enterprise Servers (http://www.hp.com/support/EnterpriseServers-Warranties)

HP Storage Products (http://www.hp.com/support/Storage-Warranties)

HP Networking Products (http://www.hp.com/support/Networking-Warranties)

Support and other resources

Before you contact HP

Be sure to have the following information available before you call HP:

- Active Health System log (HP ProLiant Gen8 or later products)

  Download and have available an Active Health System log for 7 days before the failure was detected. For more information, see the *HP iLO 4 User Guide* or *HP Intelligent Provisioning User Guide* on the HP website (http://www.hp.com/go/ilo/docs).
- Onboard Administrator SHOW ALL report (for HP BladeSystem products only)

  For more information on obtaining the Onboard Administrator SHOW ALL report, see the HP website (http://www.hp.com/go/OAlog).
- Technical support registration number (if applicable)
- Product serial number
- Product model name and number
- Product identification number
- Applicable error messages
- Add-on boards or hardware
- Third-party hardware or software
- Operating system type and revision level

HP contact information

For United States and worldwide contact information, see the Contact HP website (http://www.hp.com/go/assistance).

In the United States:

- To contact HP by phone, call 1-800-334-5144. For continuous quality improvement, calls may be recorded or monitored.
- If you have purchased a Care Pack (service upgrade), see the Support & Drivers website (http://www8.hp.com/us/en/support-drivers.html). If the problem cannot be resolved at the website, call 1-800-633-3600. For more information about Care Packs, see the HP website (http://pro-aq-sama.houston.hp.com/services/cache/10950-0-0-225-121.html).

Customer Self Repair

HP products are designed with many Customer Self Repair (CSR) parts to minimize repair time and allow for greater flexibility in performing defective parts replacement. If during the diagnosis period HP (or HP service providers or service partners) identifies that the repair can be accomplished by the use of a CSR part, HP will ship that part directly to you for replacement. There are two categories of CSR parts:

- **Mandatory**—Parts for which customer self repair is mandatory. If you request HP to replace these parts, you will be charged for the travel and labor costs of this service.
- **Optional**—Parts for which customer self repair is optional. These parts are also designed for customer self repair. If, however, you require that HP replace them for you, there may or may not be additional charges, depending on the type of warranty service designated for your product.

NOTE: Some HP parts are not designed for customer self repair. In order to satisfy the customer warranty, HP requires that an authorized service provider replace the part. These parts are identified as "No" in the Illustrated Parts Catalog.

Based on availability and where geography permits, CSR parts will be shipped for next business day delivery. Same day or four-hour delivery may be offered at an additional charge where geography permits. If assistance is required, you can call the HP Technical Support Center and a technician will help you over the telephone. HP specifies in the materials shipped with a replacement CSR part whether a defective part must be returned to HP. In cases where it is required to return the defective part to HP, you must ship the defective part back to HP within a defined period of time, normally five (5) business days. The defective part must be returned with the associated documentation in the provided shipping material. Failure to return the defective part may result in HP billing you for the replacement. With a customer self repair, HP will pay all shipping and part return costs and determine the courier/carrier to be used.

For more information about HP's Customer Self Repair program, contact your local service provider. For the North American program, refer to the HP website (http://www.hp.com/go/selfrepair).

Réparation par le client (CSR)

Les produits HP comportent de nombreuses pièces CSR (Customer Self Repair = réparation par le client) afin de minimiser les délais de réparation et faciliter le remplacement des pièces défectueuses. Si pendant la période de diagnostic, HP (ou ses partenaires ou mainteneurs agréés) détermine que la réparation peut être effectuée à l'aide d'une pièce CSR, HP vous l'envoie directement. Il existe deux catégories de pièces CSR:

Obligatoire - Pièces pour lesquelles la réparation par le client est obligatoire. Si vous demandez à HP de remplacer ces pièces, les coûts de déplacement et main d'œuvre du service vous seront facturés.

Facultatif - Pièces pour lesquelles la réparation par le client est facultative. Ces pièces sont également conçues pour permettre au client d'effectuer lui-même la réparation. Toutefois, si vous demandez à HP de remplacer ces pièces, l'intervention peut ou non vous être facturée, selon le type de garantie applicable à votre produit.

REMARQUE: Certaines pièces HP ne sont pas conçues pour permettre au client d'effectuer lui-même la réparation. Pour que la garantie puisse s'appliquer, HP exige que le remplacement de la pièce soit effectué par un Mainteneur Agréé. Ces pièces sont identifiées par la mention "Non" dans le Catalogue illustré.

Les pièces CSR sont livrées le jour ouvré suivant, dans la limite des stocks disponibles et selon votre situation géographique. Si votre situation géographique le permet et que vous demandez une livraison le jour même ou dans les 4 heures, celle-ci vous sera facturée. Pour bénéficier d'une assistance téléphonique, appelez le Centre d'assistance technique HP. Dans les documents envoyés avec la pièce de rechange CSR, HP précise s'il est nécessaire de lui retourner la pièce défectueuse. Si c'est le cas, vous devez le faire dans le délai indiqué, généralement cinq (5) jours ouvrés. La pièce et sa documentation doivent être retournées dans l'emballage fourni. Si vous ne retournez pas la pièce défectueuse, HP se réserve le droit de vous facturer les coûts de remplacement. Dans le cas d'une pièce CSR, HP supporte l'ensemble des frais d'expédition et de retour, et détermine la société de courses ou le transporteur à utiliser.

Pour plus d'informations sur le programme CSR de HP, contactez votre Mainteneur Agrée local. Pour plus d'informations sur ce programme en Amérique du Nord, consultez le site Web HP (http://www.hp.com/go/selfrepair).

Riparazione da parte del cliente

Per abbreviare i tempi di riparazione e garantire una maggiore flessibilità nella sostituzione di parti difettose, i prodotti HP sono realizzati con numerosi componenti che possono essere riparati direttamente dal cliente (CSR, Customer Self Repair). Se in fase di diagnostica HP (o un centro di servizi o di assistenza HP) identifica il guasto come riparabile mediante un ricambio CSR, HP lo spedirà direttamente al cliente per la sostituzione. Vi sono due categorie di parti CSR:

Obbligatorie – Parti che devono essere necessariamente riparate dal cliente. Se il cliente ne affida la riparazione ad HP, deve sostenere le spese di spedizione e di manodopera per il servizio.

Opzionali – Parti la cui riparazione da parte del cliente è facoltativa. Si tratta comunque di componenti progettati per questo scopo. Se tuttavia il cliente ne richiede la sostituzione ad HP, potrebbe dover sostenere spese addizionali a seconda del tipo di garanzia previsto per il prodotto.

NOTA: alcuni componenti HP non sono progettati per la riparazione da parte del cliente. Per rispettare la garanzia, HP richiede che queste parti siano sostituite da un centro di assistenza autorizzato. Tali parti sono identificate da un "No" nel Catalogo illustrato dei componenti.

In base alla disponibilità e alla località geografica, le parti CSR vengono spedite con consegna entro il giorno lavorativo seguente. La consegna nel giorno stesso o entro quattro ore è offerta con un supplemento di costo solo in alcune zone. In caso di necessità si può richiedere l'assistenza telefonica di un addetto del centro di supporto tecnico HP. Nel materiale fornito con una parte di ricambio CSR, HP specifica se il cliente deve restituire dei componenti. Qualora sia richiesta la resa ad HP del componente difettoso, lo si deve spedire ad HP entro un determinato periodo di tempo, generalmente cinque (5) giorni lavorativi. Il componente difettoso deve essere restituito con la documentazione associata nell'imballo di spedizione fornito. La mancata restituzione del componente può comportare la fatturazione del ricambio da parte di HP. Nel caso di riparazione da parte del cliente, HP sostiene tutte le spese di spedizione e resa e sceglie il corriere/vettore da utilizzare.

Per ulteriori informazioni sul programma CSR di HP contattare il centro di assistenza di zona. Per il programma in Nord America fare riferimento al sito Web HP (http://www.hp.com/go/selfrepair).

Customer Self Repair

HP Produkte enthalten viele CSR-Teile (Customer Self Repair), um Reparaturzeiten zu minimieren und höhere Flexibilität beim Austausch defekter Bauteile zu ermöglichen. Wenn HP (oder ein HP Servicepartner) bei der Diagnose feststellt, dass das Produkt mithilfe eines CSR-Teils repariert werden kann, sendet Ihnen HP dieses Bauteil zum Austausch direkt zu. CSR-Teile werden in zwei Kategorien unterteilt:

Zwingend – Teile, für die das Customer Self Repair-Verfahren zwingend vorgegeben ist. Wenn Sie den Austausch dieser Teile von HP vornehmen lassen, werden Ihnen die Anfahrt- und Arbeitskosten für diesen Service berechnet.

Optional – Teile, für die das Customer Self Repair-Verfahren optional ist. Diese Teile sind auch für Customer Self Repair ausgelegt. Wenn Sie jedoch den Austausch dieser Teile von HP vornehmen lassen möchten, können bei diesem Service je nach den für Ihr Produkt vorgesehenen Garantiebedingungen zusätzliche Kosten anfallen.

HINWEIS: Einige Teile sind nicht für Customer Self Repair ausgelegt. Um den Garantieanspruch des Kunden zu erfüllen, muss das Teil von einem HP Servicepartner ersetzt werden. Im illustrierten Teilekatalog sind diese Teile mit „No" bzw. „Nein" gekennzeichnet.

CSR-Teile werden abhängig von der Verfügbarkeit und vom Lieferziel am folgenden Geschäftstag geliefert. Für bestimmte Standorte ist eine Lieferung am selben Tag oder innerhalb von vier Stunden gegen einen Aufpreis verfügbar. Wenn Sie Hilfe benötigen, können Sie das HP technische Support Center anrufen und sich von einem Mitarbeiter per Telefon helfen lassen. Den Materialien, die mit einem CSR-Ersatzteil geliefert werden, können Sie entnehmen, ob das defekte Teil an HP zurückgeschickt werden muss. Wenn es erforderlich ist, das defekte Teil an HP zurückzuschicken, müssen Sie dies innerhalb eines vorgegebenen Zeitraums tun, in der Regel innerhalb von fünf (5) Geschäftstagen. Das defekte Teil muss mit der zugehörigen Dokumentation in der Verpackung zurückgeschickt werden, die im Lieferumfang enthalten ist. Wenn Sie das defekte Teil nicht zurückschicken, kann HP Ihnen das Ersatzteil in Rechnung stellen. Im Falle von Customer Self Repair kommt HP für alle Kosten für die Lieferung und Rücksendung auf und bestimmt den Kurier-/Frachtdienst.

Weitere Informationen über das HP Customer Self Repair Programm erhalten Sie von Ihrem Servicepartner vor Ort. Informationen über das CSR-Programm in Nordamerika finden Sie auf der HP Website unter (http://www.hp.com/go/selfrepair).

Reparaciones del propio cliente

Los productos de HP incluyen muchos componentes que el propio usuario puede reemplazar (*Customer Self Repair*, CSR) para minimizar el tiempo de reparación y ofrecer una mayor flexibilidad a la hora de realizar sustituciones de componentes defectuosos. Si, durante la fase de diagnóstico, HP (o los proveedores o socios de servicio de HP) identifica que una reparación puede llevarse a cabo mediante el uso de un componente CSR, HP le enviará dicho componente directamente para que realice su sustitución. Los componentes CSR se clasifican en dos categorías:

- **Obligatorio:** componentes para los que la reparación por parte del usuario es obligatoria. Si solicita a HP que realice la sustitución de estos componentes, tendrá que hacerse cargo de los gastos de desplazamiento y de mano de obra de dicho servicio.
- **Opcional:** componentes para los que la reparación por parte del usuario es opcional. Estos componentes también están diseñados para que puedan ser reparados por el usuario. Sin embargo, si precisa que HP realice su sustitución, puede o no conllevar costes adicionales, dependiendo del tipo de servicio de garantía correspondiente al producto.

NOTA: Algunos componentes no están diseñados para que puedan ser reparados por el usuario. Para que el usuario haga valer su garantía, HP pone como condición que un proveedor de servicios autorizado realice la sustitución de estos componentes. Dichos componentes se identifican con la palabra "No" en el catálogo ilustrado de componentes.

Según la disponibilidad y la situación geográfica, los componentes CSR se enviarán para que lleguen a su destino al siguiente día laborable. Si la situación geográfica lo permite, se puede solicitar la entrega en el mismo día o en cuatro horas con un coste adicional. Si precisa asistencia técnica, puede llamar al Centro de asistencia técnica de HP y recibirá ayuda telefónica por parte de un técnico. Con el envío de materiales para la sustitución de componentes CSR, HP especificará si los componentes defectuosos deberán devolverse a HP. En aquellos casos en los que sea necesario devolver algún componente a HP, deberá hacerlo en el periodo de tiempo especificado, normalmente cinco días laborables. Los componentes defectuosos deberán devolverse con toda la documentación relacionada y con el embalaje de envío. Si no enviara el componente defectuoso requerido, HP podrá cobrarle por el de sustitución. En el caso de todas sustituciones que lleve a cabo el cliente, HP se hará cargo de todos los gastos de envío y devolución de componentes y escogerá la empresa de transporte que se utilice para dicho servicio.

Para obtener más información acerca del programa de Reparaciones del propio cliente de HP, póngase en contacto con su proveedor de servicios local. Si está interesado en el programa para Norteamérica, visite la página web de HP siguiente (http://www.hp.com/go/selfrepair).

Customer Self Repair

Veel onderdelen in HP producten zijn door de klant zelf te repareren, waardoor de reparatieduur tot een minimum beperkt kan blijven en de flexibiliteit in het vervangen van defecte onderdelen groter is. Deze onderdelen worden CSR-onderdelen (Customer Self Repair) genoemd. Als HP (of een HP Service Partner) bij de diagnose vaststelt dat de reparatie kan worden uitgevoerd met een CSR-onderdeel, verzendt HP dat onderdeel rechtstreeks naar u, zodat u het defecte onderdeel daarmee kunt vervangen. Er zijn twee categorieën CSR-onderdelen:

Verplicht: Onderdelen waarvoor reparatie door de klant verplicht is. Als u HP verzoekt deze onderdelen voor u te vervangen, worden u voor deze service reiskosten en arbeidsloon in rekening gebracht.

Optioneel: Onderdelen waarvoor reparatie door de klant optioneel is. Ook deze onderdelen zijn ontworpen voor reparatie door de klant. Als u echter HP verzoekt deze onderdelen voor u te vervangen, kunnen daarvoor extra kosten in rekening worden gebracht, afhankelijk van het type garantieservice voor het product.

OPMERKING: Sommige HP onderdelen zijn niet ontwikkeld voor reparatie door de klant. In verband met de garantievoorwaarden moet het onderdeel door een geautoriseerde Service Partner worden vervangen. Deze onderdelen worden in de geïllustreerde onderdelencatalogus aangemerkt met "Nee".

Afhankelijk van de leverbaarheid en de locatie worden CSR-onderdelen verzonden voor levering op de eerstvolgende werkdag. Levering op dezelfde dag of binnen vier uur kan tegen meerkosten worden aangeboden, indien dit mogelijk is gezien de locatie. Indien assistentie gewenst is, belt u een HP Service Partner om via de telefoon technische ondersteuning te ontvangen. HP vermeldt in de documentatie bij het vervangende CSR-onderdeel of het defecte onderdeel aan HP moet worden geretourneerd. Als het defecte onderdeel aan HP moet worden teruggezonden, moet u het defecte onderdeel binnen een bepaalde periode, gewoonlijk vijf (5) werkdagen, retourneren aan HP. Het defecte onderdeel moet met de bijbehorende documentatie worden geretourneerd in het meegeleverde verpakkingsmateriaal. Als u het defecte onderdeel niet terugzendt, kan HP u voor het vervangende onderdeel kosten in rekening brengen. Bij reparatie door de klant betaalt HP alle verzendkosten voor het vervangende en geretourneerde onderdeel en kiest HP zelf welke koerier/transportonderneming hiervoor wordt gebruikt.

Neem contact op met een Service Partner voor meer informatie over het Customer Self Repair programma van HP. Informatie over Service Partners vindt u op de HP website (http://www.hp.com/go/selfrepair).

Reparo feito pelo cliente

Os produtos da HP são projetados com muitas peças para reparo feito pelo cliente (CSR) de modo a minimizar o tempo de reparo e permitir maior flexibilidade na substituição de peças com defeito. Se, durante o período de diagnóstico, a HP (ou fornecedores/parceiros de serviço da HP) concluir que o reparo pode ser efetuado pelo uso de uma peça CSR, a peça de reposição será enviada diretamente ao cliente. Existem duas categorias de peças CSR:

Obrigatória – Peças cujo reparo feito pelo cliente é obrigatório. Se desejar que a HP substitua essas peças, serão cobradas as despesas de transporte e mão-de-obra do serviço.

Opcional – Peças cujo reparo feito pelo cliente é opcional. Essas peças também são projetadas para o reparo feito pelo cliente. No entanto, se desejar que a HP as substitua, pode haver ou não a cobrança de taxa adicional, dependendo do tipo de serviço de garantia destinado ao produto.

OBSERVAÇÃO: Algumas peças da HP não são projetadas para o reparo feito pelo cliente. A fim de cumprir a garantia do cliente, a HP exige que um técnico autorizado substitua a peça. Essas peças estão identificadas com a marca "No" (Não), no catálogo de peças ilustrado.

Conforme a disponibilidade e o local geográfico, as peças CSR serão enviadas no primeiro dia útil após o pedido. Onde as condições geográficas permitirem, a entrega no mesmo dia ou em quatro horas pode ser feita mediante uma taxa adicional. Se precisar de auxílio, entre em contato com o Centro de suporte técnico da HP para que um técnico o ajude por telefone. A HP especifica nos materiais fornecidos com a peça CSR de reposição se a peça com defeito deve ser devolvida à HP. Nos casos em que isso for necessário, é preciso enviar a peça com defeito à HP dentro do período determinado, normalmente cinco (5) dias úteis. A peça com defeito deve ser enviada com a documentação correspondente no material de transporte fornecido. Caso não o faça, a HP poderá cobrar a reposição. Para as peças de reparo feito pelo cliente, a HP paga todas as despesas de transporte e de devolução da peça e determina a transportadora/serviço postal a ser utilizado.

Para obter mais informações sobre o programa de reparo feito pelo cliente da HP, entre em contato com o fornecedor de serviços local. Para o programa norte-americano, visite o site da HP (http://www.hp.com/go/selfrepair).

カスタマーセルフリペア

修理時間を短縮し、故障部品の交換における高い柔軟性を確保するために、HP製品には多数のCSR部品があります。診断の際に、CSR部品を使用すれば修理ができるとHP（HPまたはHP正規保守代理店）が判断した場合、HPはその部品を直接、お客様に発送し、お客様に交換していただきます。CSR部品には以下の2通りがあります。

- **必須 -** カスタマーセルフリペアが必須の部品。当該部品について、もしもお客様がHPに交換作業を依頼される場合には、その修理サービスに関する交通費および人件費がお客様に請求されます。
- **任意 -** カスタマーセルフリペアが任意である部品。この部品もカスタマーセルフリペア用です。当該部品について、もしもお客様がHPに交換作業を依頼される場合には、お買い上げの製品に適用される保証サービス内容の範囲内においては、別途費用を負担していただくことなく保証サービスを受けることができます。

注： HP製品の一部の部品は、カスタマーセルフリペア用ではありません。製品の保証を継続するためには、HPまたはHP正規保守代理店による交換作業が必須となります。部品カタログには、当該部品がカスタマーセルフリペア除外品である旨が記載されています。

部品供給が可能な場合、地域によっては、CSR部品を翌営業日に届くように発送します。また、地域によっては、追加費用を負担いただくことにより同日または4時間以内に届くように発送することも可能な場合があります。サポートが必要なときは、HPの修理受付窓口に電話していただければ、技術者が電話でアドバイスします。交換用のCSR部品または同梱物には、故障部品をHPに返送する必要があるかどうかが表示されています。故障部品をHPに返送する必要がある場合は、指定期限内（通常は5営業日以内）に故障部品をHPに返送してください。故障部品を返送する場合は、届いた時の梱包箱に関連書類とともに入れてください。故障部品を返送しない場合、HPから部品費用が請求されます。カスタマーセルフリペアの際には、HPは送料および部品返送費を全額負担し、使用する宅配便会社や運送会社を指定します。

客户自行维修

HP 产品提供许多客户自行维修 (CSR) 部件，以尽可能缩短维修时间和在更换缺陷部件方面提供更大的灵活性。如果在诊断期间 HP（或 HP 服务提供商或服务合作伙伴）确定可以通过使用 CSR 部件完成维修，HP 将直接把该部件发送给您进行更换。有两类 CSR 部件：

- **强制性的** — 要求客户必须自行维修的部件。如果您请求 HP 更换这些部件，则必须为该服务支付差旅费和人工费用。
- **可选的** — 客户可以选择是否自行维修的部件。这些部件也是为客户自行维修设计的。不过，如果您要求 HP 为您更换这些部件，则根据为您的产品指定的保修服务类型，HP 可能收取或不再收取任何附加费用。

注：某些 HP 部件的设计并未考虑客户自行维修。为了满足客户保修的需要，HP 要求授权服务提供商更换相关部件。这些部件在部件图解目录中标记为“否”。

CSR 部件将在下一个工作日发运（取决于备货情况和允许的地理范围）。在允许的地理范围内，可在当天或四小时内发运，但要收取额外费用。如果需要帮助，您可以致电 HP 技术支持中心，将会有技术人员通过电话为您提供帮助。HP 会在随更换的 CSR 部件发运的材料中指明是否必须将有缺陷的部件返还给 HP。如果要求您将有缺陷的部件返还给 HP，那么您必须在规定期限内（通常是五 (5) 个工作日）将缺陷部件发给 HP。有缺陷的部件必须随所提供的发运材料中的相关文件一起返还。如果未能送还有缺陷的部件，HP 可能会要求您支付更换费用。客户自行维修时，HP 将承担所有相关运输和部件返回费用，并指定快递商/承运商。

有关 HP 客户自行维修计划的详细信息，请与您当地的服务提供商联系。有关北美地区的计划，请访问 HP 网站 (http://www.hp.com/go/selfrepair)。

客戶自行維修

HP 產品設計了許多「客戶自行維修」(CSR) 的零件以減少維修時間，並且使得更換瑕疵零件時能有更大的彈性。如果在診斷期間 HP（或 HP 服務供應商或維修夥伴）辨認出此項維修工作可以藉由使用 CSR 零件來完成，則 HP 將直接寄送該零件給您作更換。CSR 零件分為兩種類別：

- **強制的** — 客戶自行維修所使用的零件是強制性的。如果您要求 HP 更換這些零件，HP 將會向您收取此服務所需的外出費用與勞動成本。
- **選購的** — 客戶自行維修所使用的零件是選購的。這些零件也設計用於客戶自行維修之用。不過，如果您要求 HP 為您更換，則可能需要也可能不需要負擔額外的費用，端視針對此產品指定的保固服務類型而定。

備註：某些 HP 零件沒有消費者可自行維修的設計。為符合客戶保固，HP 需要授權的服務供應商更換零件。這些零件在圖示的零件目錄中，被標示為「否」。

基於材料取得及環境允許的情況下，CSR 零件將於下一個工作日以快遞寄送。在環境的允許下當天或四小時內送達，則可能需要額外的費用。若您需要協助，可致電「HP 技術支援中心」，會有一位技術人員透過電話來協助您。不論損壞的零件是否必須退回，HP 皆會在與 CSR 替換零件一起運送的材料中註明。若要將損壞的零件退回 HP，您必須在指定的一段時間內（通常為五 (5) 個工作天），將損壞的零件寄回 HP。損壞的零件必須與寄送資料中隨附的相關技術文件一併退還。如果無法退還損壞的零件，HP 可能要向您收取替換費用。針對客戶自行維修情形，HP 將負責所有運費及零件退還費用並指定使用何家快遞/貨運公司。

如需 HP 的「客戶自行維修」方案詳細資訊，請連絡您當地的服務供應商。至於北美方案，請參閱 HP 網站 (http://www.hp.com/go/selfrepair)。

고객 셀프 수리

HP 제품은 수리 시간을 최소화하고 결함이 있는 부품 교체 시 더욱 융통성을 발휘할 수 있도록 하기 위해 고객 셀프 수리(CSR) 부품을 다량 사용하여 설계되었습니다. 진단 기간 동안 HP(또는 HP 서비스 공급업체 또는 서비스 협력업체)에서 CSR 부품을 사용하여 수리가 가능하다고 판단되면 HP는 해당 부품을 바로 사용자에게 보내어 사용자가 교체할 수 있도록 합니다. CSR 부품에는 두 가지 종류가 있습니다.

- **고객 셀프 수리가 의무 사항인 필수 부품.** 사용자가 HP에 이 부품의 교체를 요청할 경우 이 서비스에 대한 출장비 및 작업비가 청구됩니다.
- **고객 셀프 수리가 선택 사항인 부품.** 이 부품들도 고객 셀프 수리가 가능하도록 설계되었습니다. 하지만 사용자가 HP에 이 부품의 교체를 요청할 경우 사용자가 구입한 제품에 해당하는 보증 서비스 유형에 따라 추가 비용 없이 교체가 가능할 수 있습니다.

참고: 일부 HP 부품은 고객 셀프 수리가 불가능하도록 설계되었습니다. HP는 만족스러운 고객 보증을 위해 공인 서비스 제공업체를 통해 부품을 교체하도록 하고 있습니다. 이러한 부품들은 Illustrated Parts Catalog에 "No"라고 표시되어 있습니다.

CSR 부품은 재고 상태와 지리적 조건이 허용하는 경우 다음 영업일 납품이 가능하도록 배송이 이루어집니다. 지리적 조건이 허용하는 경우 추가 비용이 청구되는 조건으로 당일 또는 4시간 배송이 가능할 수도 있습니다. 도움이 필요하시면 HP 기술 지원 센터로 전화하십시오. 전문 기술자가 전화로 도움을 줄 것입니다. HP는 결함이 발생한 부품을 HP로 반환해야 하는지 여부를 CSR 교체 부품과 함께 배송된 자료에 지정합니다. 결함이 발생한 부품을 HP로 반환해야 하는 경우에는 지정된 기간 내(통상 영업일 기준 5일)에 HP로 반환해야 합니다. 이 때 결함이 발생한 부품은 제공된 포장 재료에 넣어 관련 설명서와 함께 반환해야 합니다. 결함이 발생한 부품을 반환하지 않는 경우 HP가 교체 부품에 대해 비용을 청구할 수 있습니다. 고객 셀프 수리의 경우, HP는 모든 운송 및 부품 반환 비용을 부담하며 이용할 운송업체 및 택배 서비스를 결정합니다.

HP 고객 셀프 수리 프로그램에 대한 자세한 내용은 가까운 서비스 제공업체에 문의하십시오. 북미 지역의 프로그램에 대해서는 HP 웹 사이트(http://www.hp.com/go/selfrepair)를 참조하십시오.

Acronyms and abbreviations

ACPI

Advanced Configuration and Power Interface

CIM

common information model

DVFS

dynamic voltage and frequency scaling

IOPS

input/output operations per second

LEB

Logical Erase Block

LVM

Logical Volume Manager

MD

multiple devices

MIB

management information base

NAND

Not AND

NUMA

Non-Uniform Memory Architecture

PSP

HP ProLiant Support Pack

RFC

request for comments

RHEL

Red Hat Enterprise Linux

RPM

Red Hat Package Manager

SMH

System Management Homepage

SMI-S

Storage Management Initiative Specification

TSM

technical support mode

UEK

Unbreakable Enterprise Kernel

VSL

virtual storage layer

Documentation feedback

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Index

A

Adding a file system 33
advanced NUMA configuration example 83, 85
array, mounting 31
authorized reseller 97
automatic power draw 24

B

battery replacement notice 96
before you contact HP 97
building Debian package 12
building driver from source package 10, 12
building RPM package 10

C

CIM 43
command line, installing from 18
command-line utilities 42, 58
configuration 24
configuring a ZFS pool 37
configuring IO Accelerator paging support 34, 87
configuring RAID 30, 31
configuring the IO Accelerator using ESXi 35
contact information 97
contacting HP 97
CSR (customer self repair) 97
customer self repair (CSR) 97

D

DEB package 11
defragmentation 56
Detach Device 56
determining bus number 82, 84
determining device ID 82, 84
device status 43
disabling auto attach 51, 52
disabling DVFS 39
disabling paging support 88
disabling the driver 49, 50
discard (TRIM) in Linux 27
discard (TRIM) in Windows 27
discard support 27
documentation feedback 107
downloading files 19, 97
driver loading 13, 50, 51
driver loading, controlling 13
driver unloads 15, 49

E

enabling paging support 88
enabling the driver 50, 51
error messages 92
ESXi 17, 35, 36, 43, 49, 50, 52, 62, 81
ESXi command-line installation 18
European Union notice 96

F

filesystems, mounting 30
FIO_AFFINITY parameters 64, 84
fio-attach utility 59
fio-beacon utility 60
fio-bugreport utility 60
fio-config utility 64
fio-detach utility 66
fio-firmware 67
fio-format utility 68
fio-pci-check utility 69
fio-status utility 69
fio-sure-erase utility 72
fio-update-iodrive utility 74
firmware, upgrading 23
full-slot power draw 24

H

health metrics 79
health monitoring techniques 79
health reserves percentage 45
HP IO Accelerator Management Tool 42
HP technical support 97
HP, contacting 97

I

information messages 93

init script 14, 15, 30
installation overview 8, 16, 18, 21, 94
installation wizard 95
installing on VMware ESXi 20
installing software 16, 22
installing VSL files using the command line 21
installing VSL files using vCLI 21
installing, DEB package 11
installing, RPM package 9
introduction 7

L

LED indicators 45
limiting APCI C-states 39
Linux 8, 27, 28, 30, 31, 40, 42, 47, 49, 51, 61, 82
Linux kernel upgrade 49
loading the driver 22
Logical Volume Manager 30

M

maintenance 47, 53, 56
management tools 42, 43
managing devices 42
mdadm 31, 32, 33
modifying VMware resource pool 36
module parameters 53, 76
monitoring conditions 43
monitoring devices 42
monitoring IO Accelerator health 79
mounting array 31
multipath storage 15

N

naming 33
non-paged memory pool 88
NUMA architecture 82
NUMA configuration 82
NUMA node override parameter 82, 83

O

one-time configuration 28
override parameter, enabling 25
overview 7

P

page files, setting up 34, 89
PCIe power override, enabling 24
PCIe power, enabling 24
performance and tuning 39
performance and tuning, introduction 39
persistent configuration 28
phone numbers 97

R

RAID 34, 56
RAID 0/Striped 32, 37
RAID 1/Mirrored 32, 37
RAID 10 33, 38
RAID configuration 30, 31, 34
RAM consumption for page files 87
regulatory compliance identification numbers 96
regulatory compliance notices 96
rescan 53
rescan times 53
reserving memory 36
resources 97
RPM package 9

S

series number 96
setting driver options 28, 33
setting interrupt handler affinity 40
setting NUMA affinity 40
silent installation 17
silent uninstallation 17
SMI-S (Storage Management Initiative Specification) 43
SNMP agentx subagent 43
software installation 8, 15, 16, 17, 21, 22
software overview 7
software RAID and health monitoring 80
Solaris 21, 37, 42, 49, 50, 51, 52, 63
SSH (Secure Shell) 43, 58
support 97
support and other resources 97
system log errors 85

T

Taiwan battery recycling notice 96
technical support 97
telephone numbers 97
temperature 44
transferring VSL files 19
Trim configurations 27
Trim support 27
Trim, platforms 27
troubleshooting 92

troubleshooting event log messages 92
TSM 43, 58

U

udev 14, 15
uninstalling driver DEB package 48
uninstalling driver RPM package 48
uninstalling software 47
uninstalling the driver 47, 48, 49
uninstalling utilities 47
unloading the driver 49
unmanaged shutdown issues 53
upgrading driver software 56
upgrading firmware 23
using init scripts 14, 15, 30
using module parameters 53, 76
using the IO Accelerator as a swap 28, 34, 37
using the Logical Volume Manager 30
using udev script 14, 15
utilities 22, 58
utilities reference 58

V

verbose event log parameter 92
verifying page file operation 90
viewing log files 92
Virtual Memory performance 91
VM disks 35
VMDirectPathIO 18, 81
VMDirectPathIO, working with IO Accelerators 81
VSL SDK 43

W

website, HP 97
Windows 15, 27, 33, 34, 40, 41, 42, 48, 52, 61, 64, 84, 92, 94
Windows page file management 89
Windows page files 87
wizard, manual installation 95
write (health reserves) status 45

Z

ZFS pool, configuring 37
ZFS pool, single device 37
ZFS pools 37, 38